# Linear Algebra I

## Section 1.1 Matrices

For any two positive integers $m$ and $n$, an $m \times n$ matrix $A$ is a rectangular array of $mn$ numbers:

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1j} & \cdots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{i1} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & & \vdots & & \vdots \\ a_{m1} & \cdots & a_{mj} & \cdots & a_{mn} \end{bmatrix}$$

*j-th column* (above $a_{1j}$) and *i-th row* (beside $a_{in}$)

in $m$ rows and $n$ columns. We will sometimes shorten this to $A = (a_{ij})_{m \times n}$. If $m = n$, $A$ is called a square matrix. The entry $a_{ij}$ is located in the $i$-th row and the $j$-th column.

Two matrices $A$ and $B$ are equal if they have the same size (number of rows and columns) and corresponding entries are equal: $a_{ij} = b_{ij}$ for all $i$ and $j$.

If $A = (a_{ij})_{m \times n}$ and $B = (b_{ij})_{m \times n}$ are both $m \times n$ matrices, then the **sum** $A + B$ is the $m \times n$ matrix whose $(i, j)$ entry is $a_{ij} = b_{ij}$ for each ordered pair $(i, j)$.

Definition 1.1.1 (Scalar Multiplication)

If $A$ is an $m \times n$ matrix and $\alpha$ is a scalar, then $\alpha A$ is the $m \times n$ matrix whose $(i, j)$ entry is $\alpha a_{ij}$.

If all entries are all 0, then such matrix is called zero matrix and denoted by $O$. For instance, $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ is a $2 \times 3$ zero matrix.

If $O$ represents the zero matrix, with the same size as $A$, whose, then

$$A + O = O + A = A.$$

It acts as an additive identity on the set of all $m \times n$ matrices. Furthermore, each $m \times n$ matrix $A$ has an additive inverse. Indeed,

$$A + (-1)A = O = (-1)A + A.$$

It is customary to denote the additive inverse by $-A$. Thus,

$$-A = (-1)A.$$

If $A$ and $B$ are both $m \times n$ matrices, we define $A - B$ to be $A + (-1)B$. Then it turns out that $A - B$ is formed by subtracting the corresponding entry of $B$ from each entry of $A$.

Example 1.1.2

$$3\begin{pmatrix} 2 & -1 & 7 \\ 3 & 0 & 4 \end{pmatrix} - 5\begin{pmatrix} 3 & -4 & 2 \\ 1 & 2 & -5 \end{pmatrix} = \begin{pmatrix} 3(2)-5(3) & 3(-1)-5(-4) & 3(7)-5(2) \\ 3(3)-5(1) & 3(0)-5(2) & 3(4)-5(-5) \end{pmatrix} = \begin{pmatrix} -9 & 17 & 11 \\ 4 & -10 & 37 \end{pmatrix}$$

Definition 1.1.3

If $A = (a_{ij})_{m \times k}$ is an $m \times k$ matrix and $B = (b_{ij})_{k \times n}$ is an $k \times n$ matrix, then the product $AB = C = (c_{ij})_{m \times n}$ is the $m \times n$ matrix whose entries are defined by

$$c_{ij} = \sum_{r=1}^{k} a_{ir} b_{rj}.$$

If number of column of $A$ is not number of row of $B$, then the matrix multiplication of $AB$ is not valid.

Example 1.1.4

$$AB = \begin{pmatrix} -3 & -1 & -1 & -5 & 1 \\ -3 & -3 & -4 & -5 & 3 \\ -1 & -5 & 3 & -1 & -3 \\ 3 & 2 & -1 & -4 & -4 \\ -5 & 3 & -2 & -1 & -1 \end{pmatrix} \begin{pmatrix} 0 & 5 & 3 & -3 & 0 \\ 5 & 5 & 2 & 0 & -1 \\ 3 & 0 & -4 & -1 & -4 \\ 4 & 0 & -3 & 2 & 4 \\ 4 & -2 & 0 & -1 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} -24 & -22 & 8 & -1 & -12 \\ -35 & -36 & 16 & 0 & 8 \\ -32 & -24 & -22 & 1 & -20 \\ -25 & 33 & 29 & -12 & -26 \\ 1 & -8 & 2 & 16 & -2 \end{pmatrix}$$

$$\begin{aligned} & (-3)(3)+(-1)(2)+(-1)(-4)+(-5)(-3)+(1)(0) \\ = \; & -9 + -2 + 4 + 15 + 0 \\ = \; & 8 \end{aligned}$$

Remark 1.1.5

$AB \neq BA$ even the matrix multiplication is valid. Consider $A = \begin{pmatrix} 0 & 1 \\ 0 & 2 \end{pmatrix}, B = \begin{pmatrix} 3 & 4 \\ 0 & 0 \end{pmatrix}$.

$$AB = \begin{pmatrix} 0 & 1 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 3 & 4 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \quad BA = \begin{pmatrix} 3 & 4 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 11 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

The example in Remark 1.1.5 also shows us the following:

If a product $AB$ is the zero matrix, you cannot conclude in general that either $A = O$ or $B = O$. Furthermore, if $AB = AC$ for a nonzero matrix $A$, then it is not true in general that $B = C$. If we consider $C = \begin{pmatrix} 5 & 6 \\ 0 & 0 \end{pmatrix}$,

$$AB = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = AC.$$

However $B \neq C$.

Definition 1.1.6

The **transpose** of an $m \times n$ matrix $A = (a_{ij})_{m \times n}$ is the $n \times m$ matrix, denoted by $A^T = (b_{ij})_{n \times m}$, such that

$$b_{ij} = a_{ji}$$

for $i = 1, \cdots, m$ and $j = 1, \cdots, n$. That is, rows of $A^T$ are formed from the corresponding columns of $A$.

Example 1.1.7

$$A = \begin{pmatrix} -1 & -5 \\ 3 & 2 \\ -5 & 3 \end{pmatrix} \qquad A^T = \begin{pmatrix} -1 & 3 & -5 \\ -5 & 2 & 3 \end{pmatrix}$$

$$B = \begin{pmatrix} 3 & -3 & 0 \\ 2 & 0 & -1 \end{pmatrix} \qquad B^T = \begin{pmatrix} 3 & 2 \\ -3 & 0 \\ 0 & -1 \end{pmatrix}$$

Theorem 1.1.8

Let $A$ and $B$ denote matrices whose sizes are appropriate for the following sums and products.

(i) $\left(A^T\right)^T = A$

(ii) $\left(A + B\right)^T = A^T + B^T$

(iii) For any scalar $r$, $\left(rA\right)^T = rA^T$

(iv) $\left(AB\right)^T = B^T A^T$

Example 1.1.9

$$AB = \begin{pmatrix} -1 & -5 \\ 3 & 2 \\ -5 & 3 \end{pmatrix} \begin{pmatrix} 3 & -3 & 0 \\ 2 & 0 & -1 \end{pmatrix} = \begin{pmatrix} -13 & 3 & 5 \\ 13 & -9 & -2 \\ -9 & 15 & -3 \end{pmatrix}$$

$$B^T A^T = \begin{pmatrix} 3 & 2 \\ -3 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} -1 & 3 & -5 \\ -5 & 2 & 3 \end{pmatrix} = \begin{pmatrix} -13 & 13 & -9 \\ 3 & -9 & 15 \\ 5 & -2 & -3 \end{pmatrix} = (AB)^T$$

Definition 1.1.10

An $n \times n$ matrix $A$ is said to be **symmetric** if $A^T = A$.

Example 1.1.11

$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{pmatrix}$ is symmetric but $\begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 3 & 1 \end{pmatrix}$ is not.

An $n \times n$ matrix $A$ is **diagonal** if $a_{ij} = 0$ whenever $i \neq j$. The matrices

$$\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \quad \begin{pmatrix} 5 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 7 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix}$$

are all diagonal.

Definition 1.1.12

The $n \times n$ **identity matrix** is the matrix $I_n = \left(\delta_{ij}\right)_{n \times n}$, where

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j. \end{cases}$$

$$I_1 = (1), \quad I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Just as the number 1 acts as an identity for the multiplication of real numbers, identity matrix $I_n$ acts as an identity for matrix multiplication; that is,

$$I_n A = A I_n = A$$

for any $n \times n$ matrix $A$.

An $n \times n$ matrix $A$ is said to be **upper triangular** if $a_{ij} = 0$ for $i > j$ and **lower triangular** if $a_{ij} = 0$ for $i < j$ . For instance,

$\begin{pmatrix} 3 & 2 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 5 \end{pmatrix}$ is an upper triangular matrix and $\begin{pmatrix} 1 & 0 & 0 \\ 6 & 0 & 0 \\ 1 & 4 & 3 \end{pmatrix}$ is a lower triangular matrix.

Theorem 1.1.13

Let $A$ be an $m \times n$ matrix, and let $\alpha$ and $\beta$ be scalars. For any matrices $B$ and $C$ having sizes for which the indicated sums and products are defined, we have

(i) $A + B = B + A$

(ii) $(A + B) + C = A + (B + C)$

(iii) $(AB)C = A(BC)$

(iv) $A(B + C) = AB + AC$

(v) $(A + B)C = AC + BC$

(vi) $\alpha(\beta A) = (\alpha\beta)A$

(vii) $\alpha(AB) = (\alpha A)B = A(\alpha B)$

(viii) $(\alpha + \beta)A = \alpha A + \beta A$

(ix) $\alpha(A + B) = \alpha A + \alpha B$

(x) $I_m A = AI_n = A$

Example 1.1.14

If

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 \\ -3 & 2 \end{pmatrix} \quad \text{and} \quad C = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix},$$

verify that $A(BC) = (AB)C$.

Solution

$$(AB)C = \left(\begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}\begin{pmatrix}2 & 1\\ -3 & 2\end{pmatrix}\right)\cdot\begin{pmatrix}1 & 0\\ 2 & 1\end{pmatrix} = \begin{pmatrix}-4 & 5\\ -6 & 11\end{pmatrix}\begin{pmatrix}1 & 0\\ 2 & 1\end{pmatrix} = \begin{pmatrix}6 & 5\\ 16 & 11\end{pmatrix}$$

$$A(BC) = \begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}\cdot\left(\begin{pmatrix}2 & 1\\ -3 & 2\end{pmatrix}\begin{pmatrix}1 & 0\\ 2 & 1\end{pmatrix}\right) = \begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}\begin{pmatrix}4 & 1\\ 1 & 2\end{pmatrix} = \begin{pmatrix}6 & 5\\ 16 & 11\end{pmatrix} = (AB)C$$

Since $(AB)C = A(BC)$, we may simply omit the parentheses and write $ABC$.

If $A$ is an $n\times n$ matrix and $k$ a positive integer, then $A^k$ denotes the product of $k$ copies of $A$:

$$A^k = \underbrace{A\cdots A}_{k \text{ times}}.$$

Suppose matrices $A$ and $B$ are partitioned into blocks as

$$A = \begin{pmatrix}A_{11} & A_{12} & \cdots & A_{1k}\\ A_{21} & A_{22} & \cdots & A_{2k}\\ \vdots & \vdots & \ddots & \vdots\\ A_{m1} & A_{m2} & \cdots & A_{mk}\end{pmatrix}, \qquad B = \begin{pmatrix}B_{11} & B_{12} & \cdots & B_{1n}\\ B_{21} & B_{22} & \cdots & B_{2n}\\ \vdots & \vdots & \ddots & \vdots\\ B_{k1} & B_{k2} & \cdots & B_{kn}\end{pmatrix}$$

where $A_{ij}$ and $B_{ij}$ are matrices.

Assume that the partitioning has been done in such a way that each product $A_{il}B_{lj}$ can be formed. Define

$$C_{ij} = \begin{pmatrix}A_{i1} & A_{i2} & \cdots & A_{ik}\end{pmatrix}\begin{pmatrix}B_{1j}\\ B_{2j}\\ \vdots\\ B_{kj}\end{pmatrix} = \sum_{l=1}^{k} A_{il}B_{lj}.$$

Then $C = AB$ can be partitioned into blocks as follows:

$$C = \begin{pmatrix}C_{11} & C_{12} & \cdots & C_{1n}\\ C_{21} & C_{22} & \cdots & C_{2n}\\ \vdots & \vdots & \ddots & \vdots\\ C_{m1} & C_{m2} & \cdots & C_{mn}\end{pmatrix}.$$

Example 1.1.15

An example of a $2\times 3$ block partitioned matrix is

$$A=\left(\begin{array}{cc|cc|c} -3 & -1 & -1 & -5 & 1 \\ -3 & -3 & -4 & -5 & 3 \\ \hline -1 & -5 & 3 & -1 & -3 \\ 3 & 2 & -1 & -4 & -4 \\ -5 & 3 & -2 & -1 & -1 \end{array}\right)=\left(\begin{array}{c|c|c} A_{11} & A_{12} & A_{13} \\ \hline A_{21} & A_{22} & A_{23} \end{array}\right)$$

where

$$A_{11}=\begin{pmatrix} -3 & -1 \\ -3 & -3 \end{pmatrix}, A_{12}=\begin{pmatrix} -1 & -5 \\ -4 & -5 \end{pmatrix}, A_{13}=\begin{pmatrix} 1 \\ 3 \end{pmatrix}, A_{21}=\begin{pmatrix} -1 & -5 \\ 3 & 2 \\ -5 & 3 \end{pmatrix}, A_{22}=\begin{pmatrix} 3 & -1 \\ -1 & -4 \\ -2 & -1 \end{pmatrix}, A_{23}=\begin{pmatrix} -3 \\ -4 \\ -1 \end{pmatrix}.$$

Consider a $3\times 2$ block partitioned matrix

$$B=\left(\begin{array}{cc|ccc} 0 & 5 & 3 & -3 & 0 \\ 5 & 5 & 2 & 0 & -1 \\ \hline 3 & 0 & -4 & -1 & -4 \\ 4 & 0 & -3 & 2 & 4 \\ \hline 4 & -2 & 0 & -1 & 3 \end{array}\right)=\left(\begin{array}{c|c} B_{11} & B_{12} \\ \hline B_{21} & B_{22} \\ \hline B_{31} & B_{32} \end{array}\right)$$

where $B_{11}=\begin{pmatrix} 0 & 5 \\ 5 & 5 \end{pmatrix}, B_{12}=\begin{pmatrix} 3 & -3 & 0 \\ 2 & 0 & -1 \end{pmatrix}, B_{21}=\begin{pmatrix} 3 & 0 \\ 4 & 0 \end{pmatrix}, B_{22}=\begin{pmatrix} -4 & -1 & -4 \\ -3 & 2 & 4 \end{pmatrix}, B_{31}=\begin{pmatrix} 4 & -2 \end{pmatrix}$ and $B_{32}=\begin{pmatrix} 0 & -1 & 3 \end{pmatrix}$.

Then

$$AB=\left(\begin{array}{c|c|c} A_{11} & A_{12} & A_{13} \\ \hline A_{21} & A_{22} & A_{23} \end{array}\right)\left(\begin{array}{c|c} B_{11} & B_{12} \\ \hline B_{21} & B_{22} \\ \hline B_{31} & B_{32} \end{array}\right)=\left(\begin{array}{c|c} A_{11}B_{11}+A_{12}B_{21}+A_{13}B_{31} & A_{11}B_{12}+A_{12}B_{22}+A_{13}B_{32} \\ \hline A_{21}B_{11}+A_{22}B_{21}+A_{23}B_{31} & A_{21}B_{12}+A_{22}B_{22}+A_{23}B_{32} \end{array}\right)$$

$$\begin{aligned} A_{11}B_{11}+A_{12}B_{21}+A_{13}B_{31} &= \begin{pmatrix} -3 & -1 \\ -3 & -3 \end{pmatrix}\begin{pmatrix} 0 & 5 \\ 5 & 5 \end{pmatrix}+\begin{pmatrix} -1 & -5 \\ -4 & -5 \end{pmatrix}\begin{pmatrix} 3 & 0 \\ 4 & 0 \end{pmatrix}+\begin{pmatrix} 1 \\ 3 \end{pmatrix}\begin{pmatrix} 4 & -2 \end{pmatrix} \\ &= \begin{pmatrix} -5 & -20 \\ -15 & -30 \end{pmatrix}+\begin{pmatrix} -23 & 0 \\ -32 & 0 \end{pmatrix}+\begin{pmatrix} 4 & -2 \\ 12 & -6 \end{pmatrix}=\begin{pmatrix} -24 & -22 \\ -35 & -36 \end{pmatrix} \end{aligned}$$

$$A_{11}B_{12}+A_{12}B_{22}+A_{13}B_{32}=\begin{pmatrix}-3&-1\\-3&-3\end{pmatrix}\begin{pmatrix}3&-3&0\\2&0&-1\end{pmatrix}+\begin{pmatrix}-1&-5\\-4&-5\end{pmatrix}\begin{pmatrix}-4&-1&-4\\-3&2&4\end{pmatrix}+\begin{pmatrix}1\\3\end{pmatrix}\begin{pmatrix}0&-1&3\end{pmatrix}$$
$$=\begin{pmatrix}-11&9&1\\-15&9&3\end{pmatrix}+\begin{pmatrix}19&-9&-16\\31&-6&-4\end{pmatrix}+\begin{pmatrix}0&-1&3\\0&-3&9\end{pmatrix}=\begin{pmatrix}8&-1&-12\\16&0&8\end{pmatrix}$$

$$A_{21}B_{11}+A_{22}B_{21}+A_{23}B_{31}=\begin{pmatrix}-1&-5\\3&2\\-5&3\end{pmatrix}\begin{pmatrix}0&5\\5&5\end{pmatrix}+\begin{pmatrix}3&-1\\-1&-4\\-2&-1\end{pmatrix}\begin{pmatrix}3&0\\4&0\end{pmatrix}+\begin{pmatrix}-3\\-4\\-1\end{pmatrix}\begin{pmatrix}4&-2\end{pmatrix}$$
$$=\begin{pmatrix}-25&-30\\10&25\\15&-10\end{pmatrix}+\begin{pmatrix}5&0\\-19&0\\-10&0\end{pmatrix}+\begin{pmatrix}-12&6\\-16&8\\-4&2\end{pmatrix}=\begin{pmatrix}-32&-24\\-25&33\\1&-8\end{pmatrix}$$

$$A_{21}B_{12}+A_{22}B_{22}+A_{23}B_{32}=\begin{pmatrix}-1&-5\\3&2\\-5&3\end{pmatrix}\begin{pmatrix}3&-3&0\\2&0&-1\end{pmatrix}+\begin{pmatrix}3&-1\\-1&-4\\-2&-1\end{pmatrix}\begin{pmatrix}-4&-1&-4\\-3&2&4\end{pmatrix}+\begin{pmatrix}-3\\-4\\-1\end{pmatrix}\begin{pmatrix}0&-1&3\end{pmatrix}$$
$$=\begin{pmatrix}-13&3&5\\13&-9&-2\\-9&15&-3\end{pmatrix}+\begin{pmatrix}-9&-5&-16\\16&-7&-12\\11&0&4\end{pmatrix}+\begin{pmatrix}0&3&-9\\0&4&-12\\0&1&-3\end{pmatrix}=\begin{pmatrix}-22&1&-20\\29&-12&-26\\2&16&-2\end{pmatrix}$$

Hence $AB=\left(\begin{array}{cc|ccc}-24&-22&8&-1&-12\\-35&-36&16&0&8\\\hline-32&-24&-22&1&-20\\-25&33&29&-12&-26\\1&-8&2&16&-2\end{array}\right)$, which is the same as in Example 1.1.4.

Example 1.1.16

Suppose *A* and *C* are $m\times m$ matrices, *B* and *D* are $n\times n$ matrices. Then

$$\left(\begin{array}{c|c}A&O_{m\times n}\\\hline O_{n\times m}&B\end{array}\right)\left(\begin{array}{c|c}C&O_{m\times n}\\\hline O_{n\times m}&D\end{array}\right)=\left(\begin{array}{c|c}AC&O_{m\times n}\\\hline O_{n\times m}&BD\end{array}\right).$$

In particular, for any positive integer *k*, we have

$$\left(\begin{array}{cc|ccc}\lambda&1&0&0&0\\0&\lambda&0&0&0\\\hline0&0&\lambda&1&0\\0&0&0&\lambda&1\\0&0&0&0&\lambda\end{array}\right)^k=\left(\begin{array}{c|c}\begin{pmatrix}\lambda&1\\0&\lambda\end{pmatrix}^k&\begin{matrix}0&0&0\\0&0&0\end{matrix}\\\hline\begin{matrix}0&0\\0&0\\0&0\end{matrix}&\begin{pmatrix}\lambda&1&0\\0&\lambda&1\\0&0&\lambda\end{pmatrix}^k\end{array}\right).$$

## Section 1.2 Solving system of linear equations

Given the system of linear equations

$$(*): \begin{cases} 2y - 8z = 8 \\ x - 2y + z = 0 \\ -4x + 5y + 9z = -9 \end{cases}$$

with the coefficients of each variable aligned in columns, the matrix

$$\begin{pmatrix} 0 & 2 & -8 \\ 1 & -2 & 1 \\ -4 & 5 & 9 \end{pmatrix}$$

is called the coefficient matrix of the system $(*)$, and

$$\left(\begin{array}{ccc|c} 0 & 2 & -8 & 8 \\ 1 & -2 & 1 & 0 \\ -4 & 5 & 9 & -9 \end{array}\right)$$

is called the augmented matrix of the system.

The basic strategy is to replace one system with an equivalent system (i.e., one with the same solution set) that is easier to solve.

**Strategy**

We proceed from equation to equation, from top to bottom.

(i) Rearrange the equations if the position of leading variables do not appear in the ascending order.

(ii) Suppose we get to the *i*-th equation, with leading variable $x_j$ and leading (nonzero) coefficient *c*, so that the equation will be of the form $cx_j + \cdots = b$.

(iii) Divide the *i*-th equation by *c* bring it into the form $x_j + \cdots = b/c$.

(iv) Eliminate $x_j$ from all the other equations, above and below the *i*-th equation, by subtracting suitable multiples of the *i*-th equation.

(v) If an equation zero = nonzero appears in this process, then the system has no solutions.

(vi) Proceed to the next equation.

Repeat such process until the last equation. Solve each equation for its leading variable.

Example 1.2.1

$$\begin{cases} 2y-8z=8 \\ x-2y+z=0 \\ -4x+5y+9z=-9 \end{cases} \qquad \left(\begin{array}{ccc|c} 0 & 2 & -8 & 8 \\ 1 & -2 & 1 & 0 \\ -4 & 5 & 9 & -9 \end{array}\right)$$

$$\xrightarrow{\text{1st eq}\leftrightarrow\text{2nd eq}} \begin{cases} x-2y+z=0 \\ 2y-8z=8 \\ -4x+5y+9z=-9 \end{cases} \qquad \xrightarrow{R_1\leftrightarrow R_2} \left(\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ -4 & 5 & 9 & -9 \end{array}\right)$$

$$\xrightarrow{\text{3rd eq}+4\times(\text{1st eq})\to\text{3rd eq}} \begin{cases} x-2y+z=0 \\ 2y-8z=8 \\ -3y+13z=-9 \end{cases} \qquad \xrightarrow{R_3+4R_1\to R_3} \left(\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 0 & -3 & 13 & -9 \end{array}\right)$$

$$\xrightarrow{\text{2nd eq}\div 2\to\text{2nd eq}} \begin{cases} x-2y+z=0 \\ y-4z=4 \\ -3y+13z=-9 \end{cases} \qquad \xrightarrow{R_2\div 2\to R_2} \left(\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & -3 & 13 & -9 \end{array}\right)$$

$$\xrightarrow{\text{3rd eq}+3\times(\text{2nd eq})\to\text{3rd eq}} \begin{cases} x-2y+z=0 \\ y-4z=4 \\ z=3 \end{cases} \qquad \xrightarrow{R_3+3R_2\to R_3} \left(\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & 0 & 1 & 3 \end{array}\right)$$

Using back substitution, we obtain $z=3$, $y=4+4z=16$, $x=2y-z=29$.

You may also keep elimination until the simplest form.

$$\begin{cases} x-2y+z=0 \\ y-4z=4 \\ z=3 \end{cases} \qquad \left(\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & 0 & 1 & 3 \end{array}\right)$$

$$\xrightarrow{\text{2nd eq}+4\times(\text{3rd eq})\to\text{2nd eq}} \begin{cases} x-2y+z=0 \\ y=16 \\ z=3 \end{cases} \qquad \xrightarrow{R_2+4R_3\to R_2} \left(\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{array}\right)$$

$$\xrightarrow{\text{1st eq}-\text{3rd eq}\to\text{1st eq}} \begin{cases} x-2y=-3 \\ y=16 \\ z=3 \end{cases} \qquad \xrightarrow{R_1-R_3\to R_1} \left(\begin{array}{ccc|c} 1 & -2 & 0 & -3 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{array}\right)$$

$$\xrightarrow{\text{1st eq+2×(2nd) eq}\to\text{1st eq}}\begin{cases} x & & = 29 \\ & y & = 16 \\ & & z = 3 \end{cases} \qquad \xrightarrow{R_1+2R_2\to R_1}\left(\begin{array}{ccc|c} 1 & 0 & 0 & 29 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{array}\right)$$

We still obtain $x = 29, y = 16, z = 3$.

Definition 1.2.2 (**Elementary Row Operations**)

(I) Interchange two rows.

(II) Multiply a row by a nonzero real number.

(III) Replace a row by its sum with a multiple of another row.

Remark 1.2.3

$R_3 + 4R_1 \to R_3$ is a type III row operation, but $R_3 + 4R_1 \to R_1$ is not.

Elementary row operations are equivalent to the left multiplications of an augmented matrix $A$ by elementary matrices $E$.

An elementary matrix of type I is a matrix obtained by interchanging two rows of $I_n$. For example, the matrix

$$E = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

is an elementary matrix of type I since it was obtained by interchanging the first two rows of $I_3$.

$$EA = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ a_{31} & a_{32} & \cdots & a_{3n} \end{pmatrix} = \begin{pmatrix} a_{21} & a_{22} & \cdots & a_{2n} \\ a_{11} & a_{12} & \cdots & a_{1n} \\ a_{31} & a_{32} & \cdots & a_{3n} \end{pmatrix}$$

Left multiplication of $A$ by $E$ is equivalent to the elementary row operation (type I) of interchanging the first and second row.

An elementary matrix of type II is a matrix obtained by multiplying a row of $I_n$ by a nonzero constant. For example, for any nonzero $\alpha$,

$$E=\begin{pmatrix}1&0&0\\0&\alpha&0\\0&0&1\end{pmatrix}$$

is an elementary matrix of type II.

$$EA=\begin{pmatrix}1&0&0\\0&\alpha&0\\0&0&1\end{pmatrix}\begin{pmatrix}a_{11}&a_{12}&\cdots&a_{1n}\\a_{21}&a_{22}&\cdots&a_{2n}\\a_{31}&a_{32}&\cdots&a_{3n}\end{pmatrix}=\begin{pmatrix}a_{11}&a_{12}&\cdots&a_{1n}\\\alpha a_{21}&\alpha a_{22}&\cdots&\alpha a_{2n}\\a_{31}&a_{32}&\cdots&a_{3n}\end{pmatrix}.$$

Left multiplication of $A$ by $E$ is equivalent to the elementary row operation (type II) of multiplying the second row by $\alpha$.

An elementary matrix of type III is a matrix obtained from $I_n$ by adding a multiple of one row to another row. For example, for any $\beta$,

$$E=\begin{pmatrix}1&0&\beta\\0&1&0\\0&0&1\end{pmatrix}$$

is an elementary matrix of type III since it was obtained from $I_n$ by adding a multiple $\beta$ of third row to the first row.

$$EA=\begin{pmatrix}1&0&\beta\\0&1&0\\0&0&1\end{pmatrix}\begin{pmatrix}a_{11}&a_{12}&\cdots&a_{1n}\\a_{21}&a_{22}&\cdots&a_{2n}\\a_{31}&a_{32}&\cdots&a_{3n}\end{pmatrix}=\begin{pmatrix}a_{11}+\beta a_{31}&a_{12}+\beta a_{32}&\cdots&a_{1n}+\beta a_{3n}\\a_{21}&a_{22}&\cdots&a_{2n}\\a_{31}&a_{32}&\cdots&a_{3n}\end{pmatrix}.$$

Left multiplication of $A$ by $E$ is equivalent to the elementary row operation (type III) of adding $\beta$ times the third row to the first row.

Remark 1.2.4

Right multiplication of $A$ by elementary matrices performs elementary column operations.

Example 1.2.5

$$\left(\begin{array}{ccc|c} 0 & 2 & -8 & 8 \\ 1 & -2 & 1 & 0 \\ -4 & 5 & 9 & -9 \end{array}\right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ -4 & 5 & 9 & -9 \end{array}\right) = \left(\begin{array}{ccc} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{array}\right)\left(\begin{array}{ccc|c} 0 & 2 & -8 & 8 \\ 1 & -2 & 1 & 0 \\ -4 & 5 & 9 & -9 \end{array}\right)$$

$$\xrightarrow{R_3+4R_1 \to R_3} \left(\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 0 & -3 & 13 & -9 \end{array}\right) = \left(\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 1 \end{array}\right)\left(\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ -4 & 5 & 9 & -9 \end{array}\right)$$

$$\xrightarrow{R_2 \div 2 \to R_2} \left(\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & -3 & 13 & -9 \end{array}\right) = \left(\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{array}\right)\left(\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 0 & -3 & 13 & -9 \end{array}\right)$$

$$\xrightarrow{R_3+3R_2 \to R_3} \left(\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & 0 & 1 & 3 \end{array}\right) = \left(\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{array}\right)\left(\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & -3 & 13 & -9 \end{array}\right)$$

$$\xrightarrow{R_2+4R_3 \to R_2} \left(\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{array}\right) = \left(\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{array}\right)\left(\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & 0 & 1 & 3 \end{array}\right)$$

$$\xrightarrow{R_1-R_3 \to R_1} \left(\begin{array}{ccc|c} 1 & -2 & 0 & -3 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{array}\right) = \left(\begin{array}{ccc} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right)\left(\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{array}\right)$$

$$\xrightarrow{R_1+2R_2 \to R_1} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 29 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{array}\right) = \left(\begin{array}{ccc} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right)\left(\begin{array}{ccc|c} 1 & -2 & 0 & -3 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{array}\right)$$

Definition 1.2.6

Let $A$ and $B$ be two $m \times n$ matrices. $B$ is **row equivalent** to a matrix $A$ if there exists a finite sequence $E_1, E_2, \cdots, E_k$ of elementary matrices such that

$$B = E_k E_{k-1} \cdots E_1 A.$$

In other words, $B$ is row equivalent to $A$ if $B$ can be obtained from $A$ by a finite number of row operations.

Since

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 29 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{array}\right)$$

$$= \underbrace{\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{R_1+2R_2\to R_1} \underbrace{\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{R_1-R_3\to R_1} \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{pmatrix}}_{R_2+4R_3\to R_2} \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{pmatrix}}_{R_3+3R_2\to R_3} \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{R_2\div 2\to R_2} \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 1 \end{pmatrix}}_{R_3+4R_1\to R_3} \underbrace{\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{R_1\leftrightarrow R_2} \left(\begin{array}{ccc|c} 0 & 2 & -8 & 8 \\ 1 & -2 & 1 & 0 \\ -4 & 5 & 9 & -9 \end{array}\right)$$

$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 29 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{array}\right)$ is row equivalent to $\left(\begin{array}{ccc|c} 0 & 2 & -8 & 8 \\ 1 & -2 & 1 & 0 \\ -4 & 5 & 9 & -9 \end{array}\right)$. Thus,

$\begin{cases} 2y - 8z = 8 \\ x - 2y + z = 0 \\ -4x + 5y + 9z = -9 \end{cases}$ and $\begin{cases} x = 29 \\ y = 16 \\ z = 3 \end{cases}$ have the same solution.

Definition 1.2.7

A matrix is said to be in **row echelon form** if

(i) the first nonzero entry in each nonzero row is 1,

(ii) row $k$ does not consist entirely of zeros, the number of leading zero entries in row $k + 1$ is greater than the number of leading zero entries in row $k$,

(iii) there are rows whose entries are all zero, they are below the rows having nonzero entries.

$$\begin{bmatrix} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & * & * & * & * & * & * & * & * \\ 0 & 0 & 0 & 1 & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 1 & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \end{bmatrix}$$

Definition 1.2.8

The process of using row operations I, II, and III to transform a linear system into one whose augmented matrix is in row echelon form is called **Gaussian elimination**.

Definition 1.2.9

A matrix is said to be in **reduced row echelon form** if

(i) the matrix is in row echelon form,

(ii) the first nonzero entry in each row is the only nonzero entry in its column.

$$\begin{bmatrix} 1 & 0 & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & * & 0 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 1 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 1 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \end{bmatrix}$$

The reduced row echelon form of $A$ is denoted by rref($A$).

A system of linear equations may have a unique solution, infinitely many solutions, or no solution.

If a linear system has no solution, we say that the system is **inconsistent**. If the system has at least one solution, we say that it is **consistent**.

Does the system has solutions? If a solution exists, is it the only one; that is, is the solution unique?

If the row echelon form of the augmented matrix contains a row of the form

$$\begin{pmatrix} 0 & \cdots & 0 \mid k \end{pmatrix}$$

for nonzero $k$, the system is inconsistent. Otherwise, the system will be consistent. If the system is consistent and the nonzero rows of the row echelon form of the matrix form a strictly triangular system

$$\left(\begin{array}{cccc|c} 1 & * & \cdots & * & * \\ 0 & 1 & \ddots & \vdots & * \\ \vdots & \ddots & \ddots & * & \vdots \\ 0 & \cdots & 0 & 1 & * \end{array}\right),$$

the system will have a unique solution. Otherwise, the last row of the row echelon form must be 0.

$$\left(\begin{array}{cccc|c} * & * & \cdots & * & * \\ 0 & \ddots & \ddots & \vdots & \vdots \\ \vdots & \ddots & * & * & * \\ 0 & \cdots & 0 & 0 & 0 \end{array}\right),$$

the system has infinitely many solutions.

Example 1.2.10

Consider the system of linear equations

$$(*): \begin{cases} x + \lambda y + z = \lambda \\ 3x - y + (\lambda+2)z = 7 \\ x - y + z = 3 \end{cases} \quad \text{where } \lambda \in \boldsymbol{R}.$$

Solve $(*)$ for

(a) $\lambda \neq \pm 1$,

(b) $\lambda = -1$,

(c) $\lambda = 1$.

Solution

The augmented matrix of $(*)$ is

$$\left(\begin{array}{ccc|c} 1 & \lambda & 1 & \lambda \\ 3 & -1 & \lambda+2 & 7 \\ 1 & -1 & 1 & 3 \end{array}\right) \xrightarrow[R_3 - R_1 \to R_3]{R_2 - 3R_3 \to R_2} \left(\begin{array}{ccc|c} 1 & \lambda & 1 & \lambda \\ 0 & 2 & \lambda-1 & -2 \\ 0 & -1-\lambda & 0 & 3-\lambda \end{array}\right)$$

$$\xrightarrow{R_2 \div 2 \to R_2} \left(\begin{array}{ccc|c} 1 & \lambda & 1 & \lambda \\ 0 & 1 & \dfrac{\lambda-1}{2} & -1 \\ 0 & -1-\lambda & 0 & 3-\lambda \end{array}\right)$$

$$\xrightarrow{R_3 + (\lambda+1)R_2 \to R_3} \left(\begin{array}{ccc|c} 1 & \lambda & 1 & \lambda \\ 0 & 1 & \dfrac{\lambda-1}{2} & -1 \\ 0 & 0 & \dfrac{(\lambda-1)(\lambda+1)}{2} & -2\lambda+2 \end{array}\right)$$

(a) If $\lambda \neq \pm 1$, $z = -\dfrac{4}{\lambda+1}$, $y = -1 - \dfrac{\lambda-1}{2}z = -1 + 2\dfrac{\lambda-1}{\lambda+1} = \dfrac{\lambda-3}{\lambda+1}$,

$x = \lambda - \lambda y - z = \lambda - \lambda\dfrac{\lambda-3}{\lambda+1} + \dfrac{4}{\lambda+1} = \dfrac{\lambda^2+\lambda-\lambda^2+3\lambda+4}{\lambda+1} = 4$, exactly 1 solution.

(b) If $\lambda = -1$, $\left(\begin{array}{ccc|c} 1 & \lambda & 1 & \lambda \\ 0 & 1 & \dfrac{\lambda-1}{2} & -1 \\ 0 & 0 & \dfrac{(\lambda-1)(\lambda+1)}{2} & -2\lambda+2 \end{array}\right) = \left(\begin{array}{ccc|c} 1 & -1 & 1 & -1 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 4 \end{array}\right)$.

$0z = 4$! No solution. $(*)$ is inconsistent.

(c) If $\lambda = 1$, $\left(\begin{array}{ccc|c} 1 & \lambda & 1 & \lambda \\ 0 & 1 & \dfrac{\lambda-1}{2} & -1 \\ 0 & 0 & \dfrac{(\lambda-1)(\lambda+1)}{2} & -2\lambda+2 \end{array}\right) = \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right)$.

$y = -1, x = 1 - y - z = 2 - z.$

The system has infinite many solutions. Solution set is $\{(2-t,-1,t) \in \boldsymbol{R}^3 : t \in \boldsymbol{R}\}$.

The variables corresponding to the first nonzero elements in each row of the reduced row echelon form are called **leading variables**. The remaining variables are called **free variables**.

Thus, in Example 1.2.10(c), $x$ and $y$ are the leading variables and $z$ is the free variable.

"$z$ is the free variable" means that you are free to choose any value for $z$. Once you choose, the solution of the system is determined by the choice of $z$.

Theorem 1.2.11

If a system of linear equation is consistent, then it has either

(i) infinitely many solutions (if there is at least one free variable), or

(ii) exactly one solution (if all variable are leading).

# Section 2.1 Inverse of a Matrix

Consider the systems of linear equations of $m$ equations and $n$ unknowns

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

It can be expressed as $A\boldsymbol{x} = \boldsymbol{b}$ where

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \quad \boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \quad \text{and} \quad \boldsymbol{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}.$$

Such an equation is called a matrix equation

Roughly speaking, 1 equation eliminates 1 variable. If number of equations is greater than number of unknowns ($m > n$), the system has no solution in most cases. On the other hand, if number of equations is less than number of unknowns ($m < n$), the system has infinitely many solutions in most cases. If number of equations equal number of unknowns ($m = n$), the system has 1 solution in most cases.

Question 2.1.1

Under what condition does the system of linear equations with $m = n$ give us exactly 1 solution?

(Theorem 2.1.9 and Theorem 2.1.26)

Consider an equation $ax = b$.

Case (i): If $a \neq 0,$ then the equation has exactly 1 solution, which is $x = a^{-1}b.$

Case (ii): If $a = 0$ and $b = 0$, then any value $x$ is a solution (infinitely many solutions).

Case (iii): If $a = 0$ and $b \neq 0,$ then $0 = b \neq 0!$ No solution.

Question 2.1.2

If $A$ is an $n\times n$ nonzero matrix, is it possible to find the inverse of $A$? If inverse of $A$ exists, the solution must be unique. We need the definition of inverse of a square matrix.

Definition 2.1.3

An $n\times n$ matrix $A$ is said to be **nonsingular** or **invertible** if there exists an $n\times n$ matrix $B$ such that $AB=BA=I_n$. The matrix $B$ is said to be a **multiplicative inverse** of $A$ and denoted by $A^{-1}$.

Remark 2.1.4

If $B$ and $C$ are both multiplicative inverses of $A$, then

$$B=BI_n=BAC=I_nC=C.$$

Thus, a matrix can have at most one multiplicative inverse.

Example 2.1.5

$$\begin{pmatrix}0&1&0\\1&0&0\\0&0&1\end{pmatrix}\begin{pmatrix}0&1&0\\1&0&0\\0&0&1\end{pmatrix}=\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix},\qquad \begin{pmatrix}0&1&0\\1&0&0\\0&0&1\end{pmatrix}^{-1}=\begin{pmatrix}0&1&0\\1&0&0\\0&0&1\end{pmatrix}$$

For any nonzero $\alpha$,

$$\begin{pmatrix}1&0&0\\0&\alpha&0\\0&0&1\end{pmatrix}\begin{pmatrix}1&0&0\\0&\alpha^{-1}&0\\0&0&1\end{pmatrix}=\begin{pmatrix}1&0&0\\0&\alpha^{-1}&0\\0&0&1\end{pmatrix}\begin{pmatrix}1&0&0\\0&\alpha&0\\0&0&1\end{pmatrix}=\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix},\quad \begin{pmatrix}1&0&0\\0&\alpha&0\\0&0&1\end{pmatrix}^{-1}=\begin{pmatrix}1&0&0\\0&\alpha^{-1}&0\\0&0&1\end{pmatrix}$$

For any $\beta$,

$$\begin{pmatrix}1&0&\beta\\0&1&0\\0&0&1\end{pmatrix}\begin{pmatrix}1&0&-\beta\\0&1&0\\0&0&1\end{pmatrix}=\begin{pmatrix}1&0&-\beta\\0&1&0\\0&0&1\end{pmatrix}\begin{pmatrix}1&0&\beta\\0&1&0\\0&0&1\end{pmatrix}=\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix},\quad \begin{pmatrix}1&0&\beta\\0&1&0\\0&0&1\end{pmatrix}^{-1}=\begin{pmatrix}1&0&-\beta\\0&1&0\\0&0&1\end{pmatrix}$$

Theorem 2.1.6

If $E$ is an elementary matrix, then $E$ is nonsingular and $E^{-1}$ is an elementary matrix of the same type.

Theorem 2.1.7

(a) If $A$ is invertible, then $A^{-1}$ is invertible and

$$(A^{-1})^{-1} = A.$$

(b) If $A$ and $B$ are invertible, then so is $AB$, and

$$(AB)^{-1} = B^{-1}A^{-1}$$

(c) If $A$ is invertible, then so is $A^T$ and

$$(A^T)^{-1} = (A^{-1})^T.$$

(d) If $A$ is invertible, then so is $\alpha A$ for nonzero $\alpha$ and

$$(\alpha A)^{-1} = \alpha^{-1}A^{-1}.$$

Example 2.1.8

The answer of Question 2.1.2 is NO. We can find a nonzero square matrix such that its inverse does not exist.

Consider

$$\begin{cases} x + y = 1 \\ 2x + 2y = 2 \end{cases}$$

The matrix equation is

$$\begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

$\begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}^{-1}$ does not exist. Suppose $\begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}^{-1}$ exists and we can find a $2 \times 2$ matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ such that

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

(1, 1) entry and (1, 2) entry on the left give us $a + 2b = 1$ and $a + 2b = 0$! Impossible. A nonzero square matrix does not guarantee the existence of inverse.

$$\left(\begin{array}{cc|c} 1 & 1 & 1 \\ 2 & 2 & 2 \end{array}\right) \xrightarrow{R_2-2R_1\to R_2} \left(\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 0 & 0 \end{array}\right)$$

Indeed, this system of linear equations have infinitely many solutions $\{(1-t,t): t \in \boldsymbol{R}\}$, but not one solution. It is because the reduced row echelon form of $\begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}$ is not $I_2$.

Theorem 2.1.9

An $n \times n$ matrix $A$ is invertible if and only if $A$ is row equivalent to $I_n$, that is $\operatorname{rref}(A) = I_n$.

To find the inverse of an $n \times n$ matrix $A$, form the $n \times 2n$ matrix $(A \mid I_n)$ and compute $\operatorname{rref}(A \mid I_n)$.

If $\operatorname{rref}(A \mid I_n)$ is of the form $(I_n \mid C)$, then $\operatorname{rref}(A) = I_n$ and $A$ is invertible. There exist a sequence of elementary matrices $E_1, E_2, \cdots, E_k$ such that $E_k E_{k-1} \cdots E_1 A = I_n$.

$$\begin{aligned} (A \mid I_n) &\to E_1(A \mid I_n) = (E_1 A \mid E_1) \\ &\to E_2(E_1 A \mid E_1) = (E_2 E_1 A \mid E_2 E_1) \\ &\vdots \\ &\to E_k(E_{k-1} \cdots E_1 A \mid E_{k-1} \cdots E_1) = (E_k E_{k-1} \cdots E_1 A \mid E_k E_{k-1} \cdots E_1) = (I_n \mid \underbrace{E_k E_{k-1} \cdots E_1}_{A^{-1}}) \end{aligned}$$

$A^{-1} = E_k E_{k-1} \cdots E_1$.

If $\operatorname{rref}(A \mid I_n)$ is of another form (that is, its left half fails to be $I_n$), then $A$ is not invertible. Note that the left half of $\operatorname{rref}(A \mid I_n)$ is rref($A$).

Example 2.1.10

Let $A = \begin{pmatrix} 0 & 2 & -8 \\ 1 & -2 & 1 \\ -4 & 5 & 9 \end{pmatrix}$. Find $A^{-1}$ using row operation.

Solution

$$\left(\begin{array}{ccc|ccc} 0 & 2 & -8 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 & 1 & 0 \\ -4 & 5 & 9 & 0 & 0 & 1 \end{array}\right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{ccc|ccc} 1 & -2 & 1 & 0 & 1 & 0 \\ 0 & 2 & -8 & 1 & 0 & 0 \\ -4 & 5 & 9 & 0 & 0 & 1 \end{array}\right) = \left(\begin{array}{ccc} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{array}\right)\left(\begin{array}{ccc|ccc} 0 & 2 & -8 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 & 1 & 0 \\ -4 & 5 & 9 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_3+4R_1 \to R_3} \left(\begin{array}{ccc|ccc} 1 & -2 & 1 & 0 & 1 & 0 \\ 0 & 2 & -8 & 1 & 0 & 0 \\ 0 & -3 & 13 & 0 & 4 & 1 \end{array}\right) = \left(\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 1 \end{array}\right)\left(\begin{array}{ccc|ccc} 1 & -2 & 1 & 0 & 1 & 0 \\ 0 & 2 & -8 & 1 & 0 & 0 \\ -4 & 5 & 9 & 0 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_2 \div 2 \to R_2} \left(\begin{array}{ccc|ccc} 1 & -2 & 1 & 0 & 1 & 0 \\ 0 & 1 & -4 & 1/2 & 0 & 0 \\ 0 & -3 & 13 & 0 & 4 & 1 \end{array}\right) = \left(\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{array}\right)\left(\begin{array}{ccc|ccc} 1 & -2 & 1 & 0 & 1 & 0 \\ 0 & 2 & -8 & 1 & 0 & 0 \\ 0 & -3 & 13 & 0 & 4 & 1 \end{array}\right)$$

$$\xrightarrow{R_3+3R_2 \to R_3} \left(\begin{array}{ccc|ccc} 1 & -2 & 1 & 0 & 1 & 0 \\ 0 & 1 & -4 & 1/2 & 0 & 0 \\ 0 & 0 & 1 & 3/2 & 4 & 1 \end{array}\right) = \left(\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{array}\right)\left(\begin{array}{ccc|ccc} 1 & -2 & 1 & 0 & 1 & 0 \\ 0 & 1 & -4 & 1/2 & 0 & 0 \\ 0 & -3 & 13 & 0 & 4 & 1 \end{array}\right)$$

$$\xrightarrow{R_2+4R_3 \to R_2} \left(\begin{array}{ccc|ccc} 1 & -2 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 13/2 & 16 & 4 \\ 0 & 0 & 1 & 3/2 & 4 & 1 \end{array}\right) = \left(\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{array}\right)\left(\begin{array}{ccc|ccc} 1 & -2 & 1 & 0 & 1 & 0 \\ 0 & 1 & -4 & 1/2 & 0 & 0 \\ 0 & 0 & 1 & 3/2 & 4 & 1 \end{array}\right)$$

$$\xrightarrow{R_1-R_3 \to R_1} \left(\begin{array}{ccc|ccc} 1 & -2 & 0 & -3/2 & -3 & -1 \\ 0 & 1 & 0 & 13/2 & 16 & 4 \\ 0 & 0 & 1 & 3/2 & 4 & 1 \end{array}\right) = \left(\begin{array}{ccc} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right)\left(\begin{array}{ccc|ccc} 1 & -2 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 13/2 & 16 & 4 \\ 0 & 0 & 1 & 3/2 & 4 & 1 \end{array}\right)$$

$$\xrightarrow{R_1+2R_2 \to R_1} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 23/2 & 29 & 7 \\ 0 & 1 & 0 & 13/2 & 16 & 4 \\ 0 & 0 & 1 & 3/2 & 4 & 1 \end{array}\right) = \left(\begin{array}{ccc} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right)\left(\begin{array}{ccc|ccc} 1 & -2 & 0 & -3/2 & -3 & -1 \\ 0 & 1 & 0 & 13/2 & 16 & 4 \\ 0 & 0 & 1 & 3/2 & 4 & 1 \end{array}\right)$$

$$A^{-1} = \begin{pmatrix} 23/2 & 29 & 7 \\ 13/2 & 16 & 4 \\ 3/2 & 4 & 1 \end{pmatrix}.$$

Theorem 2.1.11 (*LU* decomposition)

Let $A$ be an $n \times n$ matrix. Suppose $A$ is row equivalent to an upper triangular matrix $U$ using only elementary matrix of type III that add a multiple of one row to another row below it. In this case, there exist lower triangular elementary matrices $E_1, \cdots, E_k$ with 1's on the diagonal such that

$$E_k \cdots E_1 A = U.$$

Then $A = LU$ where $L = (E_k \cdots E_1)^{-1} = E_1^{-1} \cdots E_k^{-1}$ is a lower triangular matrix with 1's on the diagonal.

Example 2.1.12

$$\underbrace{\begin{pmatrix} 2 & 3 & 1 \\ 4 & 1 & 4 \\ 3 & 4 & 6 \end{pmatrix}}_{A} \xrightarrow{R_2 - 2R_1 \to R_2} \begin{pmatrix} 2 & 3 & 1 \\ 0 & -5 & 2 \\ 3 & 4 & 6 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{E_1} \underbrace{\begin{pmatrix} 2 & 3 & 1 \\ 4 & 1 & 4 \\ 3 & 4 & 6 \end{pmatrix}}_{A}$$

$$\xrightarrow{R_3 - 1.5R_1 \to R_3} \begin{pmatrix} 2 & 3 & 1 \\ 0 & -5 & 2 \\ 0 & -0.5 & 4.5 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1.5 & 0 & 1 \end{pmatrix}}_{E_2} \begin{pmatrix} 2 & 3 & 1 \\ 0 & -5 & 2 \\ 3 & 4 & 6 \end{pmatrix}$$

$$\xrightarrow{R_3 - 0.1R_2 \to R_3} \underbrace{\begin{pmatrix} 2 & 3 & 1 \\ 0 & -5 & 2 \\ 0 & 0 & 4.3 \end{pmatrix}}_{U} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -0.1 & 1 \end{pmatrix}}_{E_3} \begin{pmatrix} 2 & 3 & 1 \\ 0 & -5 & 2 \\ 0 & -\frac{1}{2} & \frac{9}{2} \end{pmatrix}$$

$$\underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -0.1 & 1 \end{pmatrix}}_{E_3} \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1.5 & 0 & 1 \end{pmatrix}}_{E_2} \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{E_1} \underbrace{\begin{pmatrix} 2 & 3 & 1 \\ 4 & 1 & 4 \\ 3 & 4 & 6 \end{pmatrix}}_{A} = \underbrace{\begin{pmatrix} 2 & 3 & 1 \\ 0 & -5 & 2 \\ 0 & 0 & 4.3 \end{pmatrix}}_{U}$$

$$\underbrace{\begin{pmatrix} 2 & 3 & 1 \\ 4 & 1 & 4 \\ 3 & 4 & 6 \end{pmatrix}}_{A} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1.5 & 0.1 & 1 \end{pmatrix}}_{L} \underbrace{\begin{pmatrix} 2 & 3 & 1 \\ 0 & -5 & 2 \\ 0 & 0 & 4.3 \end{pmatrix}}_{U}$$

The (2, 1) entry of $L$ comes from $R_2 - 2R_1 \to R_2$. The (3, 1) entry of $L$ comes from $R_3 - 1.5R_1 \to R_3$. The (3, 2) entry of $L$ comes from $R_3 - 0.1R_2 \to R_3$.

When $A = LU$, the equation $A\boldsymbol{x} = \boldsymbol{b}$ can be written as $L(U\boldsymbol{x}) = \boldsymbol{b}$. Writing $\boldsymbol{y}$ for $U\boldsymbol{x}$, we can find $\boldsymbol{x}$ by solving the pair of equations

$$L\boldsymbol{y} = \boldsymbol{b} \quad \text{and} \quad U\boldsymbol{x} = \boldsymbol{y}.$$

First solve $L\boldsymbol{y} = \boldsymbol{b}$ for $\boldsymbol{y}$, and then solve $U\boldsymbol{x} = \boldsymbol{y}$ for $\boldsymbol{x}$ by forward and backward substitution. Each equation is easy to solve because $L$ and $U$ are triangular.

Example 2.1.13

Solve the system the following system of linear equations by $LU$ decomposition.

$$\begin{cases} 2x_1 + 3x_2 + x_3 = -4 \\ 4x_1 + x_2 + 4x_3 = 9 \\ 3x_1 + 4x_2 + 6x_3 = 0 \end{cases}$$

Solution

We have the matrix equation

$$\begin{pmatrix} 2 & 3 & 1 \\ 4 & 1 & 4 \\ 3 & 4 & 6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -4 \\ 9 \\ 0 \end{pmatrix}$$

where $\begin{pmatrix} 2 & 3 & 1 \\ 4 & 1 & 4 \\ 3 & 4 & 6 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1.5 & 0.1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 3 & 1 \\ 0 & -5 & 2 \\ 0 & 0 & 4.3 \end{pmatrix}$ as in Example 2.1.12.

Consider $\begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1.5 & 0.1 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} -4 \\ 9 \\ 0 \end{pmatrix}$. We get

$$y_1 = -4, \quad y_2 = 9 - 2y_1 = 17, \quad y_3 = 0 - 1.5y_1 - 0.1y_2 = 4.3.$$

Consider $\begin{pmatrix} 2 & 3 & 1 \\ 0 & -5 & 2 \\ 0 & 0 & 4.3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -4 \\ 17 \\ 4.3 \end{pmatrix}$. We get

$$x_3 = 1, \quad x_2 = -\tfrac{1}{5}(17 - 2x_3) = -3, \quad x_1 = \tfrac{1}{2}(-4 - 3x_2 - x_3) = 2.$$

Definition 2.1.14

Let $A = (a_{ij})$ be an $n \times n$ matrix, and let $M_{ij}$ denote the $(n-1) \times (n-1)$ matrix obtained from $A$ by deleting the row and column containing $a_{ij}$. The determinant of $M_{ij}$ is called the **minor** of $a_{ij}$. We define the **cofactor** $A_{ij}$ of $a_{ij}$ by

$$A_{ij} = (-1)^{i+j} \det(M_{ij}).$$

$$A=\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2j} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nj} & \cdots & a_{nn} \end{bmatrix}, \quad M_{ij}=\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1j} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2j} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ \cancel{a_{i1}} & \cancel{a_{i2}} & \cancel{\cdots} & \cancel{a_{ij}} & \cancel{\cdots} & \cancel{a_{in}} \\ \vdots & \vdots & & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nj} & \cdots & a_{nn} \end{bmatrix}$$

Definition 2.1.15

The **determinant** of an $n\times n$ matrix $A$, denoted $\det(A)$ or $|A|$, is a scalar associated with the matrix $A$ that is defined inductively as

$$\det(A)=\begin{cases} a_{11} & \text{if } n=1 \\ a_{11}A_{11}+a_{12}A_{12}+\cdots+a_{1n}A_{1n} & \text{if } n>1 \end{cases}$$

where

$$A_{1j}=(-1)^{1+j}\det(M_{1j}) \quad j=1,\cdots,n$$

are cofactors associated with the entries in the first row of $A$.

The appropriate sign of cofactors of $1\times1, 2\times2, 3\times3$ matrices are $(+)$, $\begin{pmatrix} + & - \\ - & + \end{pmatrix}$, $\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$.

Example 2.1.16

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}=a_{11}a_{22}-a_{12}a_{21}$$

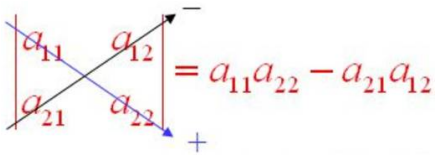
$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}=a_{11}\begin{vmatrix} \cancel{a_{11}} & \cancel{a_{12}} & \cancel{a_{13}} \\ \cancel{a_{21}} & a_{22} & a_{23} \\ \cancel{a_{31}} & a_{32} & a_{33} \end{vmatrix}-a_{12}\begin{vmatrix} \cancel{a_{11}} & \cancel{a_{12}} & \cancel{a_{13}} \\ a_{21} & \cancel{a_{22}} & a_{23} \\ a_{31} & \cancel{a_{32}} & a_{33} \end{vmatrix}+a_{13}\begin{vmatrix} \cancel{a_{11}} & \cancel{a_{12}} & \cancel{a_{13}} \\ a_{21} & a_{22} & \cancel{a_{23}} \\ a_{31} & a_{32} & \cancel{a_{33}} \end{vmatrix}$$

$$=a_{11}(a_{22}a_{33}-a_{23}a_{32})-a_{12}(a_{21}a_{33}-a_{23}a_{31})+a_{13}(a_{21}a_{32}-a_{22}a_{31})$$

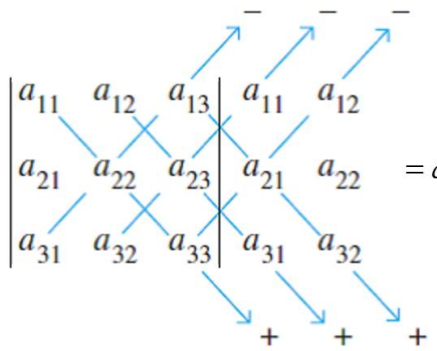
$$=a_{11}a_{22}a_{33}+a_{12}a_{23}a_{31}+a_{13}a_{21}a_{32}-a_{11}a_{23}a_{32}-a_{12}a_{21}a_{33}-a_{13}a_{22}a_{31}$$

Remark 2.1.17

Shortcut for calculating determinant of $2\times 2$ and $3\times 3$ matrices.




$$= a_{11}a_{22} - a_{21}a_{12}$$




$$= a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31}$$

Theorem 2.1.18

If $A$ is an $n\times n$ matrix with $n\geq 2$, then $\det(A)$ can be expressed as a cofactor expansion using any row or column of $A$; that is,

$$\begin{aligned}\det(A) &= a_{i1}A_{i1} + a_{i2}A_{i2} + \cdots + a_{in}A_{in} \\ &= a_{1j}A_{1j} + a_{2j}A_{2j} + \cdots + a_{nj}A_{nj}\end{aligned}$$

for $i = 1,\cdots,n$ and $j = 1,\cdots,n.$ In particular, $\det(A) = \det(A^T)$.

Example 2.1.19

Use cofactor expansion to compute $\det(A)$ for

$$A = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 9 & 1 & 3 & 0 \\ 9 & 2 & 2 & 0 \\ 5 & 0 & 0 & 3 \end{pmatrix}.$$

Solution

Looking for rows or columns with as many zeros as possible, we choose the second column:

$$\det(A) = a_{12}A_{12} + a_{22}A_{22} + a_{32}A_{32} + a_{42}A_{42}$$

$$= 1 \cdot \det \begin{pmatrix} 1 & 0 & 1 & 2 \\ 9 & 1 & 3 & 0 \\ 9 & 2 & 2 & 0 \\ 5 & 0 & 0 & 3 \end{pmatrix} - 2 \cdot \det \begin{pmatrix} 1 & 0 & 1 & 2 \\ 9 & 1 & 3 & 0 \\ 9 & 2 & 2 & 0 \\ 5 & 0 & 0 & 3 \end{pmatrix}$$

$$= \det \begin{pmatrix} 1 & 1 & 2 \\ 9 & 2 & 0 \\ 5 & 0 & 3 \end{pmatrix} - 2\det \begin{pmatrix} 1 & 1 & 2 \\ 9 & 3 & 0 \\ 5 & 0 & 3 \end{pmatrix}$$

$$= \underbrace{2\det \begin{pmatrix} 9 & 2 \\ 5 & 0 \end{pmatrix}}_{\det \begin{pmatrix} 1 & 1 & 2 \\ 9 & 2 & 0 \\ 5 & 0 & 3 \end{pmatrix}} + \underbrace{3\det \begin{pmatrix} 1 & 1 \\ 9 & 2 \end{pmatrix}}_{\det \begin{pmatrix} 1 & 1 & 2 \\ 9 & 2 & 0 \\ 5 & 0 & 3 \end{pmatrix}} - 2\left[ \underbrace{2\det \begin{pmatrix} 9 & 3 \\ 5 & 0 \end{pmatrix}}_{\det \begin{pmatrix} 1 & 1 & 2 \\ 9 & 3 & 0 \\ 5 & 0 & 3 \end{pmatrix}} + \underbrace{3\det \begin{pmatrix} 1 & 1 \\ 9 & 3 \end{pmatrix}}_{\det \begin{pmatrix} 1 & 1 & 2 \\ 9 & 3 & 0 \\ 5 & 0 & 3 \end{pmatrix}} \right]$$

$$= -20 - 21 - 2(-30 - 18) = 55$$

If we expand along the last row, we also get the same answer.

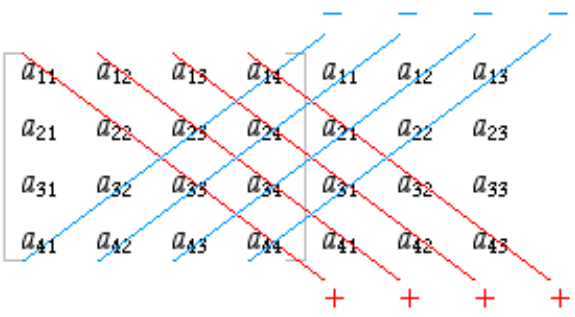
$$\det(A) = a_{41}A_{41} + a_{42}A_{42} + a_{43}A_{43} + a_{44}A_{44}$$

$$= -5\det \begin{pmatrix} 0 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 2 & 0 \end{pmatrix} + 3\det \begin{pmatrix} 1 & 0 & 1 \\ 9 & 1 & 3 \\ 9 & 2 & 2 \end{pmatrix}$$

$$= \underbrace{-10\det \begin{pmatrix} 1 & 3 \\ 2 & 2 \end{pmatrix}}_{\det \begin{pmatrix} 0 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 2 & 0 \end{pmatrix}} + 3\left[ \underbrace{\det \begin{pmatrix} 1 & 3 \\ 2 & 2 \end{pmatrix}}_{\det \begin{pmatrix} 1 & 0 & 1 \\ 9 & 1 & 3 \\ 9 & 2 & 2 \end{pmatrix}} + \underbrace{\det \begin{pmatrix} 9 & 1 \\ 9 & 2 \end{pmatrix}}_{\det \begin{pmatrix} 1 & 0 & 1 \\ 9 & 1 & 3 \\ 9 & 2 & 2 \end{pmatrix}} \right]$$

$$= 40 + 3(-4 + 9) = 55$$

This shortcut in Remark 2.1.17 is not valid for any $n \times n$ matrix for $n \geq 4$.




Incorrect!

If we use this shortcut in Example 2.1.19, we get

$$\det\begin{pmatrix} 1 & 0 & 1 & 2 \\ 9 & 1 & 3 & 0 \\ 9 & 2 & 2 & 0 \\ 5 & 0 & 0 & 3 \end{pmatrix} = (1)(1)(2)(3) - (2)(3)(2)(5) - (1)(1)(9)(3) = -81$$

which is incorrect.

Example 2.1.20

Determinant of an upper and lower triangular matrices are product of diagonal entries, that is

$$\det\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & a_{nn} \end{pmatrix} = a_{11}a_{22}\cdots a_{nn}, \quad \det\begin{pmatrix} b_{11} & 0 & \cdots & 0 \\ b_{21} & b_{22} & \ddots & \vdots \\ \vdots & \vdots & \ddots & 0 \\ b_{m1} & b_{2m} & \cdots & b_{mm} \end{pmatrix} = b_{11}b_{22}\cdots b_{mm}.$$

Remark 2.1.21

$\det(A+B) \neq \det(A) + \det(B)$ in general. Consider $A = I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $B = -I_2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$.

$\det A = \det(B) = 1$

$\det(A+B) = \det\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = 0 \neq 2 = \det(A) + \det(B)$

Theorem 2.1.22

If $A$ is an $n \times n$ matrix and $E$ is an $n \times n$ elementary matrix, then

$$\det(EA) = \det(E)\det(A)$$

where

$$\det(E) = \begin{cases} -1 & \text{if } E \text{ is of type I} \\ \alpha \neq 0 & \text{if } E \text{ is of type II} \\ 1 & \text{if } E \text{ is of type III} \end{cases}$$

Similar results hold for elementary column operations, that is,

$$\det(AE) = \det(E)\det(A)$$

Theorem 2.1.23

Let $A = \left(a_{ij}\right)_{n\times n}$. Then

(i) If $A$ has a row or column consisting entirely of zeros, then $\det(A) = 0$. For example,

$$\det\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 0 \end{pmatrix} = 0.$$

(ii) If $A$ has two identical rows or two identical columns, then $\det(A) = 0$. For example,

$$\det\begin{pmatrix} a_{31} & a_{32} & a_{33} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = 0.$$

(iii) A determinant can be expressed as the sum of two determinants by expressing every element in any row (or column) as the sum of two terms. For example,

$$\det\begin{pmatrix} a_{11}+b_{11} & a_{12}+b_{12} & a_{13}+b_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \det\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} + \det\begin{pmatrix} b_{11} & b_{12} & b_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}.$$

(iv) If we interchange any two rows (or columns) of $\det(A)$, then the result is the negative of $\det(A)$. For example,

$$\det\begin{pmatrix} a_{21} & a_{22} & a_{23} \\ a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = -\det\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}.$$

(v) If all the elements of one row (or column) of $\det(A)$ have a common factor $\alpha$, then we can take this common factor out of the determinant. For example,

$$\det\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ \alpha a_{21} & \alpha a_{22} & \alpha a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = \alpha\det\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}.$$

Hence, $\det(\alpha A) = \alpha^n \det(A)$ for any $\alpha$.

(vi) The value of the determinant remains unchanged if the elements of a row (or column) are changed by adding to them any multiple of the corresponding elements in any other row (or column). For example,

$$\det\begin{pmatrix} a_{11}+\beta a_{31} & a_{12}+\beta a_{32} & a_{13}+\beta a_{33} \\ a_{21} & a_{22} & a_{2n} \\ a_{31} & a_{32} & a_{3n} \end{pmatrix} = \det\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}.$$

Example 2.1.24

Given 1445, 2091, 6188 are multiples of 17 and 126, 4508, 518 are multiples of 7.

Show that $\begin{vmatrix} 1 & 44 & 5 \\ 2 & 9 & 1 \\ 6 & 18 & 8 \end{vmatrix}$ is divisible by $2\times 2\times 7\times 17$ without expansion.

Solution

$$\begin{vmatrix} 1 & 44 & 5 \\ 2 & 9 & 1 \\ 6 & 18 & 8 \end{vmatrix} = \begin{vmatrix} 1 & 44 & 1445 \\ 2 & 9 & 2091 \\ 6 & 18 & 6188 \end{vmatrix} \quad [C_3 + 10C_2 + 1000C_1 \to C_3]$$

$$= 17\begin{vmatrix} 1 & 44 & 85 \\ 2 & 9 & 123 \\ 6 & 18 & 364 \end{vmatrix}$$

$\begin{vmatrix} 1 & 44 & 5 \\ 2 & 9 & 1 \\ 6 & 18 & 8 \end{vmatrix}$ is divisible by 17.

$$\begin{vmatrix} 1 & 44 & 5 \\ 2 & 9 & 1 \\ 6 & 18 & 8 \end{vmatrix} = \begin{vmatrix} 1 & 44 & 5 \\ 2 & 9 & 1 \\ 126 & 4508 & 518 \end{vmatrix} \quad [R_3 + 10R_2 + 100R_1 \to R_3]$$

$$= 7\begin{vmatrix} 1 & 44 & 85 \\ 2 & 9 & 123 \\ 18 & 644 & 74 \end{vmatrix}$$

$\begin{vmatrix} 1 & 44 & 5 \\ 2 & 9 & 1 \\ 6 & 18 & 8 \end{vmatrix}$ is divisible by 7.

$$\begin{vmatrix} 1 & 44 & 5 \\ 2 & 9 & 1 \\ 6 & 18 & 8 \end{vmatrix} = \begin{vmatrix} 1 & 44 & 5 \\ 2 & 9 & 1 \\ 12 & 124 & 20 \end{vmatrix} \quad [R_3 + 2R_2 + 2R_1 \to R_3]$$

$$= 2 \times 2 \begin{vmatrix} 1 & 44 & 5 \\ 2 & 9 & 1 \\ 3 & 31 & 5 \end{vmatrix}$$

$\begin{vmatrix} 1 & 44 & 5 \\ 2 & 9 & 1 \\ 6 & 18 & 8 \end{vmatrix}$ is divisible by $2 \times 2$.

Since $2 \times 2, 7, 17$ are relatively prime, $\begin{vmatrix} 1 & 44 & 5 \\ 2 & 9 & 1 \\ 6 & 18 & 8 \end{vmatrix}$ is divisible by $2 \times 2 \times 7 \times 17$.

Indeed, $\begin{vmatrix} 1 & 44 & 5 \\ 2 & 9 & 1 \\ 6 & 18 & 8 \end{vmatrix} = -2 \times 2 \times 7 \times 17.$

Theorem 2.1.25

An $n \times n$ matrix $A$ is singular if and only if $\det(A) = 0$.

Proof

Let $U$ = rref($A$). Then there exist a sequence of elementary matrices $E_1, E_2, \cdots, E_k$ such that $E_k E_{k-1} \cdots E_1 A = U$. By Theorem 2.1.22, we have

$$\det(E_k)\det(E_{k-1}) \cdots \det(E_1)\det(A) = \det(E_k E_{k-1} \cdots E_1 A) = \det(U).$$

Since $\det(E_i) \neq 0$ for $i = 1, \cdots, k$, $\det(A) = 0$ if and only if $\det(U) = 0$.

By Theorem 2.1.9, if $A$ is nonsingular, then $U = I_n$. $\det(U) = 1$ implies $\det(A) \neq 0$.

By Theorem 2.1.9, if $A$ is singular, the last row of $U$ must be 0, that is,

$$U = \begin{pmatrix} * & \cdots & * & * \\ 0 & \ddots & \vdots & \vdots \\ \vdots & \ddots & * & * \\ 0 & \cdots & 0 & 0 \end{pmatrix}.$$

Then $\det(U) = 0$. Hence $\det(A) = 0$.

Theorem 2.1.26

The system of linear equation $A\boldsymbol{x} = \boldsymbol{b}$ of $n$ linear equations in $n$ unknowns has exactly 1 solution if and only if $A$ is nonsingular. Equivalently, $\det(A) \neq 0$. Such solution is $\boldsymbol{x} = A^{-1}\boldsymbol{b}$.

Theorem 2.1.27

Let $A$ and $B$ be two $n \times n$ matrices. Then $\det(AB) = \det(A)\det(B)$. If $A$ is invertible ( $\det(A) \neq 0$ ), then $\det(A^{-1}) = \dfrac{1}{\det(A)}$.

Theorem 2.1.28

If $A$ is an $n \times n$ matrix where $n$ is odd and $A = -A^T$, then $\det A = 0$.

Solution

$$\begin{aligned} \det A &= \det\left(-A^T\right) \\ &= \det\left((-I_n) \cdot A^T\right) \\ &= \det(-I_n) \det A^T \\ &= (-1)^n \det A \\ &= -\det A \end{aligned}$$

Therefore, $\det A = 0$.

Clearly, $\begin{vmatrix} 0 & 1 & -2 \\ -1 & 0 & 3 \\ 2 & -3 & 0 \end{vmatrix} = 0.$

Example 2.1.29

Factorize

(a) $\begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$

(b) $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$. Hence factorize $\begin{vmatrix} 1 & 1 & 1 \\ \cos a & \cos b & \cos c \\ \cos 2a & \cos 2b & \cos 2c \end{vmatrix}$.

(c) $\begin{vmatrix} a+b & b+c & c+a \\ a^2+b^2 & b^2+c^2 & c^2+a^2 \\ a^3+b^3 & b^3+c^3 & c^3+a^3 \end{vmatrix}$

Solution

(a) $\begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix} = \begin{vmatrix} a+b+c & a+b+c & a+b+c \\ c & a & b \\ b & c & a \end{vmatrix}$ $\left[R_1 + R_2 + R_3 \to R_1\right]$

$$= (a+b+c)\begin{vmatrix} 1 & 1 & 1 \\ c & a & b \\ b & c & a \end{vmatrix}$$

$$= (a+b+c)\begin{vmatrix} 1 & 0 & 0 \\ c & a-c & b-c \\ b & c-b & a-b \end{vmatrix} \quad \left[C_2 - C_1 \to C_2, C_3 - C_1 \to C_3\right]$$

$$= (a+b+c)\begin{vmatrix} a-c & b-c \\ c-b & a-b \end{vmatrix}$$

$$= (a+b+c)\left((a-c)(a-b)-(b-c)(c-b)\right)$$

$$= (a+b+c)\left((a^2-ac-ab+bc)-(-b^2+2bc-c^2)\right)$$

$$= (a+b+c)\left(a^2+b^2+c^2-ac-ab-bc\right)$$

(b) $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ a & b-a & c-a \\ a^2 & b^2-a^2 & c^2-a^2 \end{vmatrix}$ $\left[C_2 - C_1 \to C_2, C_3 - C_1 \to C_3\right]$

$$= \begin{vmatrix} b-a & c-a \\ b^2-a^2 & c^2-a^2 \end{vmatrix} \quad \left[\text{expand along the 1st column}\right]$$

$$= (b-a)(c-a)\begin{vmatrix} 1 & 1 \\ b+a & c+a \end{vmatrix} \quad [\text{take out the common factor in each column}]$$

$$= (b-a)(c-a)(c-b)$$

$$\begin{vmatrix} 1 & 1 & 1 \\ \cos a & \cos b & \cos c \\ \cos 2a & \cos 2b & \cos 2c \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ \cos a & \cos b & \cos c \\ 2\cos^2 a - 1 & 2\cos^2 b - 1 & 2\cos^2 c - 1 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 & 1 \\ \cos a & \cos b & \cos c \\ 2\cos^2 a & 2\cos^2 b & 2\cos^2 c \end{vmatrix} \quad [R_3 + R_1 \to R_3]$$

$$= 2\begin{vmatrix} 1 & 1 & 1 \\ \cos a & \cos b & \cos c \\ \cos^2 a & \cos^2 b & \cos^2 c \end{vmatrix}$$

$$= 2(\cos c - \cos b)(\cos b - \cos a)(\cos c - \cos a)$$

(c) Clearly, $\begin{pmatrix} a+b & b+c & c+a \\ a^2+b^2 & b^2+c^2 & c^2+a^2 \\ a^3+b^3 & b^3+c^3 & c^3+a^3 \end{pmatrix} = \begin{pmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{pmatrix}\begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$.

$$\begin{vmatrix} a+b & b+c & c+a \\ a^2+b^2 & b^2+c^2 & c^2+a^2 \\ a^3+b^3 & b^3+c^3 & c^3+a^3 \end{vmatrix} = \begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix}\begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix}$$

$$= 2abc\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$$

$$= 2abc(b-a)(c-a)(c-b)$$

Theorem 2.1.30

Let $\left(\begin{array}{c|c} A_{k\times k} & B_{k\times l} \\ \hline C_{l\times k} & D_{l\times l} \end{array}\right)$ be a $(k+l)\times(k+l)$ block matrix where $A_{k\times k}, B_{k\times l}, C_{l\times k}$ and $D_{l\times l}$ are $k\times k$, $k\times l$, $l\times k$ and $l\times l$ matrices respectively. If $C_{l\times k} = \mathbf{0}_{l\times k}$, then

$$\det\left(\begin{array}{c|c} A_{k\times k} & B_{k\times l} \\ \hline \mathbf{0}_{l\times k} & D_{l\times l} \end{array}\right) = \det A_{k\times k} \det D_{l\times l}.$$

Similarly, if $B_{k\times l}=\mathbf{0}_{k\times l}$, then

$$\det\left(\begin{array}{c|c} A_{k\times k} & \mathbf{0}_{k\times l} \\ \hline C_{l\times k} & D_{l\times l} \end{array}\right)=\det A_{k\times k}\det D_{l\times l}.$$

Proof

$$\begin{aligned}\det\left(\begin{array}{c|c} A_{k\times k} & B_{k\times l} \\ \hline \mathbf{0}_{l\times k} & D_{l\times l} \end{array}\right)&=\det\left(\left(\begin{array}{c|c} I_{k\times k} & \mathbf{0}_{k\times l} \\ \hline \mathbf{0}_{l\times k} & D_{l\times l} \end{array}\right)\left(\begin{array}{c|c} A_{k\times k} & B_{k\times l} \\ \hline \mathbf{0}_{l\times k} & I_{l\times l} \end{array}\right)\right)\\ &=\det\left(\begin{array}{c|c} I_{k\times k} & \mathbf{0}_{k\times l} \\ \hline \mathbf{0}_{l\times k} & D_{l\times l} \end{array}\right)\det\left(\begin{array}{c|c} A_{k\times k} & B_{k\times l} \\ \hline \mathbf{0}_{l\times k} & I_{l\times l} \end{array}\right)\\ &=\det A_{k\times k}\det D_{l\times l}.\end{aligned}$$

The proof of $\det\left(\begin{array}{c|c} A_{k\times k} & \mathbf{0}_{k\times l} \\ \hline C_{l\times k} & D_{l\times l} \end{array}\right)=\det A_{k\times k}\det D_{l\times l}$ is left as an exercise.

Example 2.1.31

Find $\det\left(\begin{array}{cc|cc} 1 & 2 & 5 & 6 \\ 3 & 4 & 7 & 8 \\ \hline 0 & 0 & 9 & 10 \\ 0 & 0 & 11 & 12 \end{array}\right)$.

Solution

It is not easy to find the determinant directly. However, by Theorem 2.1.30,

$$\det\left(\begin{array}{cc|cc} 1 & 2 & 5 & 6 \\ 3 & 4 & 7 & 8 \\ \hline 0 & 0 & 9 & 10 \\ 0 & 0 & 11 & 12 \end{array}\right)=\det\begin{pmatrix}1&2\\3&4\end{pmatrix}\cdot\det\begin{pmatrix}9&10\\11&12\end{pmatrix}=(1\times4-2\times3)(9\times12-10\times11)=4.$$

In general, $\det\left(\begin{array}{c|c} A_{k\times k} & B_{k\times k} \\ \hline C_{k\times k} & D_{k\times k} \end{array}\right)$, $\det\left(A_{k\times k}D_{k\times k}-B_{k\times k}C_{k\times k}\right)$ and $\det A_{k\times k}\det D_{k\times k}-\det B_{k\times k}\det C_{k\times k}$ are not equal to each other.

Example 2.1.32

Consider $\left(\begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \hline 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right)$.

$$\det\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \det\begin{pmatrix} 0 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \det\begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix} = -1.$$

$$\det\left(\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\right) = \det\left(\begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\right) = -2$$

$$\det\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\det\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} - \det\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}\det\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = 0$$

Theorem 2.1.33

Let $\left(\begin{array}{c|c} A_{k\times k} & B_{k\times k} \\ \hline C_{k\times k} & D_{k\times k} \end{array}\right)$ be a $2k\times 2k$ block matrix where $A_{k\times k}, B_{k\times k}, C_{k\times k}$ and $D_{k\times k}$ are $k\times k$ matrices. Suppose $C_{k\times k}D_{k\times k} = D_{k\times k}C_{k\times k}$ and $\det D_{k\times k} \neq 0$. Then

$$\det\left(\begin{array}{c|c} A_{k\times k} & B_{k\times k} \\ \hline C_{k\times k} & D_{k\times k} \end{array}\right) = \det\left(A_{k\times k}D_{k\times k} - B_{k\times k}C_{k\times k}\right).$$

Proof

$$\left(\begin{array}{c|c} A_{k\times k} & B_{k\times k} \\ \hline C_{k\times k} & D_{k\times k} \end{array}\right)\left(\begin{array}{c|c} D_{k\times k} & \mathbf{0}_{k\times k} \\ \hline -C_{k\times k} & I_k \end{array}\right) = \left(\begin{array}{c|c} A_{k\times k}D_{k\times k} - B_{k\times k}C_{k\times k} & B_{k\times k} \\ \hline C_{k\times k}D_{k\times k} - D_{k\times k}C_{k\times k} & D_{k\times k} \end{array}\right) = \left(\begin{array}{c|c} A_{k\times k}D_{k\times k} - B_{k\times k}C_{k\times k} & B_{k\times k} \\ \hline \mathbf{0}_{k\times k} & D_{k\times k} \end{array}\right)$$

$$\det\left(\left(\begin{array}{c|c} A_{k\times k} & B_{k\times k} \\ \hline C_{k\times k} & D_{k\times k} \end{array}\right)\left(\begin{array}{c|c} D_{k\times k} & \mathbf{0}_{k\times k} \\ \hline -C_{k\times k} & I_k \end{array}\right)\right) = \det\left(\begin{array}{c|c} A_{k\times k}D_{k\times k} - B_{k\times k}C_{k\times k} & B_{k\times k} \\ \hline \mathbf{0}_{k\times k} & D_{k\times k} \end{array}\right)$$

$$\det\left(\begin{array}{c|c} A_{k\times k} & B_{k\times k} \\ \hline C_{k\times k} & D_{k\times k} \end{array}\right)\det D_{k\times k} = \det\left(A_{k\times k}D_{k\times k} - B_{k\times k}C_{k\times k}\right)\det D_{k\times k}$$

$$\det\left(\begin{array}{c|c} A_{k\times k} & B_{k\times k} \\ \hline C_{k\times k} & D_{k\times k} \end{array}\right) = \det\left(A_{k\times k}D_{k\times k} - B_{k\times k}C_{k\times k}\right)$$

Definition 2.1.34

Let $A$ be an $n \times n$ matrix. We define a new matrix called the **adjoint** of $A$ by

$$\text{adj}(A) = \begin{pmatrix} A_{11} & A_{21} & \cdots & A_{n1} \\ A_{12} & A_{22} & \cdots & A_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ A_{1n} & A_{2n} & \cdots & A_{nn} \end{pmatrix}.$$

Thus, to form the adjoint, we must replace each term by its cofactor and then transpose the resulting matrix.

Theorem 2.1.35

Let $A$ be an $n \times n$ matrix. If $A_{ij}$ denotes the cofactor of $a_{ij}$, then

$$a_{i1}A_{j1} + a_{i2}A_{j2} + \cdots + a_{in}A_{jn} = \begin{cases} \det(A) & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

That is,

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}\begin{pmatrix} A_{11} & A_{21} & \cdots & A_{n1} \\ A_{12} & A_{22} & \cdots & A_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ A_{1n} & A_{2n} & \cdots & A_{nn} \end{pmatrix} = \begin{pmatrix} A_{11} & A_{21} & \cdots & A_{n1} \\ A_{12} & A_{22} & \cdots & A_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ A_{1n} & A_{2n} & \cdots & A_{nn} \end{pmatrix}\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$$

$$= \begin{pmatrix} \det(A) & 0 & \cdots & 0 \\ 0 & \det(A) & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \det(A) \end{pmatrix}$$

or $A(\text{adj}(A)) = (\text{adj}(A))A = \det(A)I_n$.

Theorem 2.1.36

Let $A$ be an invertible $n \times n$ matrix, that is, $\det(A) \neq 0$. Then

$$A^{-1} = \frac{1}{\det(A)}\text{adj}(A).$$

Example 2.1.37

Find the inverse of $A=\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, where $\det(A)\neq 0$.

Solution

Clearly, $\det(A)=ad-bc$.

$A_{11}=d,\ A_{12}=-c,\ A_{21}=-b,\ A_{22}=a$

$\operatorname{adj}(A)=\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.

$$A^{-1}=\frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Example 2.1.38

Let $A=\begin{pmatrix} 0 & 2 & -8 \\ 1 & -2 & 1 \\ -4 & 5 & 9 \end{pmatrix}$. Find $A^{-1}$ using adjoint of $A$. Hence solve $\underbrace{\begin{pmatrix} 0 & 2 & -8 \\ 1 & -2 & 1 \\ -4 & 5 & 9 \end{pmatrix}}_{A}\underbrace{\begin{pmatrix} x \\ y \\ z \end{pmatrix}}_{x}=\underbrace{\begin{pmatrix} 8 \\ 0 \\ -9 \end{pmatrix}}_{b}$.

Solution

Clearly, $\det\begin{pmatrix} 0 & 2 & -8 \\ 1 & -2 & 1 \\ -4 & 5 & 9 \end{pmatrix}=-40-8-(-64+18)=-2\neq 0$. Then $A$ is invertible.

The 9 cofactors are

$$A_{11}=\begin{vmatrix} -2 & 1 \\ 5 & 9 \end{vmatrix}=-23,\quad A_{12}=-\begin{vmatrix} 1 & 1 \\ -4 & 9 \end{vmatrix}=-13,\quad A_{13}=\begin{vmatrix} 1 & -2 \\ -4 & 5 \end{vmatrix}=-3,$$

$$A_{21}=-\begin{vmatrix} 2 & -8 \\ 5 & 9 \end{vmatrix}=-58,\quad A_{22}=\begin{vmatrix} 0 & -8 \\ -4 & 9 \end{vmatrix}=-32,\quad A_{23}=-\begin{vmatrix} 0 & 2 \\ -4 & 5 \end{vmatrix}=-8,$$

$$A_{31}=\begin{vmatrix} 2 & -8 \\ -2 & 1 \end{vmatrix}=-14,\quad A_{32}=-\begin{vmatrix} 0 & -8 \\ 1 & 1 \end{vmatrix}=-8,\quad A_{33}=\begin{vmatrix} 0 & 2 \\ 1 & -2 \end{vmatrix}=-2.$$

$$A^{-1} = \frac{1}{\det(A)}\operatorname{adj}(A) = \frac{1}{-2}\begin{pmatrix} -23 & -58 & -14 \\ -13 & -32 & -8 \\ -3 & -8 & -2 \end{pmatrix} = \begin{pmatrix} 23/2 & 29 & 7 \\ 13/2 & 16 & 4 \\ 3/2 & 4 & 1 \end{pmatrix}.$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 & 2 & -8 \\ 1 & -2 & 1 \\ -4 & 5 & 9 \end{pmatrix}^{-1}\begin{pmatrix} 8 \\ 0 \\ -9 \end{pmatrix} = \begin{pmatrix} 23/2 & 29 & 7 \\ 13/2 & 16 & 4 \\ 3/2 & 4 & 1 \end{pmatrix}\begin{pmatrix} 8 \\ 0 \\ -9 \end{pmatrix} = \begin{pmatrix} 29 \\ 16 \\ 3 \end{pmatrix}.$$

Theorem 2.1.39 (Cramer's Rule)

Let $A$ be an $n \times n$ nonsingular matrix, and let $\boldsymbol{b} \in \boldsymbol{R}^n$. Let $A_i$ be the matrix obtained by replacing the $i$-th column of $A$ by $\boldsymbol{b}$. If $\boldsymbol{x}$ is the unique solution of $A\boldsymbol{x} = \boldsymbol{b}$, then

$$x_i = \frac{\det(A_i)}{\det(A)} \quad \text{for } i = 1, \cdots, n.$$

Example 2.1.40

Consider Example 2.1.38. $\det(A) = -2$.

$$\det(A_1) = \det\begin{pmatrix} 8 & 2 & -8 \\ 0 & -2 & 1 \\ -9 & 5 & 9 \end{pmatrix} = -144 - 18 - (-144 + 40) = -58$$

$$\det(A_2) = \det\begin{pmatrix} 0 & 8 & -8 \\ 1 & 0 & 1 \\ -4 & -9 & 9 \end{pmatrix} = 72 - 32 - 72 = -32$$

$$\det(A_3) = \det\begin{pmatrix} 0 & 2 & 8 \\ 1 & -2 & 0 \\ -4 & 5 & -9 \end{pmatrix} = 40 - (64 - 18) = -6$$

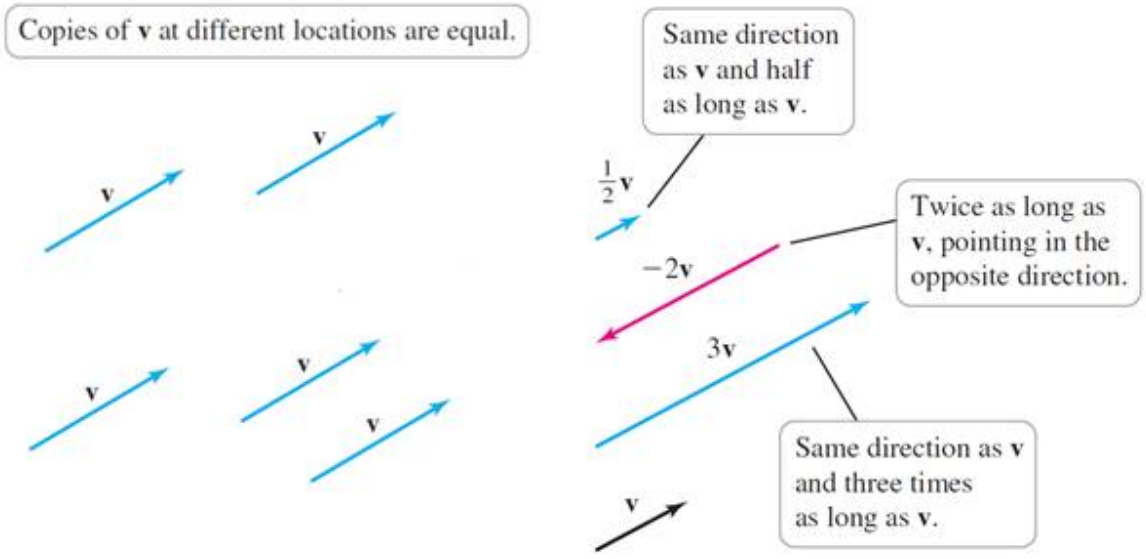
We obtain $x = \frac{\det(A_1)}{\det(A)} = 29$, $y = \frac{\det(A_2)}{\det(A)} = 16$, $z = \frac{\det(A_3)}{\det(A)} = 3$.

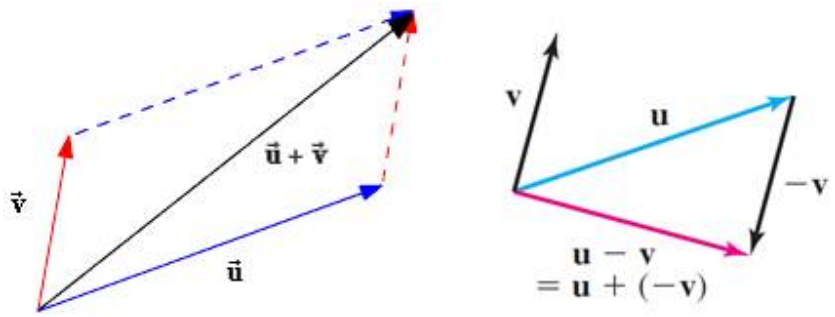
## Section 2.2 Dot Products and Cross Products

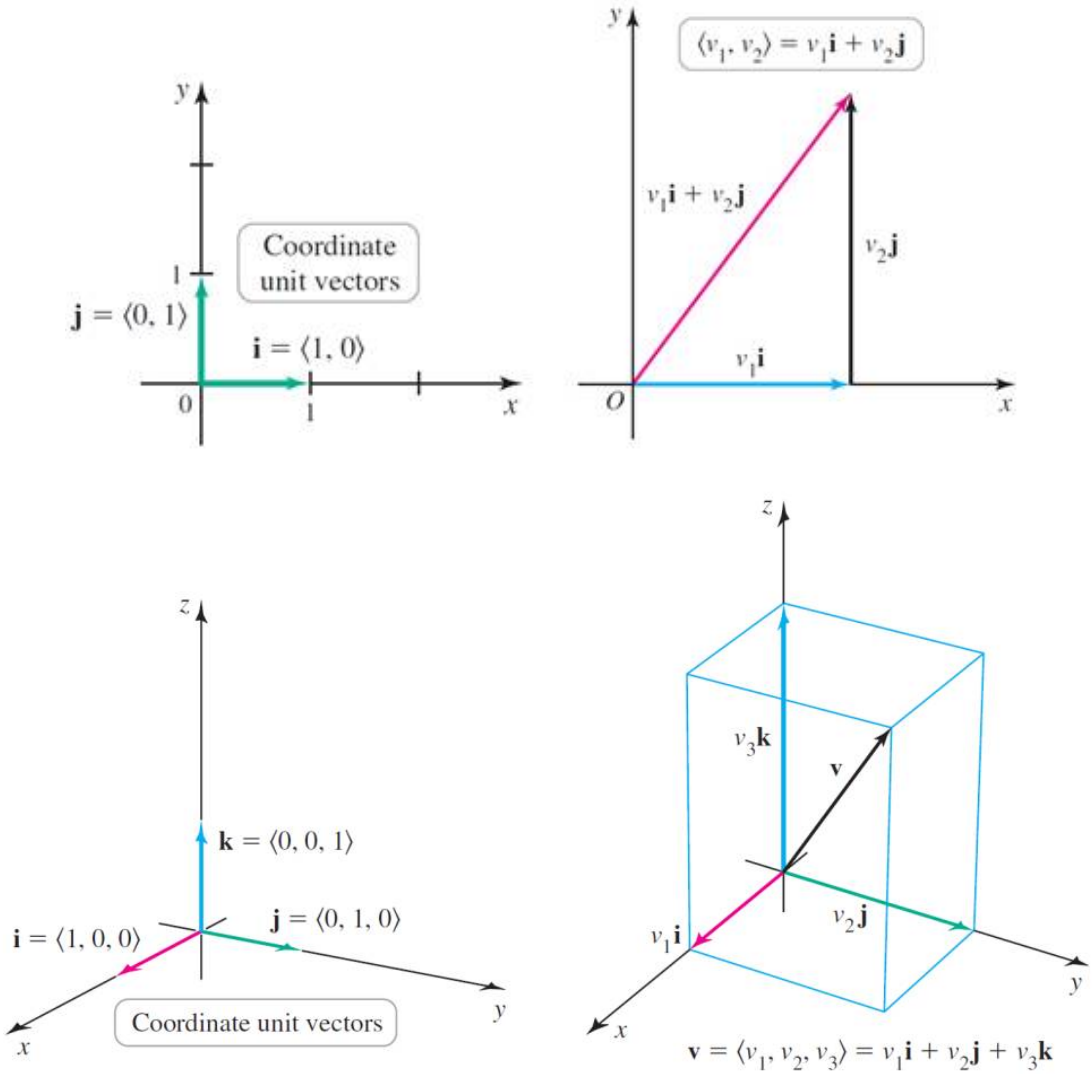
**Vectors** are quantities that have both **length** (or **magnitude**) and **direction**. Two vectors are **equal** if they have the same magnitude and direction. Quantities having magnitude but no direction are called **scalars**. One exception is the **zero vector**, denote **0**: It has length 0 and no direction.

### Definition 2.2.1

Given a scalar $c$ and a vector $\boldsymbol{v}$, the **scalar multiple** of $c\boldsymbol{v}$ is a vector whose magnitude is $|c|$ multiplied by the magnitude of $\boldsymbol{v}$. If $c > 0$, then $c\boldsymbol{v}$ has the same direction as $\boldsymbol{v}$. If $c < 0$, then $c\boldsymbol{v}$ and $\boldsymbol{v}$ point in opposite directions. Two vectors are **parallel** if they are scalar multiples of each other.






Definition 2.2.2

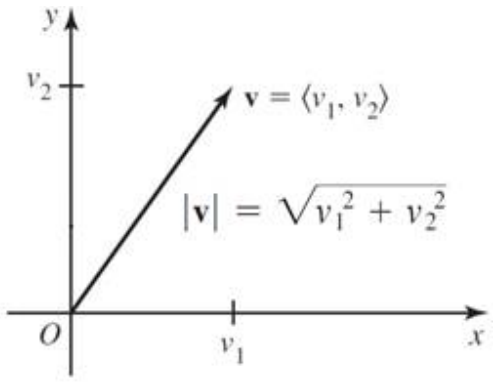
Let $c$ be a scalar, $\boldsymbol{u}=(u_1,u_2)$ and $\boldsymbol{v}=(v_1,v_2)$ be two vectors in $\boldsymbol{R}^2$. Then

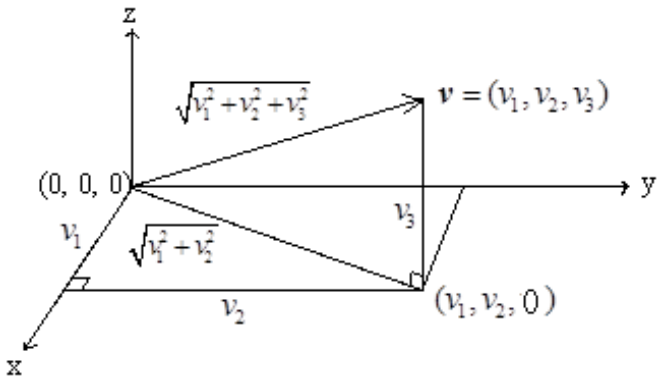
$$\boldsymbol{u}+\boldsymbol{v}=(u_1+v_1,u_2+v_2),\quad \boldsymbol{u}-\boldsymbol{v}=(u_1-v_1,u_2-v_2),\quad c\boldsymbol{u}=(cu_1,cu_2).$$

Definition 2.2.3

Let $c$ be a scalar, $\boldsymbol{u}=(u_1,u_2,u_3)$ and $\boldsymbol{v}=(v_1,v_2,v_3)$ be two vectors in $\boldsymbol{R}^3$. Then

$$\boldsymbol{u}+\boldsymbol{v}=(u_1+v_1,u_2+v_2,u_3+v_3),\quad \boldsymbol{u}-\boldsymbol{v}=(u_1-v_1,u_2-v_2,u_3-v_3),\quad c\boldsymbol{u}=(cu_1,cu_2,cu_3).$$



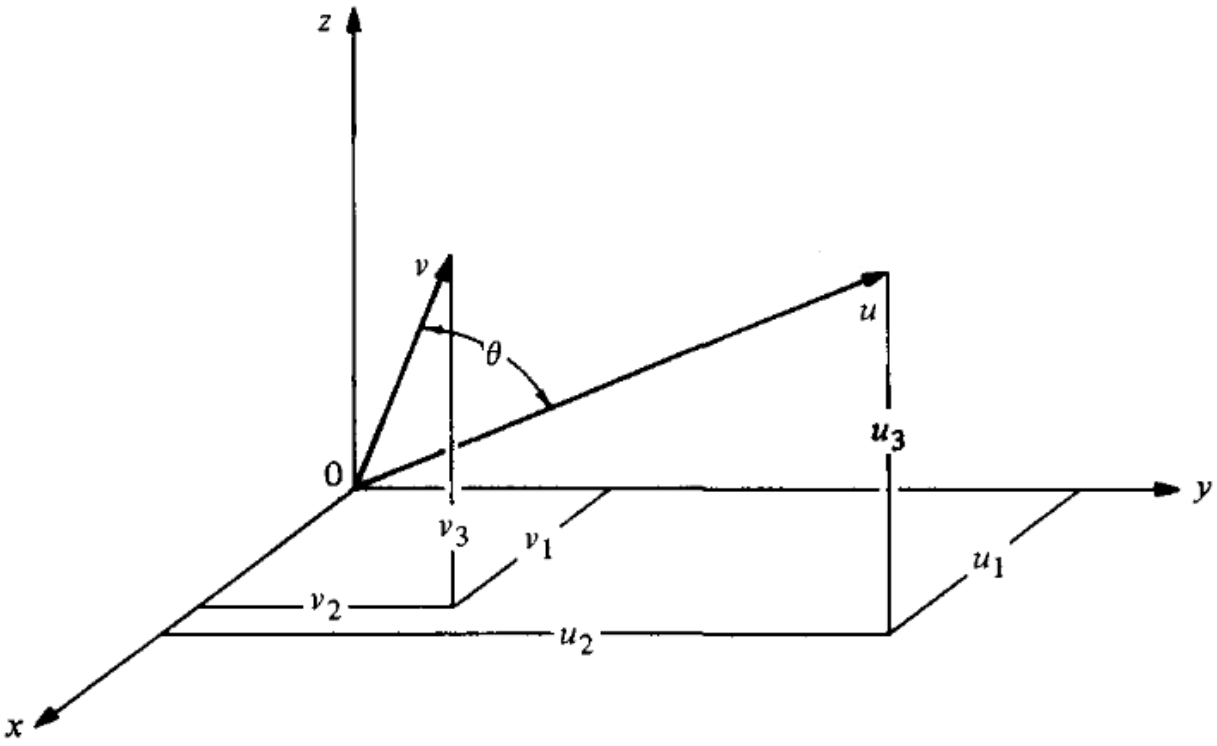




Definition 2.2.4 (Dot Product)

Given two nonzero vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ in $\boldsymbol{R}^2$ and $\boldsymbol{R}^3$, their **dot product** is defined by

$$\boldsymbol{u} \cdot \boldsymbol{v} = |\boldsymbol{u}||\boldsymbol{v}| \cos\theta,$$

where $\theta$ is the angle between $\boldsymbol{u}$ and $\boldsymbol{v}$ with $0 \le \theta \le \pi$. If $\boldsymbol{u} = \mathbf{0}$ or $\boldsymbol{v} = \mathbf{0}$, then $\boldsymbol{u} \cdot \boldsymbol{v} = 0$, and $\theta$ is undefined.

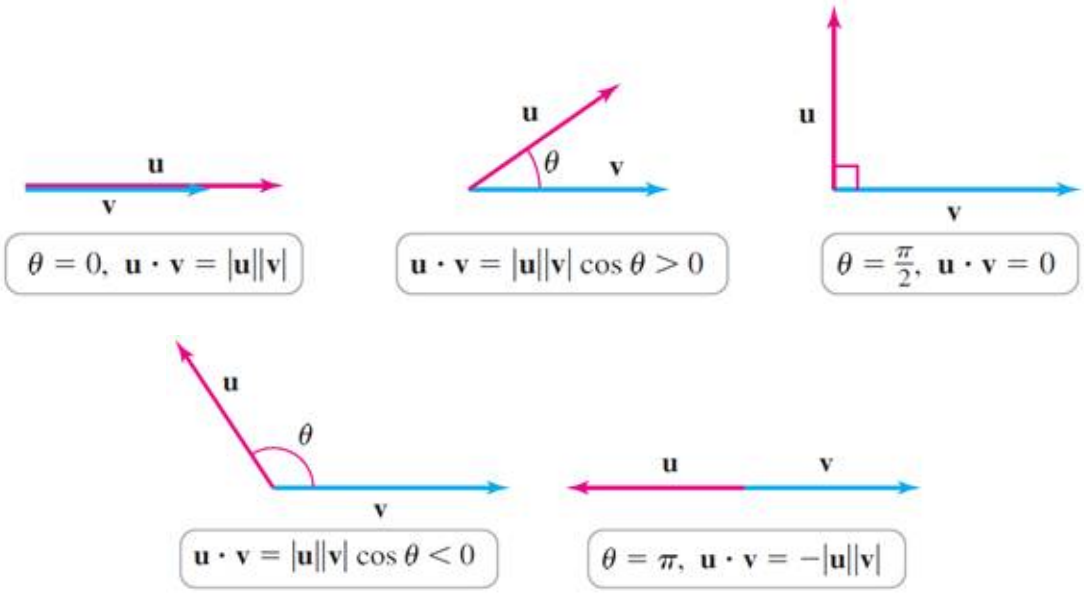


Remark 2.2.5

(i) $\boldsymbol{u}$ and $\boldsymbol{v}$ are parallel ( $\theta = 0$ or $\theta = \pi$ ) if and only if $\boldsymbol{u} \cdot \boldsymbol{v} = \pm |\boldsymbol{u}||\boldsymbol{v}|$ respectively.

(ii) $\boldsymbol{u}$ and $\boldsymbol{v}$ are perpendicular ( $\theta = \pi / 2$ ) if and only if $\boldsymbol{u} \cdot \boldsymbol{v} = 0$.

(iii) $\boldsymbol{u} \cdot \boldsymbol{u} = u_1^2 + u_2^2 = |\boldsymbol{u}|^2$ if $\boldsymbol{u} \in \boldsymbol{R}^2$ and $\boldsymbol{u} \cdot \boldsymbol{u} = u_1^2 + u_2^2 + u_3^2 = |\boldsymbol{u}|^2$ if $\boldsymbol{u} \in \boldsymbol{R}^3$.



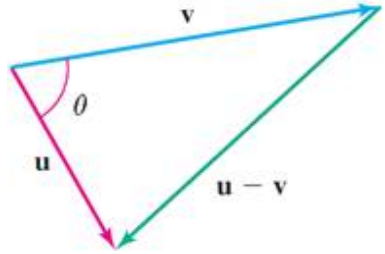

Theorem 2.2.6

Let $\boldsymbol{u} = (u_1, u_2, u_3)$ and $\boldsymbol{v} = (v_1, v_2, v_3)$. Then $\boldsymbol{u} \cdot \boldsymbol{v} = u_1 v_1 + u_2 v_2 + u_3 v_3$.

Proof

Suppose $\theta$ is the angle between $\boldsymbol{u}$ and $\boldsymbol{v}$. The vector $\boldsymbol{u} - \boldsymbol{v}$ forms the third side of a triangle. By the Law of Cosines,

$$|\boldsymbol{u} - \boldsymbol{v}|^2 = |\boldsymbol{u}|^2 + |\boldsymbol{v}|^2 - 2|\boldsymbol{u}||\boldsymbol{v}| \cos\theta.$$



Since $|\boldsymbol{u}|^2 = u_1^2 + u_2^2 + u_3^2$, $|\boldsymbol{v}|^2 = v_1^2 + v_2^2 + v_3^2$, and $|\boldsymbol{u} - \boldsymbol{v}|^2 = (u_1 - v_1)^2 + (u_2 - v_2)^2 + (u_3 - v_3)^2$,

$$\begin{aligned}
\boldsymbol{u} \cdot \boldsymbol{v} &= |\boldsymbol{u}||\boldsymbol{v}|\cos\theta \\
&= \frac{1}{2}\left(|\boldsymbol{u}|^2 + |\boldsymbol{v}|^2 - |\boldsymbol{u} - \boldsymbol{v}|^2\right) \\
&= \frac{1}{2}\left(u_1^2 + u_2^2 + u_3^2 + v_1^2 + v_2^2 + v_3^2 - (u_1 - v_1)^2 - (u_2 - v_2)^2 - (u_3 - v_3)^2\right) \\
&= \frac{1}{2}\left(u_1^2 + u_2^2 + u_3^2 + v_1^2 + v_2^2 + v_3^2 - u_1^2 + 2u_1v_1 - v_1^2 - u_2^2 + 2u_2v_2 - v_2^2 - u_3^2 + 2u_3v_3 - v_3^2\right) \\
&= u_1v_1 + u_2v_2 + u_3v_3
\end{aligned}$$

Remark 2.2.7

(i) Let $\boldsymbol{u} = (u_1, u_2)$ and $\boldsymbol{v} = (v_1, v_2)$ be two vectors in $\boldsymbol{R}^2$. Extend $\boldsymbol{u}$ to $\tilde{\boldsymbol{u}} = (u_1, u_2, 0)$ and $\boldsymbol{v}$ to $\tilde{\boldsymbol{v}} = (v_1, v_2, 0)$ in $\boldsymbol{R}^3$. Then $\boldsymbol{u} \cdot \boldsymbol{v} = \tilde{\boldsymbol{u}} \cdot \tilde{\boldsymbol{v}} = u_1v_1 + u_2v_2$.

(ii) If $\boldsymbol{u}$ and $\boldsymbol{v}$ are both nonzero, then

$$-1 \le \cos\theta = \frac{u_1v_1 + u_2v_2 + u_3v_3}{|\boldsymbol{u}||\boldsymbol{v}|} = \frac{\boldsymbol{u} \cdot \boldsymbol{v}}{|\boldsymbol{u}| \cdot |\boldsymbol{v}|} \le 1.$$

Theorem 2.2.8

Suppose $\boldsymbol{u}$, $\boldsymbol{v}$ and $\boldsymbol{w}$ are vectors and let $c$ be a scalar.

(i) $\boldsymbol{u} \cdot \boldsymbol{v} = \boldsymbol{v} \cdot \boldsymbol{u}$

(ii) $c(\boldsymbol{u} \cdot \boldsymbol{v}) = (c\boldsymbol{u}) \cdot \boldsymbol{v} = \boldsymbol{u} \cdot (c\boldsymbol{v})$

(iii) $\boldsymbol{u} \cdot (\boldsymbol{v} + \boldsymbol{w}) = \boldsymbol{u} \cdot \boldsymbol{v} + \boldsymbol{u} \cdot \boldsymbol{w}$

(iv) $(\boldsymbol{u} + \boldsymbol{v}) \cdot \boldsymbol{w} = \boldsymbol{u} \cdot \boldsymbol{w} + \boldsymbol{v} \cdot \boldsymbol{w}$

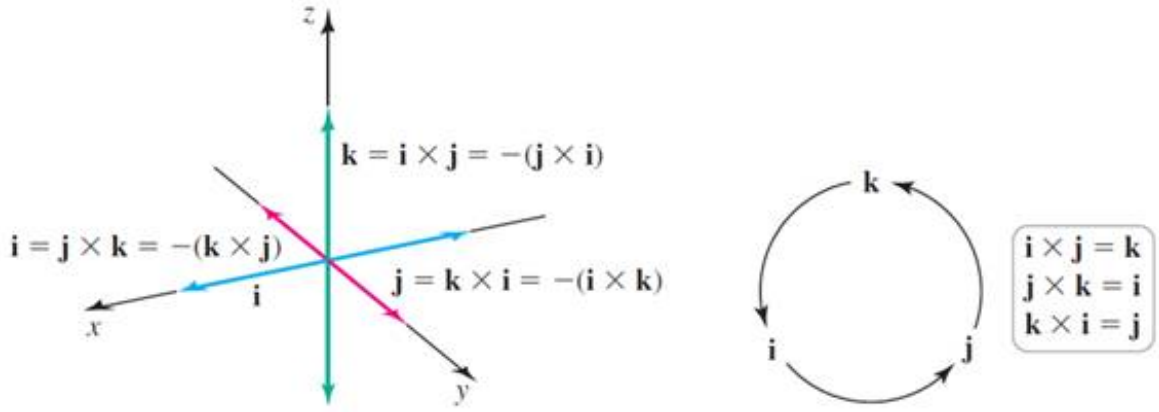
Definition 2.2.9 (Cross Product)

Let $\boldsymbol{i}$, $\boldsymbol{j}$, and $\boldsymbol{k}$ be unit vectors in $\mathbb{R}^3$ on positive $x$, $y$ and $z$-axis respectively. Let $\boldsymbol{u} = u_1\boldsymbol{i} + u_2\boldsymbol{j} + u_3\boldsymbol{k}$ and $\boldsymbol{v} = v_1\boldsymbol{i} + v_2\boldsymbol{j} + v_3\boldsymbol{k}$. The **cross product** $\boldsymbol{u} \times \boldsymbol{v}$ of $\boldsymbol{u}$ and $\boldsymbol{v}$ is the vector

$$\begin{aligned}\boldsymbol{u}\times\boldsymbol{v} &= \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} \\ &= (u_2v_3 - u_3v_2)\boldsymbol{i} + (u_3v_1 - u_1v_3)\boldsymbol{j} + (u_1v_2 - u_2v_1)\boldsymbol{k} \\ &= (u_2v_3 - u_3v_2, u_3v_1 - u_1v_3, u_1v_2 - u_2v_1)\end{aligned}$$

Theorem 2.2.10

$$\boldsymbol{i}\times\boldsymbol{j} = -(\boldsymbol{j}\times\boldsymbol{i}) = \boldsymbol{k}, \quad \boldsymbol{j}\times\boldsymbol{k} = -(\boldsymbol{k}\times\boldsymbol{j}) = \boldsymbol{i}$$
$$\boldsymbol{k}\times\boldsymbol{i} = -(\boldsymbol{i}\times\boldsymbol{k}) = \boldsymbol{j}, \quad \boldsymbol{i}\times\boldsymbol{i} = \boldsymbol{j}\times\boldsymbol{j} = \boldsymbol{k}\times\boldsymbol{k} = \boldsymbol{0}$$




Theorem 2.2.11

Let $\boldsymbol{u}$, $\boldsymbol{v}$, and $\boldsymbol{w}$ be nonzero vectors in $\mathbb{R}^3$, and let $a$ and $b$ be scalars.

(i) $\boldsymbol{u}\times\boldsymbol{v} = -(\boldsymbol{v}\times\boldsymbol{u})$

(ii) $(a\boldsymbol{u})\times(b\boldsymbol{v}) = ab(\boldsymbol{u}\times\boldsymbol{v})$

(iii) $\boldsymbol{u}\times(\boldsymbol{v}+\boldsymbol{w}) = (\boldsymbol{u}\times\boldsymbol{v}) + (\boldsymbol{u}\times\boldsymbol{w})$

(iv) $(\boldsymbol{u}+\boldsymbol{v})\times\boldsymbol{w} = (\boldsymbol{u}\times\boldsymbol{w}) + (\boldsymbol{v}\times\boldsymbol{w})$

Q? Is $(\boldsymbol{u}\times\boldsymbol{v})\times\boldsymbol{w} = \boldsymbol{u}\times(\boldsymbol{v}\times\boldsymbol{w})$ for every vector $\boldsymbol{u}$, $\boldsymbol{v}$ and $\boldsymbol{w}$?

A. No. $(\boldsymbol{i}\times\boldsymbol{i})\times\boldsymbol{j} = \boldsymbol{0}$, $\boldsymbol{i}\times(\boldsymbol{i}\times\boldsymbol{j}) = \boldsymbol{i}\times\boldsymbol{k} = -\boldsymbol{j}$.

Theorem 2.2.12

Let $\boldsymbol{u} = u_1\boldsymbol{i} + u_2\boldsymbol{j} + u_3\boldsymbol{k}$, $\boldsymbol{v} = v_1\boldsymbol{i} + v_2\boldsymbol{j} + v_3\boldsymbol{k}$ and $\boldsymbol{w} = w_1\boldsymbol{i} + w_2\boldsymbol{j} + w_3\boldsymbol{k}$. Then

$$\begin{aligned}(\boldsymbol{u}\times\boldsymbol{v})\cdot\boldsymbol{w} &= (u_2v_3 - u_3v_2, u_3v_1 - u_1v_3, u_1v_2 - u_2v_1)\cdot(w_1, w_2, w_3)\\ &= (u_2v_3 - u_3v_2)w_1 + (u_3v_1 - u_1v_3,)w_2 + (u_1v_2 - u_2v_1)w_3\\ &= \begin{vmatrix} u_1 & u_2 & u_3\\ v_1 & v_2 & v_3\\ w_1 & w_2 & w_3\end{vmatrix}\end{aligned}$$

Clearly, $\boldsymbol{u}\cdot(\boldsymbol{v}\times\boldsymbol{w}) = (\boldsymbol{v}\times\boldsymbol{w})\cdot\boldsymbol{u} = \begin{vmatrix} v_1 & v_2 & v_3\\ w_1 & w_2 & w_3\\ u_1 & u_2 & u_3\end{vmatrix} = \begin{vmatrix} u_1 & u_2 & u_3\\ v_1 & v_2 & v_3\\ w_1 & w_2 & w_3\end{vmatrix} = (\boldsymbol{u}\times\boldsymbol{v})\cdot\boldsymbol{w}$

Suppose $\boldsymbol{u}\times\boldsymbol{v} \neq \boldsymbol{0}$, i.e, $\boldsymbol{u}$ and $\boldsymbol{v}$ are not parallel. Clearly, $(\boldsymbol{u}\times\boldsymbol{v})\cdot\boldsymbol{u} = \begin{vmatrix} u_1 & u_2 & u_3\\ v_1 & v_2 & v_3\\ u_1 & u_2 & u_3\end{vmatrix} = 0$ and

$(\boldsymbol{u}\times\boldsymbol{v})\cdot\boldsymbol{v} = \begin{vmatrix} u_1 & u_2 & u_3\\ v_1 & v_2 & v_3\\ v_1 & v_2 & v_3\end{vmatrix} = 0$. $\boldsymbol{u}\times\boldsymbol{v}$ is perpendicular to both $\boldsymbol{u}$ and $\boldsymbol{v}$.

Furthermore, we also have

$$(\boldsymbol{u}\times\boldsymbol{v})\cdot(\boldsymbol{u}\times\boldsymbol{v}) = |\boldsymbol{u}\times\boldsymbol{v}|^2 > 0.$$

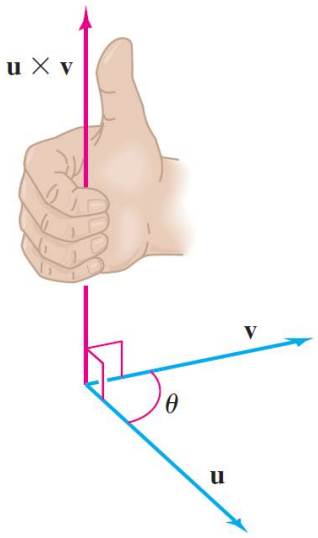
On the other hand, $\boldsymbol{i}\cdot(\boldsymbol{j}\times\boldsymbol{k}) = \boldsymbol{i}\cdot\boldsymbol{i} = 1 > 0$. $\boldsymbol{u}$, $\boldsymbol{v}$, $\boldsymbol{u}\times\boldsymbol{v}$ has the same orientation as $\boldsymbol{i}$, $\boldsymbol{j}$, $\boldsymbol{k}$.

The direction of $\boldsymbol{u}\times\boldsymbol{v}$ is given by the right hand rule:

When you put the vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ tail to tail and let the fingers of you right hand curl from $\boldsymbol{u}$ to $\boldsymbol{v}$, the direction of $\boldsymbol{u}\times\boldsymbol{v}$ is the direction of your thumb, perpendicular to both $\boldsymbol{u}$ and $\boldsymbol{v}$ (see the below figure). However switching the order of $\boldsymbol{u}$ and $\boldsymbol{v}$ gives the opposite direction but in the same magnitude, that means

$$\boldsymbol{v}\times\boldsymbol{u} = -(\boldsymbol{u}\times\boldsymbol{v}).$$

When $\boldsymbol{u}\times\boldsymbol{v} = \boldsymbol{0}$, the direction of $\boldsymbol{u}\times\boldsymbol{v}$ is undefined.




Theorem 2.2.13

Given two nonzero vectors $\boldsymbol{u}=(u_1,u_2,u_3)$, and $\boldsymbol{v}=(v_1,v_2,v_3)$ in $\mathbb{R}^3$, the magnitude of $\boldsymbol{u}\times\boldsymbol{v}$ is

$$|\boldsymbol{u}\times\boldsymbol{v}|=|\boldsymbol{u}||\boldsymbol{v}|\sin\theta,$$

where $\theta$ is the angle between $\boldsymbol{u}$ and $\boldsymbol{v}$ with $0\le\theta\le\pi$.

Solution

$$\begin{aligned}
|\boldsymbol{u}\times\boldsymbol{v}|^2 &= (u_2v_3-u_3v_2)^2+(u_3v_1-u_1v_3)^2+(u_1v_2-u_2v_1)^2\\
&=(u_2v_3)^2-2(u_2v_3)(u_3v_2)+(u_3v_2)^2\\
&+(u_3v_1)^2-2(u_3v_1)(u_1v_3)+(u_1v_3)^2\\
&+(u_1v_2)^2-2(u_1v_2)(u_2v_1)+(u_2v_1)^2\\
&=(u_1v_1)^2+(u_1v_2)^2+(u_1v_3)^2+(u_2v_1)^2+(u_2v_2)^2+(u_2v_3)^2+(u_3v_1)^2+(u_3v_2)^2+(u_3v_3)^2\\
&-\left[(u_1v_1)^2+(u_2v_2)^2+(u_3v_3)^2+2(u_1v_1)(u_2v_2)+2(u_2v_2)(u_3v_3)+2(u_1v_1)(u_3v_3)\right]\\
&=\left(u_1^2+u_2^2+u_3^2\right)\left(v_1^2+v_2^2+v_3^2\right)-(u_1v_1+u_2v_2+u_3v_3)^2\\
&=|\boldsymbol{u}|^2|\boldsymbol{v}|^2-(\boldsymbol{u}\cdot\boldsymbol{v})^2\\
&=|\boldsymbol{u}|^2|\boldsymbol{v}|^2-|\boldsymbol{u}|^2|\boldsymbol{v}|^2\cos^2\theta\\
&=|\boldsymbol{u}|^2|\boldsymbol{v}|^2\sin^2\theta
\end{aligned}$$

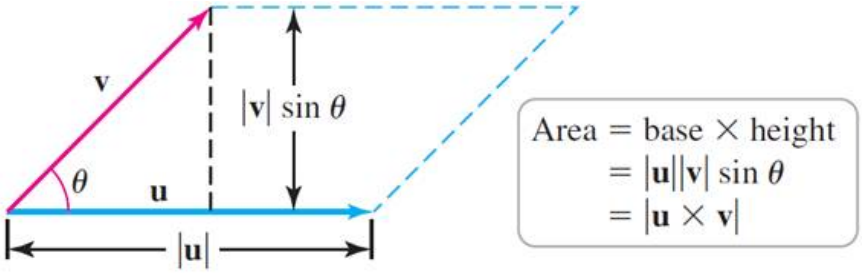
$|\boldsymbol{u}\times\boldsymbol{v}|=|\boldsymbol{u}||\boldsymbol{v}|\sin\theta$ where $0\le\theta\le\pi$.

Corollary 2.2.14

Let $\boldsymbol{u}$ and $\boldsymbol{v}$ be two nonzero vectors in $\mathbb{R}^3$, If $\boldsymbol{u}$ and $\boldsymbol{v}$ are two sides of a parallelogram, then the area of the parallelogram is

$$|\boldsymbol{u}\times\boldsymbol{v}| = |\boldsymbol{u}||\boldsymbol{v}|\sin\theta,$$

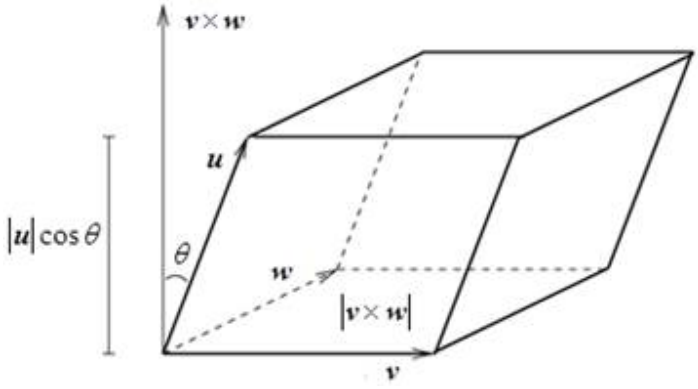
where $0 \le \theta \le \pi$ is the angle between $\boldsymbol{u}$ and $\boldsymbol{v}$.




Theorem 2.2.15

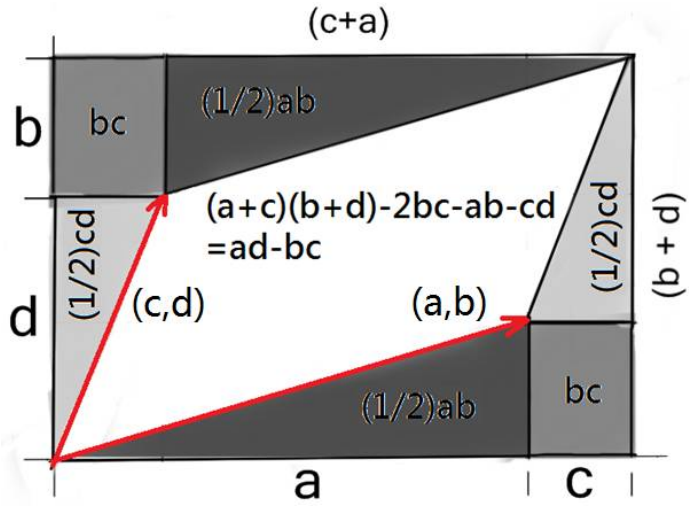
The volume of the parallelepiped generated by the vectors generated by $\boldsymbol{u} = (u_1, u_2, u_3)$, $\boldsymbol{v} = (v_1, v_2, v_3)$ and $\boldsymbol{w} = (w_1, w_2, w_3)$ is the base area times its height

$$|\boldsymbol{v}\times\boldsymbol{w}||\boldsymbol{u}|\cos\theta = \boldsymbol{u}\cdot(\boldsymbol{v}\times\boldsymbol{w}) = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}.$$




Theorem 2.2.16

The area of the parallelogram generated by the vectors $(a, b)$ and $(c, d)$ is $\left|\det\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right|$.



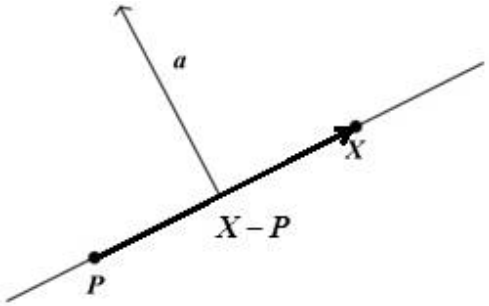

Every straight line $L$ in the plane is defined by a linear equation

$$a_1 x + a_2 y = c.$$

Let $\boldsymbol{a} = (a_1, a_2)$, $\boldsymbol{X} = (x, y)$ and $\boldsymbol{P} = (p, q)$ be two points on $L$. Then $\boldsymbol{a} \cdot \boldsymbol{X} = \boldsymbol{a} \cdot \boldsymbol{P} = c$. Hence

$$\boldsymbol{a} \cdot (\boldsymbol{X} - \boldsymbol{P}) = 0.$$

That means $\boldsymbol{a}$ and $\boldsymbol{X} - \boldsymbol{P}$ are perpendicular. $\boldsymbol{a}$ is a normal vector of the straight line $L$.

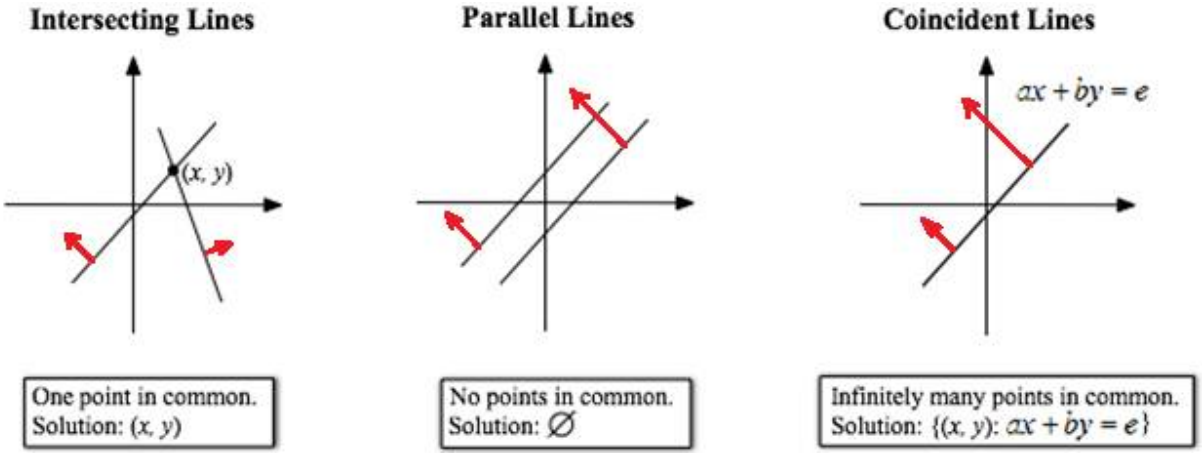



Changing $c$ will move the straight line in parallel.

Consider the system of linear equations

$$\begin{cases} ax + by = e \\ cx + dy = f \end{cases}$$

There are 2 straight lines in $\boldsymbol{R}^2$ with normal vectors $(a, b)$ and $(c, d)$.

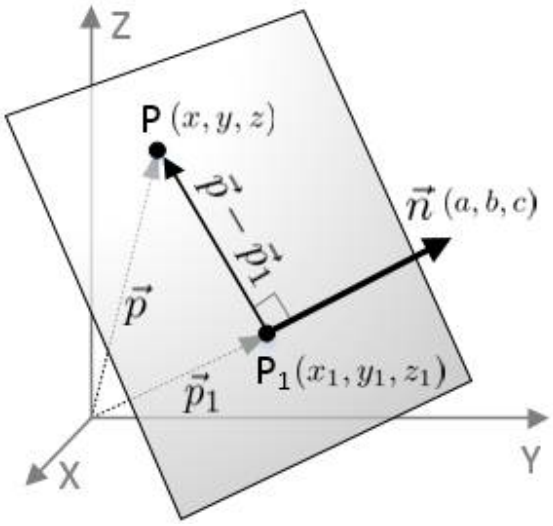



The system of linear equations has exactly one solution if and only if the area of the parallelogram generated by the normal vectors of two straight lines is nonzero. Equivalently,

$$\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} \neq 0.$$

If $\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = 0,$ whether the system has infinitely solutions depends on $e$ and $f$.

A plane in $\mathbb{R}^3$ can be represented by a linear equation in variables $x, y, z$: $ax + by + cz = d$.




Choose two distinct points $P = (x, y, z)$ and $P_1 = (x_1, y_1, z_1)$ on the plane.

$$(a, b, c) \cdot (x - x_1, y - y_1, z - z_1) = ax + by + cz - (ax_1 + by_1 + cz_1) = 0$$

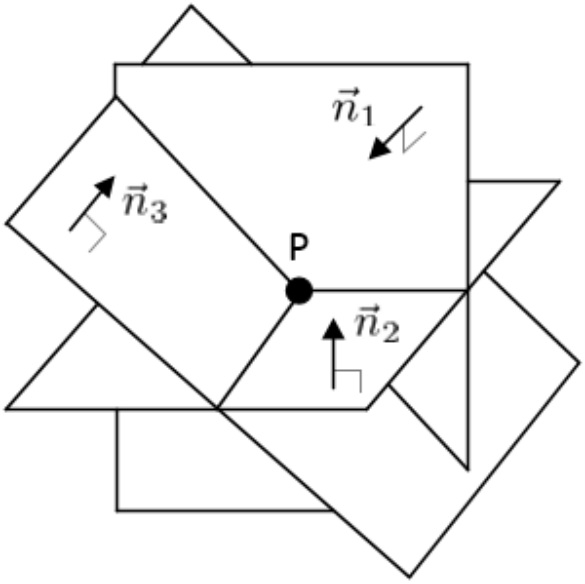
Any vector in the plane is perpendicular to vector ($a$, $b$, $c$). ($a$, $b$, $c$) is the normal vector of the plane.

Changing $d$ will move the plane in parallel.

Consider the system of linear equations

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3 \end{cases}$$

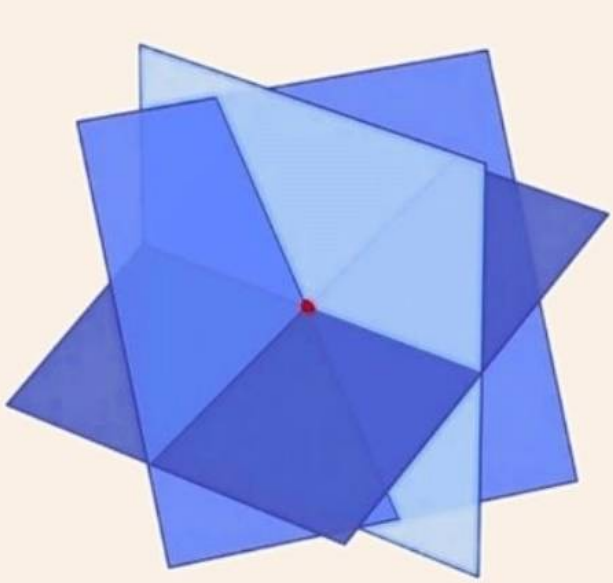
There are 3 planes in $\boldsymbol{R}^3$ with normal vectors $\vec{\boldsymbol{n}}_1 = (a_{11}, a_{12}, a_{13}), \vec{\boldsymbol{n}}_2 = (a_{21}, a_{22}, a_{23})$ and $\vec{\boldsymbol{n}}_3 = (a_{31}, a_{32}, a_{33})$.



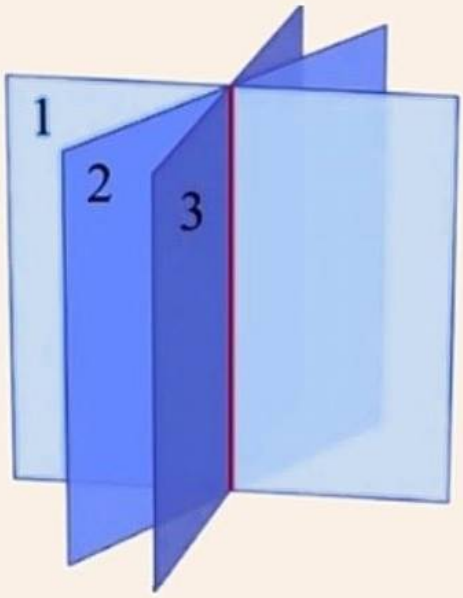

The system of linear equations has exactly one solution if and only if the volume of the parallelepiped generated by the normal vectors of three planes is nonzero. Equivalently,

$$\det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \neq 0.$$

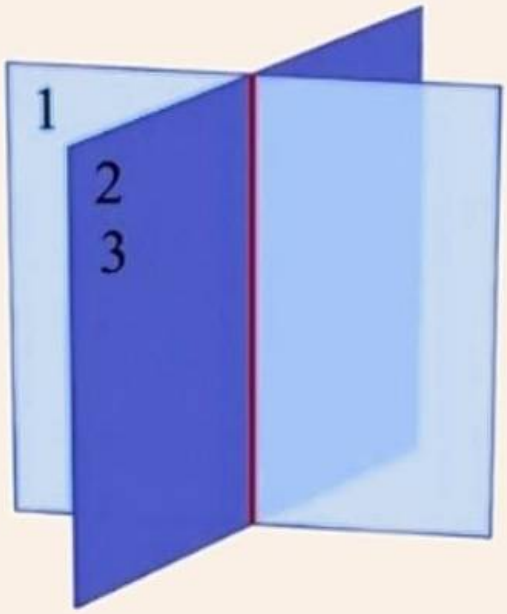
If $\det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = 0,$ whether the system has infinitely solutions depends on $b_1, b_2, b_3$.



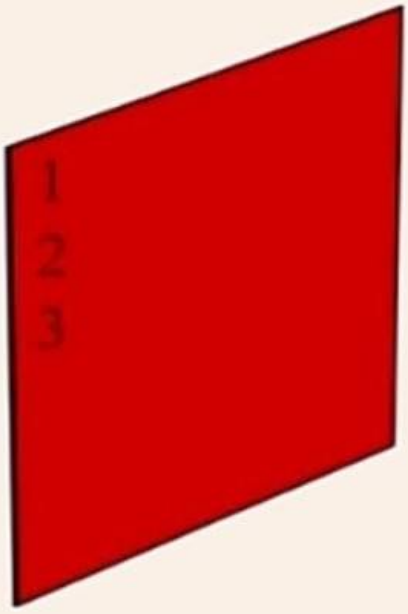
One solution
(Intersection is a point)



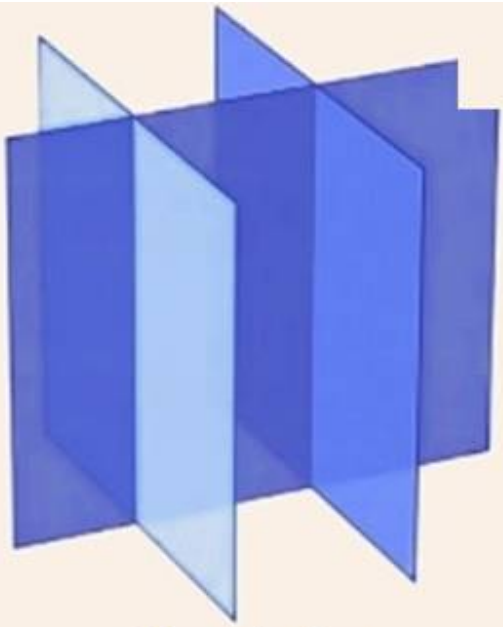

Infinitely many solutions
(Intersection is a line)



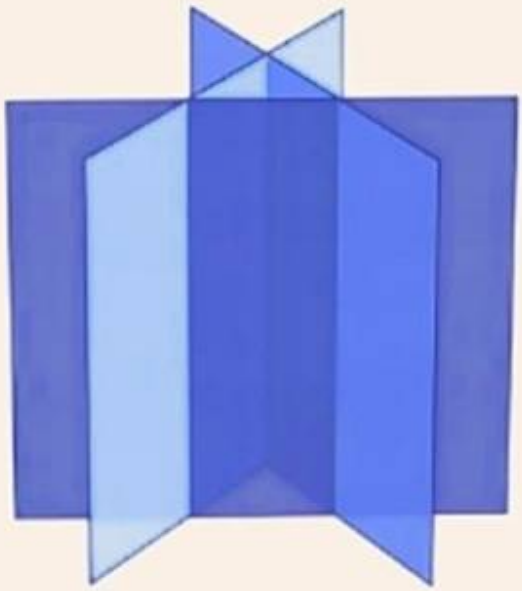

Infinitely many solutions
(two coincident planes;
intersection is a line)



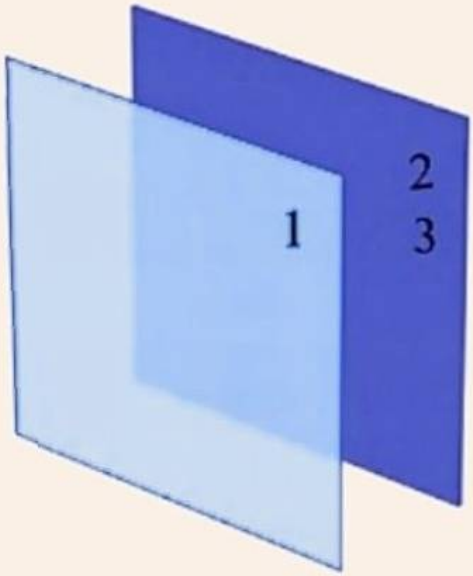

Infinitely many solutions
(planes are all coincident;
intersection is a plane)



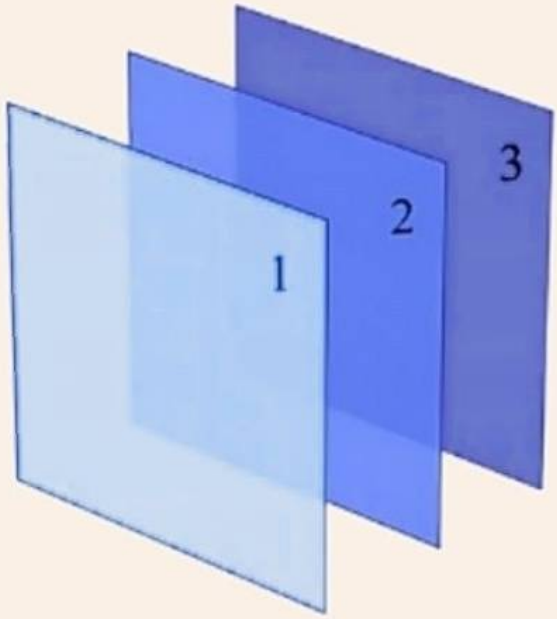
No solution
(two parallel planes;
no common intersection)



No solution
(no common intersection)




No solution
(two coindicent planes
parallel to the third;
no common intersection)



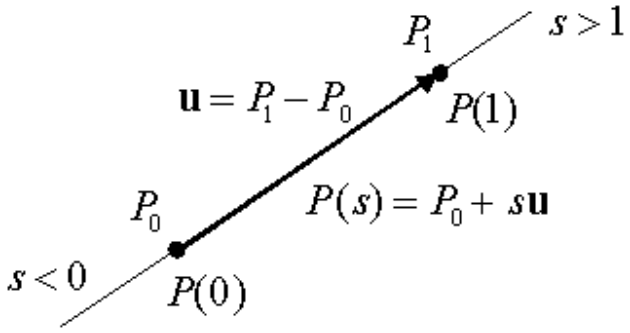

No solution
(three parallel planes;
no common intersection)

Let $\boldsymbol{u}$, $\boldsymbol{v}$ be two nonzero vectors in $\boldsymbol{R}^2$. $\boldsymbol{u}//\boldsymbol{v}$ (that is $\boldsymbol{u} = k\boldsymbol{v}$ for some $k \neq 0$) if and only if $\det(\boldsymbol{u} \mid \boldsymbol{v}) = 0$ (no area in Theorem 2.2.16). Let $\boldsymbol{u}$, $\boldsymbol{v}$ be two nonzero vectors in $\boldsymbol{R}^3$. $\boldsymbol{u}//\boldsymbol{v}$ (that is $\boldsymbol{u} = k\boldsymbol{v}$ for some $k \neq 0$) if and only if $\boldsymbol{u} \times \boldsymbol{v} = \boldsymbol{0}$ (no area in Corollary 2.2.14).

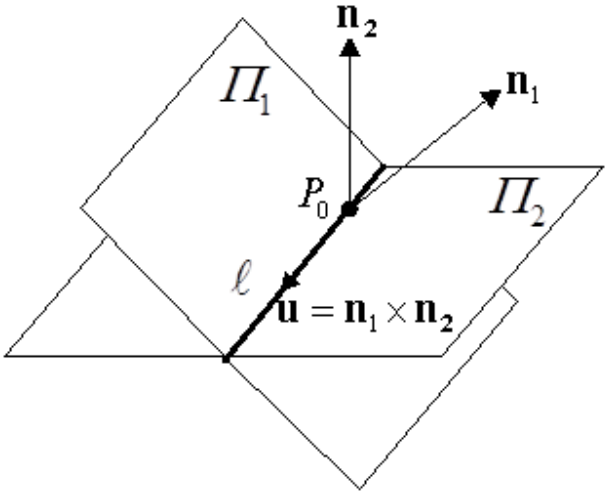
In any dimension, the parametric equation of a line defined by two points $P_0$ and $P_1$ can be represented as:

$$P(s) = P_0 + s(P_1 - P_0) = P_0 + s\boldsymbol{u},$$

where the parameter $s$ is a real number and $\boldsymbol{u} = P_1 - P_0$ is a direction vector. $P(s) = sP_1 + (1-s)P_0$ is a point on the finite segment $P_0P_1$ when $0 \le s \le 1$. Further, if $s < 0$ then $P(s)$ is outside the segment on the $P_0$ side, and if $s > 1$ then $P(s)$ is outside the segment on the $P_1$ side. Particularly in $\boldsymbol{R}^3$, the equation of $P(s)$ can be expressed as $(P(s) - P_0) \times \boldsymbol{u} = \boldsymbol{0}$ since $P(s) - P_0 // \boldsymbol{u}$.




A straight line $\ell$ in $\boldsymbol{R}^3$ can be expressed as intersection of 2 non-parallel planes $\Pi_1$ and $\Pi_2$.



Let $\boldsymbol{n}_1$ and $\boldsymbol{n}_2$ be normal vectors of $\Pi_1$ and $\Pi_2$ respectively. $\boldsymbol{u} = \boldsymbol{n}_1 \times \boldsymbol{n}_2$ is a direction vector for the intersection line $\ell$ since $\boldsymbol{u}$ is perpendicular to both $\boldsymbol{n}_1$ and $\boldsymbol{n}_2$.

Example 2.2.17

Let $\boldsymbol{r} = (x, y, z)$. The planes $\Pi_1$ and $\Pi_2$ have equations

$$\boldsymbol{r} \cdot (2,1,7) = 10 \quad \text{and} \quad \boldsymbol{r} \cdot (3,1,-4) = 7.$$

Express $\boldsymbol{r}$ in the form of $P(s) = P_0 + s\boldsymbol{u}$ and $(P(s) - P_0) \times \boldsymbol{u} = \boldsymbol{0}$.

Solution

We need to solve the system of linear equations

$$\begin{cases} 2x + y + 7z = 10 \\ 3x + y - 4z = 7 \end{cases}$$

$$\left(\begin{array}{ccc|c} 2 & 1 & 7 & 10 \\ 3 & 1 & -4 & 7 \end{array}\right) \xrightarrow{R_2 - R_1 \to R_2} \left(\begin{array}{ccc|c} 2 & 1 & 7 & 10 \\ 1 & 0 & -11 & -3 \end{array}\right) \xrightarrow{R_1 - 2R_2 \to R_1} \left(\begin{array}{ccc|c} 0 & 1 & 29 & 16 \\ 1 & 0 & -11 & -3 \end{array}\right)$$

Then $\dfrac{x+3}{11} = \dfrac{y-16}{-29} = z$, $(x, y, z) = (11s - 3, -29s + 16, s) = (-3,16,0) + s(11,-29,1)$ and

$((x, y, z) - (-3,16,0)) \times (11,-29,1) = (0,0,0)$.

$(2,1,7) \times (3,1,-4) = (-11,29,-1)$ is parallel to the direction vector $(11,-29,1)$.

Example 2.2.18

Find the equation of a plane defined by $(x, y, z) = (2,1,1) + s(3,1,2) + t(4,-1,1)$.

Solution

The normal vector of the plane is $(3,1,2) \times (4,-1,1) = (3,5,-7)$

The equation of the plane is $3(x-2) + 5(y-1) - 7(z-1) = 0$ or $3x + 5y - 7z = 4$.

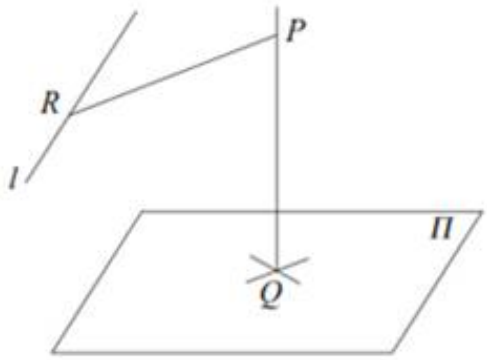
Example 2.2.19

Let $Q$ = (7, 4, 6), the plane $\Pi$ defined by $(x,y,z)\cdot(2,1,3)=36,$ and the line $l$ defined by $(x,y,z)=(20,-8,1)+\mu(-7,5,3).$

(a) Show that $Q$ lies in $\Pi$.

(b) Show that $l$ is parallel to $\Pi$.

(c) The diagram shows the point $P$, which lies on the normal to $\Pi$ that passes through $Q$. The point $R$ is the point on $l$ which is closest to $P$, and $PQ = PR$. Determine the coordinate of $P$.




Solution

(a) Since $(7,4,6)\cdot(2,1,3)=36,$ $Q$ lies in $\Pi$.

(b) Since $(-7,5,3)\cdot(2,1,3)=0,$ $l$ is perpendicular to normal of $\Pi$. $l$ is parallel to $\Pi$.

(c) $P=Q+\lambda\boldsymbol{n}=(7,4,6)+\lambda(2,1,3)$ for some $\lambda$ where $\boldsymbol{n}$ is normal of $\Pi$. $\left|\overrightarrow{PQ}\right|=|\lambda|\sqrt{14}$

$$\overrightarrow{PR}=\overrightarrow{OR}-\overrightarrow{OP}=(20,-8,1)+\mu(-7,5,3)-\big((7,4,6)+\lambda(2,1,3)\big)=(13,-12,-5)+\mu(-7,5,3)-\lambda(2,1,3)$$

Since $\overrightarrow{PR}$ is perpendicular to $l$, $\overrightarrow{PR}\cdot(-7,5,3)=0.$ We have $\mu=2.$

Hence $\overrightarrow{PR}=(-1,-2,1)-\lambda(2,1,3).$

Since $\left|\overrightarrow{PQ}\right|=\left|\overrightarrow{PR}\right|,$ we get

$$(1+2\lambda)^2+(2+\lambda)^2+(1-3\lambda)^2=14\lambda^2$$
$$1+4\lambda+4\lambda^2+4+4\lambda+\lambda^2+1-6\lambda+9\lambda^2=14\lambda^2$$
$$\lambda=-3$$

$P=(7,4,6)-3(2,1,3)=(1,1,-3).$

Example 2.2.20 (distance between 2 skew lines)

Show that 2 lines $P(s)=(-3,-8,7)+(0,-1,1)s$ and $Q(t)=(6,3,0)+(-4,-3,0)t$ never meet. Find the distance between these 2 skew lines.

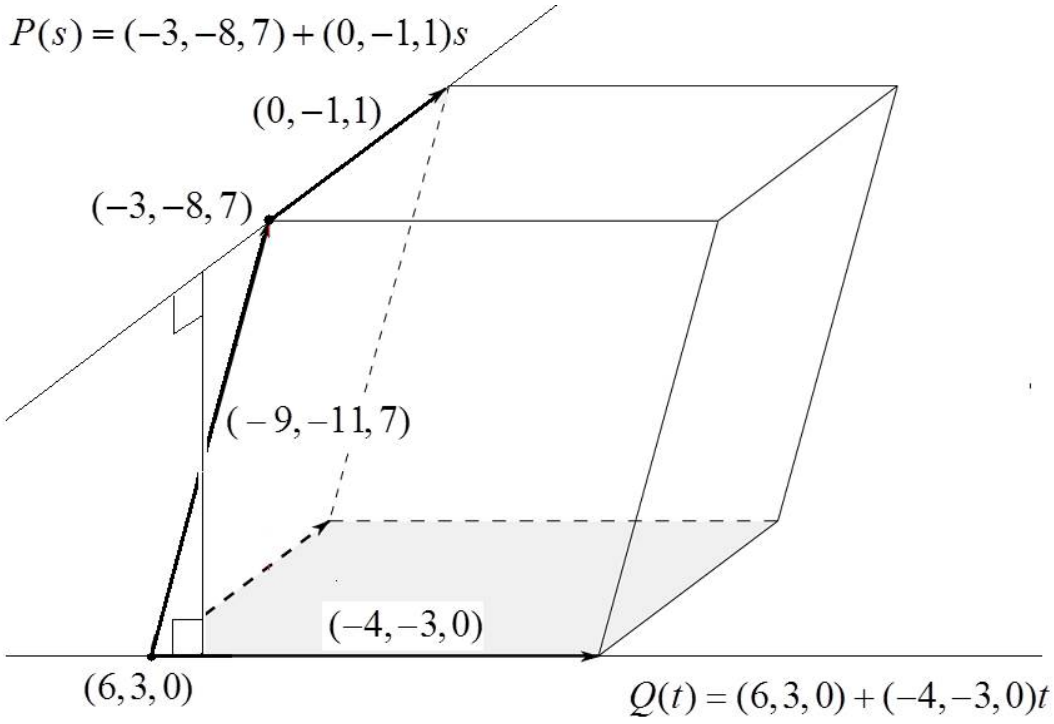
Solution

Suppose $(-3,-8,7)+(0,-1,1)s=(6,3,0)+(-4,-3,0)t$ for some $s$ and $t$. That means

$$(-9,-11,7)+(0,-1,1)s-(-4,-3,0)t=(0,0,0) \quad \text{or} \quad \begin{pmatrix}-4&0&-9\\-3&-1&-11\\0&1&7\end{pmatrix}\begin{pmatrix}-t\\s\\1\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix}$$

However, $\det\begin{pmatrix}-4&0&-9\\-3&-1&-11\\0&1&7\end{pmatrix}=11\neq 0$. Hence $\begin{pmatrix}-t\\s\\1\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix}$ impossible.

Distance between these 2 skew lines is height of parallelepiped.




$$d=\frac{\text{volume of parallelopiped}}{\text{base area of parallelogram}}=\frac{\left|\det\begin{pmatrix}-4&-3&0\\0&-1&1\\-9&-11&7\end{pmatrix}\right|}{|(-4,-3,0)\times(0,-1,1)|}=\frac{11}{|(-3,4,4)|}=\frac{11}{\sqrt{41}}$$

## Section 3.1 Vector Spaces and Subspaces

Definition 3.1.1

A vector space $V$ over $\boldsymbol{R}$ consists of a set $V$ on which two operations (called **addition** + and scalar **multiplication** $\cdot$, respectively) are defined so that for each pair of elements $\boldsymbol{u},\boldsymbol{v}\in V$, $\boldsymbol{u}+\boldsymbol{v}\in V$, and for each $\alpha\in\boldsymbol{R}$ and each element $\boldsymbol{v}\in V$, $\alpha\cdot\boldsymbol{v}\in V$, such that the following conditions hold:

(1) $\boldsymbol{u}+\boldsymbol{v}=\boldsymbol{v}+\boldsymbol{u}$ for all $\boldsymbol{u},\boldsymbol{v}\in V$.

(2) $\boldsymbol{u}+(\boldsymbol{v}+\boldsymbol{w})=(\boldsymbol{u}+\boldsymbol{v})+\boldsymbol{w}$ for all $\boldsymbol{u},\boldsymbol{v},\boldsymbol{w}\in V$.

(3) there exists an elements $\boldsymbol{e}\in V$ such that $\boldsymbol{u}+\boldsymbol{e}=\boldsymbol{u}$ for all $\boldsymbol{u}\in V$.

(4) for each element $\boldsymbol{u}\in V$ there exists an elements $\boldsymbol{v}\in V$ such that $\boldsymbol{u}+\boldsymbol{v}=\boldsymbol{e}$.

(5) for each element $\boldsymbol{u}\in V, 1\cdot\boldsymbol{u}=\boldsymbol{u}$.

(6) for each pair of scalar $a,b\in\boldsymbol{R}$ and each $\boldsymbol{u}\in V, (ab)\cdot\boldsymbol{u}=a\cdot(b\boldsymbol{u})$.

(7) for each scalar $a\in\boldsymbol{R}$ and each pair of elements $\boldsymbol{u},\boldsymbol{v}\in V, a\cdot(\boldsymbol{u}+\boldsymbol{v})=a\cdot\boldsymbol{u}+a\cdot\boldsymbol{v}$.

(8) for each pair of scalar $a,b\in\boldsymbol{R}$ and each $\boldsymbol{u}\in V, (a+b)\cdot\boldsymbol{u}=a\cdot\boldsymbol{u}+b\cdot\boldsymbol{u}$.

Remark 3.1.2

The vector $\boldsymbol{e}$ in (3) is unique, called zero vector. The vector $\boldsymbol{v}$ in (4) is unique, called the additive inverse of $\boldsymbol{u}$. However the zero vector $\boldsymbol{e}$ is not necessary 0 in all positions. (Example 3.1.4)

Example 3.1.3

Let $\boldsymbol{R}^n=\{(a_1,\cdots,a_n):a_1,\cdots,a_n\in\boldsymbol{R}\}$. Define the standard addition and scalar multiplication on $\boldsymbol{R}^n$ by

$$(a_1,\cdots,a_n)+(b_1,\cdots,b_n)=(a_1+b_1,\cdots,a_n+b_n)$$

and

$$\alpha(a_1,\cdots,a_n)=(\alpha a_1,\cdots,\alpha a_n)$$

for any $(a_1,\cdots,a_n),(b_1,\cdots,b_n)\in\boldsymbol{R}^n$ and $\alpha\in\boldsymbol{R}$. In this case, the zero vector is $(0,\cdots,0)$.

It is easily verified that all the vector space axioms hold. Thus, $\boldsymbol{R}^n$, with the standard addition and scalar multiplication, is a vector space.

Example 3.1.4

Let $V = \{(a,1) : a \in \boldsymbol{R}\} \subset \boldsymbol{R}^2$. Define a new addition $+_{\text{new}}$ and scalar multiplication $\cdot_{\text{new}}$ of $V$ by

$$(a,1) +_{\text{new}} (b,1) = (a+b,1)$$

and

$$\alpha \cdot_{\text{new}} (a,1) = (\alpha a,1)$$

for any $(a,1),(b,1) \in V$ and $\alpha \in \boldsymbol{R}$. In this case, the zero vector is (0, 1), but not (0, 0).

$\boldsymbol{R}^n$ is not the only vector space over $\boldsymbol{R}$. Besides $\boldsymbol{R}^n$, there are a lot of vector spaces over $\boldsymbol{R}$.

Example 3.1.5

Let $P_n$ denote the set of all polynomials of degree less than *n*, that is

$$P_n = \{a_0 + a_1 x + \cdots + a_{n-1}x^{n-1} : a_0, a_1, \cdots, a_{n-1} \in \boldsymbol{R}\}.$$

Define the standard addition and scalar multiplication of polynomials by

$$p(x) + q(x) = (a_0 + b_0) + (a_1 + b_1)x + \cdots + (a_{n-1} + b_{n-1})x^{n-1}$$

and

$$\alpha\, p(x) = \alpha a_0 + \alpha a_1 x + \cdots + \alpha a_{n-1}x^{n-1}$$

for any $p(x) = a_0 + a_1 x + \cdots + a_{n-1}x^{n-1}$, $q(x) = b_0 + b_1 x + \cdots + b_{n-1}x^{n-1} \in P_n$ and $\alpha \in \boldsymbol{R}$.

In this case, the zero vector is the zero polynomial 0.

It is easily verified that all the vector space axioms hold. Thus, $P_n$, with the standard addition and scalar multiplication of polynomials, is a vector space.

Example 3.1.6

Let $V$ denote the set of all polynomials of degree equal to $n \geq 1$, that is

$$V = \{a_0 + a_1 x + \cdots + a_n x^n : a_0, a_1, \cdots, a_n \in \boldsymbol{R}, a_n \neq 0\}.$$

Define the standard addition and scalar multiplication of polynomials by

$$p(x) + q(x) = (a_0 + b_0) + (a_1 + b_1)x + \cdots + (a_n + b_n)x^n$$

and

$$\alpha\, p(x) = \alpha a_0 + \alpha a_1 x + \cdots + \alpha a_n x^n$$

for any $p(x) = a_0 + a_1 x + \cdots + a_n x^n, q(x) = b_0 + b_1 x + \cdots + b_n x^n \in V$ and $\alpha \in \boldsymbol{R}$.

However, $V$ is not a vector space because $\underbrace{(1 + x^n)}_{\deg n} + \underbrace{(-x^n)}_{\deg n} = 1 \notin V$.

Example 3.1.7

Let $M_{m\times n}$ denote the set of all $m \times n$ matrix with real entries, that is

$$M_{m\times n} = \left\{ \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} : a_{ij} \in \boldsymbol{R} \right\}.$$

Define the standard addition and scalar multiplication of matrices by

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{pmatrix} = \begin{pmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1n} + b_{1n} \\ a_{21} + b_{121} & a_{22} + b_{22} & \cdots & a_{2n} + b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \cdots & a_{mn} + b_{mn} \end{pmatrix}$$

and

$$\alpha \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} = \begin{pmatrix} \alpha a_{11} & \alpha a_{12} & \cdots & \alpha a_{1n} \\ \alpha a_{21} & \alpha a_{22} & \cdots & \alpha a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha a_{m1} & \alpha a_{m2} & \cdots & \alpha a_{mn} \end{pmatrix}$$

for any $\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{pmatrix} \in M_{m\times n}$ and $\alpha \in \boldsymbol{R}$.

In this case, the zero vector is the zero matrix $\begin{pmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}_{m\times n}$.

It is easily verified that all the vector space axioms hold. Thus, $M_{m\times n}$, with the standard addition and scalar multiplication of matrices, is a vector space.

Example 3.1.8

Let $C(I)$ be the set of all continuous functions on $I$. Define the standard addition and scalar multiplication of functions by

$$(f+g)(x) = f(x) + g(x)$$

and

$$(\alpha f)(x) = \alpha f(x).$$

for all $x \in I$. In this case, the zero vector is the zero function $f(x) = 0$ for all $x \in I$.

It is easily verified that all the vector space axioms hold. Thus, $C(I)$, with the standard addition and scalar multiplication of functions, is a vector space.

Example 3.1.9

Define the standard addition on $\boldsymbol{R}^2$ by

$$(a_1, a_2) + (b_1, b_2) = (a_1 + b_1, a_2 + b_2)$$

and a new scalar multiplication $\cdot_{\text{new}}$ on $\boldsymbol{R}^2$ by

$$\alpha \cdot_{\text{new}} (a_1, a_2) = (a_1, 0).$$

Is $\boldsymbol{R}^2$ a vector space over $\boldsymbol{R}$ with these operations? Justify your answer.

Solution

It is not a vector because condition (5) fails: $1 \cdot_{\text{new}} (1,1) = (1,0) \neq (1,1)$.

Definition 3.1.10

A subset $W$ of a vector space $V$ is said to be a **subspace** of $V$ if

(i) $\boldsymbol{e} \in W$,

(ii) $\alpha \boldsymbol{v} \in W$ for any vector $\boldsymbol{v} \in W$ and scalar $\alpha \in \boldsymbol{R}$,

(iii) $\boldsymbol{u} + \boldsymbol{v} \in W$ for any vectors $\boldsymbol{u}, \boldsymbol{v} \in W$.

Condition (i), (ii) and (iii) guarantee that a subspace $W$ of $V$ is itself a vector space, under the vector space operations already defined in $V$. Fortunately, it is not necessary to verify all of the vector space properties to prove that $W$ is a vector space. Conditions 1, 2, 5, 6, 7, 8 hold for all vectors in $V$, they also hold for the vectors in $W$. Condition 3 is condition (i). Substitute $\alpha = -1$ in condition (ii), we have condition (4).

Example 3.1.11

Let $W = \{\text{all polynomial } p(x)\}$. $W$ is a subset of $C(-\infty, \infty)$. Clearly, zero polynomial $0 \in W$. $W$ satisfies condition (ii) and (iii). Thus, $W$, with the standard addition and scalar multiplication of polynomials, is a subspace of $C(-\infty, \infty)$.

Example 3.1.12

Let $W = \{f : f'' - 3f' + 2f = 0\}$. $W$ is a subset of $C(-\infty, \infty)$. Clearly, zero function $0 \in W$.

For any $f, g \in W$ and $\alpha \in \boldsymbol{R}$, we have $f'' - 3f' + 2f = 0$ and $g'' - 3g' + 2g = 0$.

$$(f+g)'' - 3(f+g)' + 2(f+g) = f'' + g'' - 3f' - 3g' + 2f + 2g = 0$$
$$(\alpha f)'' - 3(\alpha f)' + 2(\alpha f) = \alpha(f'' - 3f' + 2f) = 0$$

Thus, $W$, with the standard addition and scalar multiplication of functions, is a subspace of $C(-\infty, \infty)$.

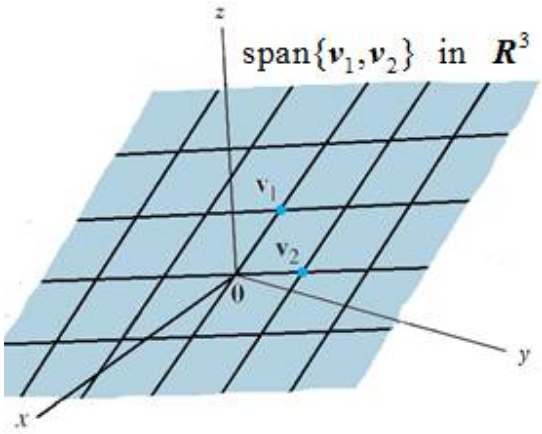
## Section 3.2 Span and linear independence

Definition 3.2.1

Let $\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_n$ be vectors in a vector space $V$. A sum of the form $\alpha_1\boldsymbol{v}_1 + \alpha_2\boldsymbol{v}_2 + \cdots + \alpha_n\boldsymbol{v}_n$, where $\alpha_1, \alpha_2, \cdots, \alpha_n$ are scalars, is called a **linear combination** of $\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_n$. The set of all linear combinations of $\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_n$

$$\{\alpha_1\boldsymbol{v}_1 + \alpha_2\boldsymbol{v}_2 + \cdots + \alpha_n\boldsymbol{v}_n : \alpha_1, \alpha_2, \cdots, \alpha_n \in \boldsymbol{R}\}$$

is called the **span** of $\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_n$ and denoted by $\text{span}\{\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_n\}$.




Theorem 3.2.2

Let $\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_n$ be vectors in a vector space $V$. Then $\text{span}\{\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_n\}$ is a subspace of $V$.

Example 3.2.3

Show that $\text{span}\left\{(2,-3,5)^T, (8,-12,20)^T, (1,0,-2)^T, (0,2,-1)^T, (7,2,0)^T\right\} = \boldsymbol{R}^3$.

Solution

For any $(x,y,z)^T \in \boldsymbol{R}^3$, if we can find $a$, $b$, $c$, $d$, $e$ such that

$$a\begin{pmatrix}2\\-3\\5\end{pmatrix} + b\begin{pmatrix}8\\-12\\20\end{pmatrix} + c\begin{pmatrix}1\\0\\-2\end{pmatrix} + d\begin{pmatrix}0\\2\\-1\end{pmatrix} + e\begin{pmatrix}7\\2\\0\end{pmatrix} = \begin{pmatrix}x\\y\\z\end{pmatrix} \qquad (*)$$

then $\text{span}\left\{(2,-3,5)^T,(8,-12,20)^T,(1,0,-2)^T,(0,2,-1)^T,(7,2,0)^T\right\}=\boldsymbol{R}^3$.

If not, then $\text{span}\left\{(2,-3,5)^T,(8,-12,20)^T,(1,0,-2)^T,(0,2,-1)^T,(7,2,0)^T\right\}\neq\boldsymbol{R}^3$.

Equation (*) becomes

$$\begin{pmatrix}2&8&1&0&7\\-3&-12&0&2&2\\5&20&-2&-1&0\end{pmatrix}\begin{pmatrix}a\\b\\c\\d\\e\end{pmatrix}=\begin{pmatrix}x\\y\\z\end{pmatrix}$$

$$\left(\begin{array}{ccccc|c}2&8&1&0&7&x\\-3&-12&0&2&2&y\\5&20&-2&-1&0&z\end{array}\right)\xrightarrow{R_1\div 2\to R_1}\left(\begin{array}{ccccc|c}1&4&1/2&0&7/2&x/2\\-3&-12&0&2&2&y\\5&20&-2&-1&0&z\end{array}\right)$$

$$\xrightarrow{R_2+3R_1\to R_2,\,R_3-5R_1\to R_3}\left(\begin{array}{ccccc|c}1&4&1/2&0&7/2&x/2\\0&0&3/2&2&25/2&y+3x/2\\0&0&-9/2&-1&-35/2&z-5x/2\end{array}\right)$$

$$\xrightarrow{R_2\times 2/3\to R_2}\left(\begin{array}{ccccc|c}1&4&1/2&0&7/2&x/2\\0&0&1&4/3&25/3&2y/3+x\\0&0&-9/2&-1&-35/2&z-5x/2\end{array}\right)$$

$$\xrightarrow{R_3+(9/2)R_2\to R_3}\left(\begin{array}{ccccc|c}1&4&1/2&0&7/2&x/2\\0&0&1&4/3&25/3&2y/3+x\\0&0&0&5&20&z+3y+2x\end{array}\right)$$

$$\xrightarrow{R_3\div 5\to R_3}\left(\begin{array}{ccccc|c}1&4&1/2&0&7/2&x/2\\0&0&1&4/3&25/3&2y/3+x\\0&0&0&1&4&z/5+3y/5+2x/5\end{array}\right)$$

$$\xrightarrow{R_2-(4/3)R_3\to R_2}\left(\begin{array}{ccccc|c}1&4&1/2&0&7/2&x/2\\0&0&1&0&3&7x/15-2y/15-4z/15\\0&0&0&1&4&z/5+3y/5+2x/5\end{array}\right)$$

$$\xrightarrow{R_1-(1/2)R_2\to R_1}\left(\begin{array}{ccccc|c}1&4&0&0&2&4x/15+y/15+2z/15\\0&0&1&0&3&7x/15-2y/15-4z/15\\0&0&0&1&4&2x/5+3y/5+z/5\end{array}\right)$$

Choose $a=4x/15+y/15+2z/15,\ b=0,\ c=7x/15-2y/15-4z/15,\ d=2x/5+3y/5+z/5,\ e=0,$
then $\text{span}\left\{(2,-3,5)^T,(8,-12,20)^T,(1,0,-2)^T,(0,2,-1)^T,(7,2,0)^T\right\}=\boldsymbol{R}^3$.

Example 3.2.4

Let $V = \{(a_1, a_2, a_3, a_4)^T \in \boldsymbol{R}^4 : a_1 + a_2 + a_3 + a_4 = 0\}$. Clearly, $(0,0,0,0) \in V$.

$$\underbrace{a_1 + a_2 + a_3 + a_4 = 0}_{(a_1,a_2,a_3,a_4)^T \in V}, \underbrace{b_1 + b_2 + b_3 + b_4 = 0}_{(b_1,b_2,b_3,b_4)^T \in V} \Rightarrow \underbrace{a_1 + b_1 + a_2 + b_2 + a_3 + b_3 + a_4 + b_4 = 0}_{(a_1,a_2,a_3,a_4)^T + (b_1,b_2,b_3,b_4)^T \in V}$$

$$\underbrace{a_1 + a_2 + a_3 + a_4 = 0}_{(a_1,a_2,a_3,a_4)^T \in V} \Rightarrow \underbrace{\alpha a_1 + \alpha a_2 + \alpha a_3 + \alpha a_4 = 0}_{\alpha (a_1,a_2,a_3,a_4)^T \in V}$$

for any $(a_1, a_2, a_3, a_4)^T, (b_1, b_2, b_3, b_4)^T \in V$ and $\alpha \in \boldsymbol{R}$. $V$ is a subspace of $\boldsymbol{R}^4$.

For any $(a_1, a_2, a_3, a_4)^T \in V$,

$$\begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ -a_1 - a_2 - a_3 \end{pmatrix} = a_1 \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix} + a_2 \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix} + a_3 \begin{pmatrix} 0 \\ 0 \\ 1 \\ -1 \end{pmatrix}$$

Hence $\begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ -1 \end{pmatrix}$ spans $V$, that is, $\text{span}\left\{\begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ -1 \end{pmatrix}\right\} = V$.

Definition 3.2.5

The vectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_n$ in a vector space $V$ are said to be **linearly independent** if

$$c_1 \boldsymbol{v}_1 + c_2 \boldsymbol{v}_2 + \cdots + c_n \boldsymbol{v}_n = \boldsymbol{0}$$

implies that all the scalars $c_1, c_2, \cdots, c_n$ must equal 0.

Definition 3.2.6

The vectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_n$ in a vector space $V$ are said to be **linearly dependent** if there exist scalars $c_1, c_2, \cdots, c_n$, not all zero, such that

$$c_1 \boldsymbol{v}_1 + c_2 \boldsymbol{v}_2 + \cdots + c_n \boldsymbol{v}_n = \boldsymbol{0}$$

Example 3.2.7

Show that $(1,1,0,0)^T,(0,1,1,0)^T,(0,0,1,1)^T,(1,0,0,1)^T$ are linearly dependent in $\boldsymbol{R}^4$.

Solution

Suppose

$$a\begin{pmatrix}1\\1\\0\\0\end{pmatrix}+b\begin{pmatrix}0\\1\\1\\0\end{pmatrix}+c\begin{pmatrix}0\\0\\1\\1\end{pmatrix}+d\begin{pmatrix}1\\0\\0\\1\end{pmatrix}=\begin{pmatrix}0\\0\\0\\0\end{pmatrix},\quad \text{that is,}\quad \begin{pmatrix}1&0&0&1\\1&1&0&0\\0&1&1&0\\0&0&1&1\end{pmatrix}\begin{pmatrix}a\\b\\c\\d\end{pmatrix}=\begin{pmatrix}0\\0\\0\\0\end{pmatrix}$$

If $a=b=c=d=0$ is the only solution, then $(1,1,0,0)^T,(0,1,1,0)^T,(0,0,1,1)^T,(1,0,0,1)^T$ are linearly independent. If not, they are linearly dependent.

You may use row operation (D.I.Y.). By observation,

$$\begin{pmatrix}1\\1\\0\\0\end{pmatrix}+\begin{pmatrix}0\\0\\1\\1\end{pmatrix}=\begin{pmatrix}1\\1\\1\\1\end{pmatrix}=\begin{pmatrix}0\\1\\1\\0\end{pmatrix}+\begin{pmatrix}1\\0\\0\\1\end{pmatrix}\quad\text{or}\quad\begin{pmatrix}1\\1\\0\\0\end{pmatrix}-\begin{pmatrix}0\\1\\1\\0\end{pmatrix}+\begin{pmatrix}0\\0\\1\\1\end{pmatrix}-\begin{pmatrix}1\\0\\0\\1\end{pmatrix}=\begin{pmatrix}0\\0\\0\\0\end{pmatrix}$$

$(1,1,0,0)^T,(0,1,1,0)^T,(0,0,1,1)^T,(1,0,0,1)^T$ are linearly dependent in $\boldsymbol{R}^4$.

Example 3.2.8

Show that $(x-1)(x-2),(x-1),1$ are linearly independent in $P_3$.

Solution

Suppose

$$a(x-1)(x-2)+b(x-1)+c=0.$$

At $x=1$, $c=0$. Then we have $a(x-1)(x-2)+b(x-1)=0$.

At $x=2$, $b=0$. Then we have $a(x-1)(x-2)=0$.

At $x=3$, $a=0$.

Therefore, $(x-1)(x-2),(x-1),1$ are linearly independent in $P_3$.

Example 3.2.9

Consider $e^x$ and $e^{2x}$. Then $\underset{c_1}{e^x}(\underset{v_1}{e^x})+\underset{c_2}{(-1)}(\underset{v_2}{e^{2x}})=0.$ Are they linearly dependent in $C(-\infty,\infty)$?

No. It is because $c_1=e^x$ is not a constant. Indeed, they are linearly independent. Consider

$$c_1e^x+c_2e^{2x}=0$$

Taking derivative, we have $c_1e^x+2c_2e^{2x}=0.$ Then

$$\begin{pmatrix}e^x & e^{2x}\\ e^x & 2e^{2x}\end{pmatrix}\begin{pmatrix}c_1\\ c_2\end{pmatrix}=\begin{pmatrix}0\\ 0\end{pmatrix}$$

Since $\det\begin{pmatrix}e^x & e^{2x}\\ e^x & 2e^{2x}\end{pmatrix}=2e^{2x}e^x-e^{2x}e^x=e^{3x}\neq 0,$ $\begin{pmatrix}c_1\\ c_2\end{pmatrix}=\begin{pmatrix}e^x & e^{2x}\\ e^x & 2e^{2x}\end{pmatrix}^{-1}\begin{pmatrix}0\\ 0\end{pmatrix}=\begin{pmatrix}0\\ 0\end{pmatrix}.$

$e^x$ and $e^{2x}$ are linearly independent.

Definition 3.2.10

The **Wronskian** of *n* differentiable functions $f_1,\cdots,f_n$ is defined by

$$W(f_1,\cdots,f_n)(x)=\begin{vmatrix}f_1(x) & f_2(x) & \cdots & f_n(x)\\ f_1'(x) & f_2'(x) & \cdots & f_n'(x)\\ \vdots & \vdots & \ddots & \vdots\\ f_1^{(n-1)}(x) & f_2^{(n-1)}(x) & \cdots & f_n^{(n-1)}(x)\end{vmatrix}$$

Theorem 3.2.11

If $W(f_1,\cdots,f_n)(x)\neq 0$ then $f_1,\cdots,f_n$ are linearly independent.

Example 3.2.12

Since $\begin{vmatrix}(x-1)(x-2) & x-1 & 1\\ 2x-3 & 1 & 0\\ 2 & 0 & 0\end{vmatrix}=-2\neq 0,$ $(x-1)(x-2),(x-1),1$ are linearly independent in $P_3.$

## Section 3.3 Basis and Dimension

Definition 3.3.1

The vectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_n$ form a **basis** for a vector space $V$ if

(i) $\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_n$ are linearly independent.

(ii) $\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_n$ span $V$.

A basis is an "efficient" spanning set that contains no unnecessary vectors. In fact, a basis can be constructed from a spanning set by discarding unneeded vectors.

Theorem 3.3.2

Let $S = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_n\}$ be a spanning set for a vector space $V$, that is, $\text{span}\{\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_n\} = V$. If one of the vectors in $S$, say $\boldsymbol{v}_k$, is a combination of the remaining vectors in $S$, then the set formed from $S$ by removing $\boldsymbol{v}_k$ still span $V$, that is,

$$\text{span}\{\boldsymbol{v}_1, \cdots, \boldsymbol{v}_{k-1}, \boldsymbol{v}_{k+1}, \cdots, \boldsymbol{v}_n\} = V.$$

If the original spanning set $S$ is linear independent, then it is already a basis for $V$. Otherwise, one of the vectors in $S$ depends on the others and can be deleted. We can repeat this process until the spanning set is linearly independent and hence is a basis for $V$.

Example 3.3.3

Let $S = \left\{(2,-3,5)^T, (8,-12,20)^T, (1,0,-2)^T, (0,2,-1)^T, (7,2,0)^T\right\}$. By Example 3.2.3, $\text{span}(S) = \boldsymbol{R}^3$.

Since $(8,-12,20)^T = 4(2,-3,5)^T$, by Theorem 3.3.2

$$\text{span}\left\{(2,-3,5)^T, (1,0,-2)^T, (0,2,-1)^T, (7,2,0)^T\right\} = \boldsymbol{R}^3.$$

Since $(7,2,0)^T = 2(2,-3,5)^T + 3(1,0,-2)^T + 4(0,2,-1)^T$, by Theorem 3.3.2

$$\text{span}\left\{(2,-3,5)^T,(1,0,-2)^T,(0,2,-1)^T\right\} = \boldsymbol{R}^3.$$

Suppose $\alpha(2,-3,5)^T + \beta(1,0,-2)^T + \gamma(0,2,-1)^T = (0,0,0)^T$, that is

$$\begin{pmatrix} 2 & 1 & 0 \\ -3 & 0 & 2 \\ 5 & -2 & -1 \end{pmatrix}\begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \alpha\begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix} + \beta\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} + \gamma\begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Since $\det\begin{pmatrix} 2 & 1 & 0 \\ -3 & 0 & 2 \\ 5 & -2 & -1 \end{pmatrix} = 15 \neq 0, \alpha = \beta = \gamma = 0.$ $\left\{(2,-3,5)^T,(1,0,-2)^T,(0,2,-1)^T\right\}$ are linearly independent. We conclude that $\left\{(2,-3,5)^T,(1,0,-2)^T,(0,2,-1)^T\right\}$ is a basis for $\boldsymbol{R}^3$.

A basis is also a linearly independent set that is as large as possible. Let $S$ be a linearly independent subset of a vector space $V$. If $\text{span}(S) \subset_{\neq} V$, then choose a vector $\boldsymbol{v} \notin \text{span}(S)$. Clearly,

$$\text{span}(S) \subset_{\neq} \text{span}(S \cup \boldsymbol{v}).$$

Since $\boldsymbol{v}$ is not a linear combination of any vector in $S$, and $S \cup \boldsymbol{v}$ is linearly independent. If $\text{span}(S \cup \boldsymbol{v}) = V$, then $S \cup \boldsymbol{v}$ is a basis for $V$. If not, repeat the process, enlarge $S$ by one more vector until the spanning set spans $V$.

Example 3.3.4

Let $S = \left\{(2,-3,5)^T\right\}$ be subset of $\boldsymbol{R}^3$. Clearly, $\text{span}(S) \neq \boldsymbol{R}^3$. Choose a vector, say $(1,0,-2)^T \notin \text{span}(S)$. Then $(2,-3,5)^T$ and $(1,0,-2)^T$ are linearly independent and

$$\text{span}\left\{(2,-3,5)^T,(1,0,-2)^T\right\} \subset_{\neq} \boldsymbol{R}^3.$$

Choose a vector, say $(0,2,-1)^T \notin \text{span}\left\{(2,-3,5)^T,(1,0,-2)^T\right\}$. Then $(2,-3,5)^T,(1,0,-2)^T$ and $(0,2,-1)^T$ are linearly independent (Example 3.3.3). For any $(x,y,z)^T \in \boldsymbol{R}^3$, there exists $(\alpha,\beta,\gamma)^T \in \boldsymbol{R}^3$ such that

$$\alpha\begin{pmatrix}2\\-3\\5\end{pmatrix}+\beta\begin{pmatrix}1\\0\\-2\end{pmatrix}+\gamma\begin{pmatrix}0\\2\\-1\end{pmatrix}=\underbrace{\begin{pmatrix}2&1&0\\-3&0&2\\5&-2&-1\end{pmatrix}}_{\det\neq 0}\begin{pmatrix}\alpha\\\beta\\\gamma\end{pmatrix}=\begin{pmatrix}x\\y\\z\end{pmatrix}.$$

That is, $\text{span}\{(2,-3,5)^T,(1,0,-2)^T,(0,2,-1)^T\}=\boldsymbol{R}^3$. Then $\{(2,-3,5)^T,(1,0,-2)^T,(0,2,-1)^T\}$ is a basis for $\boldsymbol{R}^3$.

Theorem 3.3.5

Let $V$ be a vector space having a finite basis. Then every basis for $V$ contains the same number of vectors.

Definition 3.3.6

A vector space is called **finite-dimensional** if it has a basis consisting of a finite number of vectors. The unique number of vectors in each basis for $V$ is called the **dimension** of $V$ and is denoted by $\dim(V)$. A vector space that is not finite-dimensional is called **infinite-dimensional**.

Example 3.3.7

For convenience, we define the basis of the zero-dimensional vector space is $\phi$. The vector space $\{\mathbf{0}\}$ has dimension zero.

Example 3.3.8

Clearly, the set $\{(1,0,\cdots 0),(0,1,\cdots,0),\cdots,(0,\cdots,0,1)\}$ is basis for $\boldsymbol{R}^n$. Then $\dim(\boldsymbol{R}^n)=n$.

Example 3.3.9

Consider $M_{2\times 3}$, the set of all $2\times 3$ matrix with real entries.

$$\left\{\begin{pmatrix}1&0&0\\0&0&0\end{pmatrix},\begin{pmatrix}0&0&0\\1&0&0\end{pmatrix},\begin{pmatrix}0&1&0\\0&0&0\end{pmatrix},\begin{pmatrix}0&0&0\\0&1&0\end{pmatrix},\begin{pmatrix}0&0&1\\0&0&0\end{pmatrix},\begin{pmatrix}0&0&0\\0&0&1\end{pmatrix}\right\}$$

is a basis for $M_{2\times 3}$. Hence $\dim(M_{2\times 3})=2\times 3=6$. In general, $\dim(M_{m\times n})=mn$.

Example 3.3.10

Clearly, the set $\{1, x, \cdots, x^{n-1}\}$ is basis for $P_n$. Then $\dim(P_n) = n$.

Example 3.3.11

Let $V = \{\text{all polynomial } p(x)\}$. The set $\{1, x, x^2, \cdots\}$ is basis for $V$. $V$ is an infinite dimensional vector space.

Example 3.3.12

Find the basis and dimension for $V = \{p(x) \in P_n : p(1) = 0\}$.

Solution

Clearly, the set $\{x-1, x(x-1), \cdots, x^{n-2}(x-1)\}$ is basis for $V$. Then $\dim(V) = n-1$.

Theorem 3.3.13

Let $V$ be an $n$-dimensional vector space, $n \geq 1$. Any linearly independent set of exactly $n$ elements in $V$ is a basis for $V$. Any set of exactly $n$ vectors that spans $V$ is a basis for $V$.

Corollary 3.3.14

If $V$ is a vector space of dimension $n > 0$, then

(i) any set of fewer than $n$ vectors cannot span $V$.

(ii) any set of fewer than $n$ linearly independent vectors can be extended to form a basis for $V$.

(iii) any collection of $m$ vectors in $V$, where $m > n$, is linearly dependent.

Example 3.3.15

Show that $\{(x-1)(x-2), (x-1)(x-3), (x-2)(x-3)\}$ forms a basis for $P_3$.

Solution

Suppose

$$a(x-1)(x-2)+b(x-1)(x-3)+c(x-2)(x-3)=0.$$

At $x=3$, $a=0$. At $x=2$, $b=0$. At $x=1$, $c=0$.

Hence $(x-1)(x-2),(x-1)(x-3),(x-2)(x-3)$ are linearly independent. By Theorem 3.3.13 and $\dim(P_3)=3$, $\{(x-1)(x-2),(x-1)(x-3),(x-2)(x-3)\}$ forms a basis for $P_3$.

Theorem 3.3.16

Let $W$ be a subspace of a finite-dimensional vector space $V$. Then $W$ is finite-dimensional and $\dim(W)\le\dim(V)$. Moreover, if $\dim(W)=\dim(V)$, then $V=W$.

Example 3.3.17

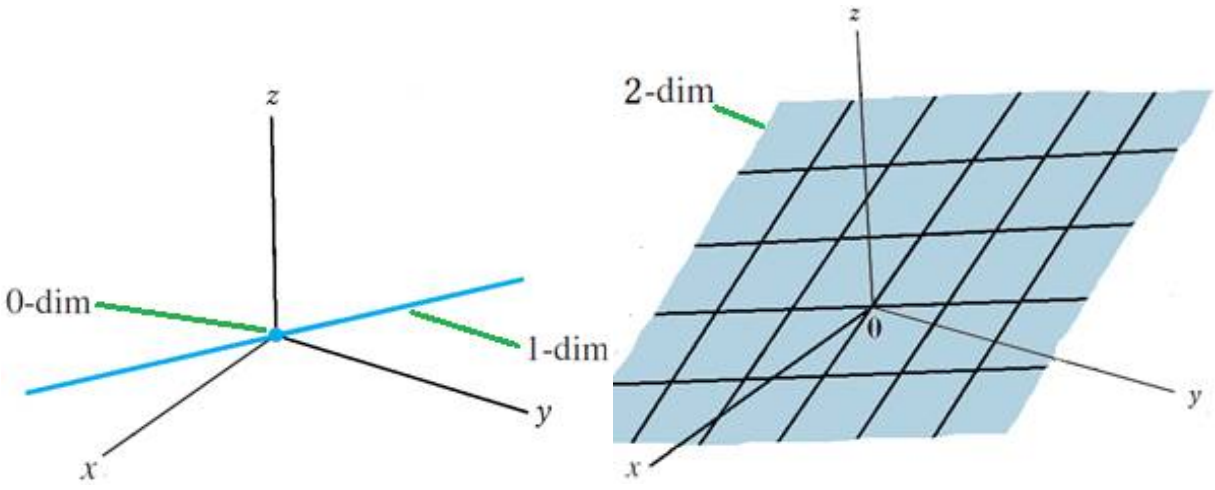
The subspaces of $\boldsymbol{R}^3$ can be classified by dimension.

0-dimensional subspaces: Only the zero subspace.

1-dimensional subspaces: Any subspace spanned by a single nonzero vector. Such subspaces are lines through the origin.

2-dimensional subspaces: Any subspace spanned by two linearly independent vectors. Such subspaces are planes through the origin.

3-dimensional subspaces: Only $\boldsymbol{R}^3$ itself. Any three linearly independent vectors in $\boldsymbol{R}^3$ span all of $\boldsymbol{R}^3$.



## Section 3.4 Change of basis

Theorem 3.4.1

Let $\beta = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_n\}$ be a basis for a vector space $V$. Then for each $\boldsymbol{v} \in V$, there exists a unique set of scalar $c_1, c_2, \cdots, c_n \in \boldsymbol{R}$ such that

$$\boldsymbol{v} = c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 + \cdots + c_n\boldsymbol{v}_n.$$

Let $\beta = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_n\}$ be a basis for a vector space $V$. If we order the basis vector in $\beta$, it is called an ordered basis for $V$. Let us order the basis elements so that $\boldsymbol{v}_i$ is the $i$-th basis vector.

Definition 3.4.2

Suppose $\beta = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_n\}$ be an ordered basis for a vector space $V$ and $\boldsymbol{v} \in V$. By Theorem 3.4.1, there exists a unique set of scalar $c_1, c_2, \cdots, c_n \in \boldsymbol{R}$ such that

$$\boldsymbol{v} = c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 + \cdots + c_n\boldsymbol{v}_n.$$

Then

$$\begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} \in \boldsymbol{R}^n$$

is called the **coordinate vector** of $\boldsymbol{v}$ with respect to the ordered basis $\beta$ and is denoted $[\boldsymbol{v}]_\beta$.

If we switch the order of $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$, we have different ordered basis $\gamma = \{\boldsymbol{v}_2, \boldsymbol{v}_1, \cdots, \boldsymbol{v}_n\}$. Then $\boldsymbol{v} = c_2\boldsymbol{v}_2 + c_1\boldsymbol{v}_1 + \cdots + c_n\boldsymbol{v}_n$. The entry of the coordinate vector is switched in the same order, i.e.,

$$[\boldsymbol{v}]_\gamma = \begin{pmatrix} c_2 \\ c_1 \\ \vdots \\ c_n \end{pmatrix}.$$

Example 3.4.3

Consider two bases $\beta = \{\boldsymbol{e}_1, \boldsymbol{e}_2\}$ and $\gamma = \{\boldsymbol{b}_1, \boldsymbol{b}_2\}$ for $\boldsymbol{R}^2$, where $\boldsymbol{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \boldsymbol{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \boldsymbol{b}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\boldsymbol{b}_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$. Suppose $\boldsymbol{x} = \begin{pmatrix} 1 \\ 6 \end{pmatrix}$. $\boldsymbol{x}$ can be expressed as

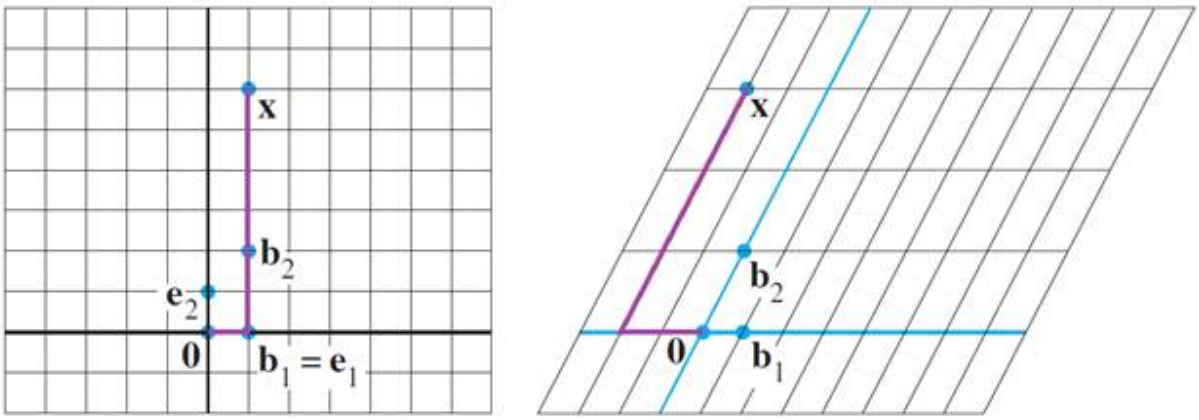
$$\boldsymbol{x} = \begin{pmatrix} 1 \\ 6 \end{pmatrix} = 1\begin{pmatrix} 1 \\ 0 \end{pmatrix} + 6\begin{pmatrix} 0 \\ 1 \end{pmatrix} = 1\boldsymbol{e}_1 + 6\boldsymbol{e}_2.$$

Hence $[\boldsymbol{x}]_\beta = \begin{pmatrix} 1 \\ 6 \end{pmatrix}$. On the other hand, $\boldsymbol{x}$ can be expressed as

$$\boldsymbol{x} = \begin{pmatrix} 1 \\ 6 \end{pmatrix} = -2\begin{pmatrix} 1 \\ 0 \end{pmatrix} + 3\begin{pmatrix} 1 \\ 2 \end{pmatrix} = -2\boldsymbol{b}_1 + 3\boldsymbol{b}_2.$$

Hence $[\boldsymbol{x}]_\gamma = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$.

Different basis gives the same vector different coordinate vector.




Theorem 3.4.4

If $\beta$ is a basis of a vector space $V$, then

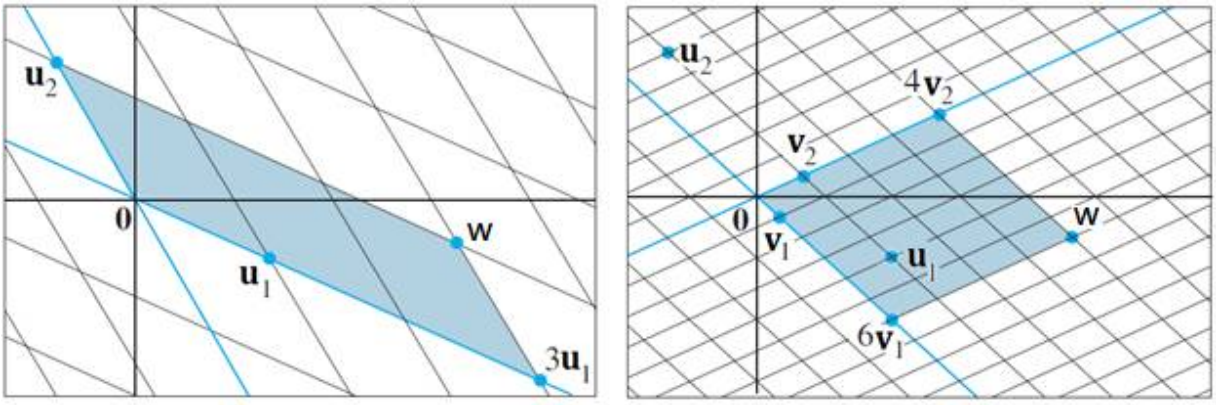
(i) $[\boldsymbol{x} + \boldsymbol{y}]_\beta = [\boldsymbol{x}]_\beta + [\boldsymbol{y}]_\beta$, for all vectors $\boldsymbol{x}, \boldsymbol{y} \in V$, and

(ii) $[k\boldsymbol{x}]_\beta = k[\boldsymbol{x}]_\beta$, for all vectors $\boldsymbol{x} \in V$ and all scalars $k \in \boldsymbol{R}$.

Example 3.4.5

Consider two bases $\beta = \{\boldsymbol{u}_1, \boldsymbol{u}_2\}$ and $\gamma = \{\boldsymbol{v}_1, \boldsymbol{v}_2\}$ for a vector space $V$. Let $\boldsymbol{w} \in V$ such that

$$\boldsymbol{w} = 3\boldsymbol{u}_1 + \boldsymbol{u}_2 \text{ and } \boldsymbol{w} = 6\boldsymbol{v}_1 + 4\boldsymbol{v}_2.$$




What is the relation between the two coordinate vectors $[\boldsymbol{w}]_\beta$ and $[\boldsymbol{w}]_\gamma$?

Clearly, $[\boldsymbol{w}]_\beta = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and $[\boldsymbol{w}]_\gamma = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$.

Let $\boldsymbol{u}_1 = a\boldsymbol{v}_1 + b\boldsymbol{v}_2$ and $\boldsymbol{u}_2 = c\boldsymbol{v}_1 + d\boldsymbol{v}_2$. Then

$$\boldsymbol{w} = 3\boldsymbol{u}_1 + \boldsymbol{u}_2 = 3(a\boldsymbol{v}_1 + b\boldsymbol{v}_2) + (c\boldsymbol{v}_1 + d\boldsymbol{v}_2) = (3a + c)\boldsymbol{v}_1 + (3b + d)\boldsymbol{v}_2.$$

We have

$$\begin{pmatrix} 6 \\ 4 \end{pmatrix} = [\boldsymbol{w}]_\gamma = \begin{pmatrix} 3a + c \\ 3b + d \end{pmatrix} = \begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \left( [\boldsymbol{u}_1]_\gamma \;\vdots\; [\boldsymbol{u}_2]_\gamma \right) [\boldsymbol{w}]_\beta.$$

Theorem 3.4.6

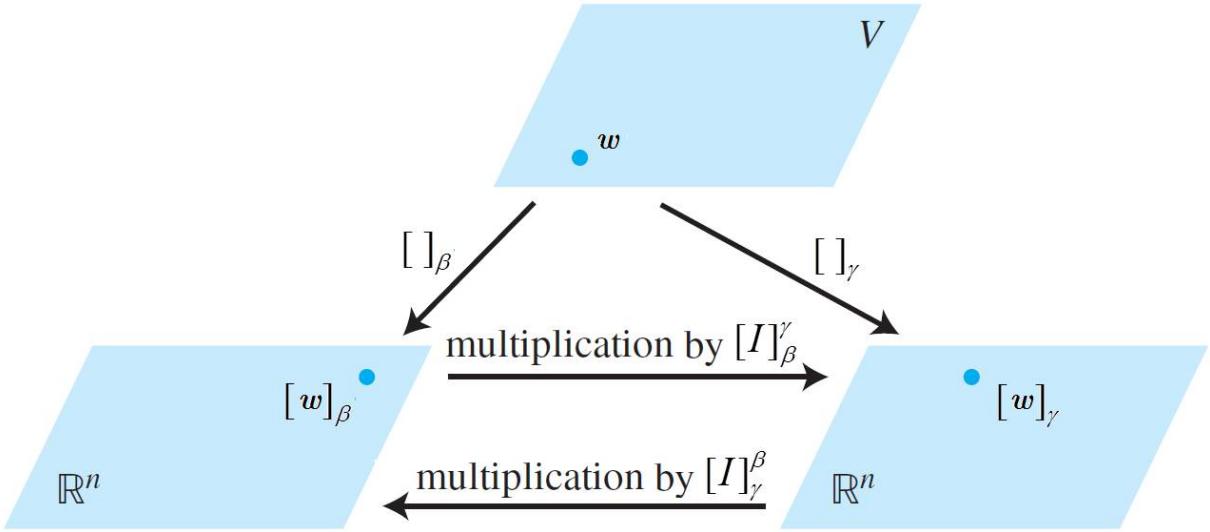
Let $\beta = \{\boldsymbol{u}_1, \cdots, \boldsymbol{u}_n\}$ and $\gamma = \{\boldsymbol{v}_1, \cdots, \boldsymbol{v}_n\}$ be two bases of a vector space $V$. Let $\boldsymbol{w} \in V$. Then

$$[\boldsymbol{w}]_\gamma = \left( [\boldsymbol{u}_1]_\gamma \;\vdots\; \cdots \;\vdots\; [\boldsymbol{u}_n]_\gamma \right) [\boldsymbol{w}]_\beta.$$

The $n \times n$ matrix $\left( [\boldsymbol{u}_1]_\gamma \;\vdots\; \cdots \;\vdots\; [\boldsymbol{u}_n]_\gamma \right)$ is called the transition matrix from $\beta$ to $\gamma$, denoted by $[I]_\beta^\gamma$.

Theorem 3.4.7

Let $\beta$ and $\gamma$ be two bases of a vector space $V$. Then $[I]_\beta^\gamma$ is invertible and $\left([I]_\beta^\gamma\right)^{-1} = [I]_\gamma^\beta$.




Example 3.4.8

Let $\beta = \{(x-1)(x-2),(x-1)(x-3),(x-2)(x-3)\}$ and $\gamma = \{(x-1)(x-2),(x-1),1\}$ be two bases of $P_3$. Find $[I]_\beta^\gamma$, $[I]_\gamma^\beta$, $\left[x^2\right]_\beta$ and $\left[x^2\right]_\gamma$. Verify Theorem 3.4.6 and 3.4.7.

Solution

Clearly,

$$[I]_\beta^\gamma = \begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & -2 \\ 0 & 0 & 2 \end{pmatrix},\quad [I]_\gamma^\beta = \begin{pmatrix} 1 & 1 & 1/2 \\ 0 & -1 & -1 \\ 0 & 0 & 1/2 \end{pmatrix},\quad \left[x^2\right]_\beta = \begin{pmatrix} 9/2 \\ -4 \\ 1/2 \end{pmatrix},\quad \left[x^2\right]_\gamma = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}$$

$$[I]_\beta^\gamma\left[x^2\right]_\beta = \begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & -2 \\ 0 & 0 & 2 \end{pmatrix}\begin{pmatrix} 9/2 \\ -4 \\ 1/2 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} = \left[x^2\right]_\gamma,\quad [I]_\gamma^\beta\left[x^2\right]_\gamma = \begin{pmatrix} 1 & 1 & 1/2 \\ 0 & -1 & -1 \\ 0 & 0 & 1/2 \end{pmatrix}\begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 9/2 \\ -4 \\ 1/2 \end{pmatrix} = \left[x^2\right]_\beta$$

$$\left([I]_\beta^\gamma\right)^{-1} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & -2 \\ 0 & 0 & 2 \end{pmatrix}^{-1} = [I]_\gamma^\beta = \begin{pmatrix} 1 & 1 & 1/2 \\ 0 & -1 & -1 \\ 0 & 0 & 1/2 \end{pmatrix}$$

## Section 3.5 Row Spaces and Column Spaces

Definition 3.5.1

Let $A$ be an $m \times n$ matrix. The set of all linear combinations of the row vectors is called the **row space** of $A$ and is denoted by Row($A$). Each row has $n$ entries, so Row($A$) is a subspace of $\boldsymbol{R}^n$.

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}. \quad \text{Row}(A) = \text{span}\left\{(a_{11}, a_{12} \cdots, a_{1n}), (a_{21}, a_{22} \cdots, a_{2n}), \cdots, (a_{m1}, a_{m2} \cdots, a_{mn})\right\}$$

The set of all linear combinations of the column vectors is called the **column space** of $A$ and is denoted by Col ($A$). Each column has $m$ entries, so Col ($A$) is a subspace of $\boldsymbol{R}^m$.

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}. \quad \text{Col}(A) = \text{span}\left\{\begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix}, \begin{pmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{pmatrix}, \cdots, \begin{pmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{pmatrix}\right\}$$

Theorem 3.5.2

If two matrices $A$ and $B$ are row equivalent, then their row spaces are the same. If $U$ is in row echelon form, the nonzero rows of $U$ form a basis for the row space of $A$ as well as for that of $U$.

We can use the row echelon form $U$ of $A$ to find a basis for the column space of $A$. We need only determine the columns of $U$ that correspond to the leading 1's. These same columns of $A$ will be linearly independent and form a basis for the column space of $A$.

Remark 3.5.3

The row echelon form $U$ tells us only which columns of $A$ to use to form a basis. We cannot use the column vectors from $U$, since, in general, $U$ and $A$ have different column spaces.

Example 3.5.4

Let

$$A=\begin{pmatrix}1&-2&1&1&2\\-1&3&0&2&-2\\0&1&1&3&4\\1&2&5&13&5\end{pmatrix}$$

$$\begin{pmatrix}1&-2&1&1&2\\-1&3&0&2&-2\\0&1&1&3&4\\1&2&5&13&5\end{pmatrix}\xrightarrow{R_2+R_1\to R_2,\,R_4-R_1\to R_4}\begin{pmatrix}1&-2&1&1&2\\0&1&1&3&0\\0&1&1&3&4\\0&4&4&12&3\end{pmatrix}$$

$$\xrightarrow{R_3-R_2\to R_3,\,R_4-4R_2\to R_4}\begin{pmatrix}1&-2&1&1&2\\0&1&1&3&0\\0&1&1&3&4\\0&4&4&12&3\end{pmatrix}$$

$$\xrightarrow{R_3-R_2\to R_3,\,R_4-4R_2\to R_4}\begin{pmatrix}1&-2&1&1&2\\0&1&1&3&0\\0&0&0&0&4\\0&0&0&0&3\end{pmatrix}$$

$$\xrightarrow{R_3\div 4\to R_3}\begin{pmatrix}1&-2&1&1&2\\0&1&1&3&0\\0&0&0&0&1\\0&0&0&0&3\end{pmatrix}$$

$$\xrightarrow{R_4-3R_3\to R_4}\begin{pmatrix}1&-2&1&1&2\\0&1&1&3&0\\0&0&0&0&1\\0&0&0&0&0\end{pmatrix}$$

$(1,-2,1,1,2),(0,1,1,3,0),(0,0,0,0,1)$ form a basis of Row($A$).

$\begin{pmatrix}1\\-1\\0\\1\end{pmatrix},\begin{pmatrix}-2\\3\\1\\2\end{pmatrix},\begin{pmatrix}2\\-2\\4\\5\end{pmatrix}$ form a basis of Col($A$) but $\begin{pmatrix}1\\0\\0\\0\end{pmatrix},\begin{pmatrix}-2\\1\\0\\0\end{pmatrix},\begin{pmatrix}2\\0\\1\\0\end{pmatrix}$ do not.

Number of basis vectors in Row($A$) equal number of basis vectors in dim Col($A$)

Theorem 3.5.5

If $A$ is an $m \times n$ matrix, then the dimension of the row space of $A$ equals the dimension of the column space of $A$.

Definition 3.5.6

The **rank** of a matrix $A$, denoted rank($A$), is the dimension of the row space of $A$.

Definition 3.5.7

Let $A$ be an $m \times n$ matrix. Let $N(A)$ denote the set of all solutions of system of linear equations $A\boldsymbol{x} = \boldsymbol{0}$, that is,

$$N(A) = \{x \in \boldsymbol{R}^n : A\boldsymbol{x} = \boldsymbol{0}\}$$

Theorem 3.5.8

Let $A$ be an $m \times n$ matrix. $N(A)$ is a subspace of $\boldsymbol{R}^n$.

Proof

Clearly, $\boldsymbol{0} \in N(A)$. If $\boldsymbol{x} \in N(A)$ and $\alpha \in \boldsymbol{R}$, then

$$A(\alpha \boldsymbol{x}) = \alpha A\boldsymbol{x} = \boldsymbol{0}$$

and hence $\alpha \boldsymbol{x} \in N(A)$. If $\boldsymbol{x}$ and $\boldsymbol{y}$ are elements of $N(A)$, then

$$A(\boldsymbol{x} + \boldsymbol{y}) = A\boldsymbol{x} + A\boldsymbol{y} = \boldsymbol{0} + \boldsymbol{0} = \boldsymbol{0}.$$

Therefore, $\boldsymbol{x} + \boldsymbol{y} \in N(A)$. It then follows that $N(A)$ is a subspace of $\boldsymbol{R}^n$.

Definition 3.5.9

Let $A$ be an $m \times n$ matrix. $N(A)$ is called the **null space** of $A$. The dimension of the null space of a matrix is called the **nullity** of the matrix.

Theorem 3.5.10 (Rank Nullity Theorem)

If $A$ is an $m \times n$ matrix, then the rank of $A$ + the nullity of $A = n$.

Outline of the proof

If rank($A$) = $r$, number of leading variables equal $r$. Thus, number of free variables equal $n - r$, which is nullity of $A$.

Example 3.5.11

Consider Example 3.5.4, the reduced row echelon form of $A$ is

$$\begin{pmatrix} 1 & -2 & 1 & 1 & 2 \\ 0 & 1 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 - 2R_3 \to R_1} \begin{pmatrix} 1 & -2 & 1 & 1 & 0 \\ 0 & 1 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 + 2R_2 \to R_2} \begin{pmatrix} 1 & 0 & 3 & 7 & 0 \\ 0 & 1 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$N(A) = \left\{ \boldsymbol{x} = (x_1, x_2, x_3, x_4, x_5)^T : A\boldsymbol{x} = \boldsymbol{0} \right\}$$

$$= \left\{ (x_1, x_2, x_3, x_4, x_5)^T : \begin{pmatrix} 1 & 0 & 3 & 7 & 0 \\ 0 & 1 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \right\}$$

$$= \left\{ (x_1, x_2, x_3, x_4, x_5)^T : x_1 + 3x_3 + 7x_4 = 0, x_2 + x_3 + 3x_4 = 0, x_5 = 0 \right\}$$

$$= \left\{ \begin{pmatrix} -3x_3 - 7x_4 \\ -x_3 - 3x_4 \\ x_3 \\ x_4 \\ 0 \end{pmatrix} \right\}$$

$$= \left\{ x_3 \begin{pmatrix} -3 \\ -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} -7 \\ -3 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right\}$$

rank($A$) = 3 (3 leading variables $x_1, x_2, x_5$), dim $N(A)$ = 2 (2 free variables $x_3, x_4$), 3 + 2 = $n$ = 5.

Theorem 3.5.12

Let $A = \begin{pmatrix} \boldsymbol{a}_1 & \boldsymbol{a}_2 & \cdots & \boldsymbol{a}_n \end{pmatrix}$ and $\boldsymbol{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$. A linear system $A\boldsymbol{x} = \boldsymbol{b}$ is consistent if and only if $\boldsymbol{b}$ is in the column space of $A$, i.e.,

$$\boldsymbol{b} = x_1\boldsymbol{a}_1 + x_2\boldsymbol{a}_2 + \cdots + x_n\boldsymbol{a}_n$$

Consider $A = \begin{pmatrix} 0 & 2 & -8 \\ 1 & -2 & 1 \\ -4 & 5 & 9 \end{pmatrix}$ in Example 1.2.1. We have $rref(A) = \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix}$ in Example 1.2.5, an invertible $A$ in Example 2.1.10 and $\det A = -2 \neq 0$ in Example 2.1.38.

Furthermore, $\begin{pmatrix} 8 \\ 0 \\ -9 \end{pmatrix} = 29\begin{pmatrix} 0 \\ 1 \\ -4 \end{pmatrix} + 16\begin{pmatrix} 2 \\ -2 \\ 5 \end{pmatrix} + 3\begin{pmatrix} -8 \\ 1 \\ 9 \end{pmatrix} \in \text{Col}(A)$ in Example 2.1.38.

Summary 3.5.13

For an $n \times n$ matrix $A$, the following statements are equivalent:

(i) $A$ is invertible.

(ii) $\det(A) \neq 0$.

(ii) The linear system $A\boldsymbol{x} = \boldsymbol{b}$ has a unique solution $\boldsymbol{x}$, for all $\boldsymbol{b}$ in $\boldsymbol{R}^n$.

(iii) $rref(A) = I_n$.

(iv) $\text{rank}(A) = n$.

(v) $\text{Col}(A) = \boldsymbol{R}^n$.

(vi) $N(A) = \{\boldsymbol{0}\}$.

(vii) The column vectors of $A$ form a basis of $\boldsymbol{R}^n$.

(viii) The column vectors of $A$ span $\boldsymbol{R}^n$.

(ix) The column vectors of A are linearly independent.

## Section 4.1 Linear Transformations

Definition 4.1.1

A mapping $T : V \to W$ from a vector space $V$ into a vector space $W$ is said to be a **linear transformation** over $\boldsymbol{R}$ if

$$T(\alpha \boldsymbol{v}_1 + \beta \boldsymbol{v}_2) = \alpha T(\boldsymbol{v}_1) + \beta T(\boldsymbol{v}_2) \tag{1}$$

for all $\boldsymbol{v}_1, \boldsymbol{v}_2 \in V$ and for all scalars $\alpha, \beta \in \boldsymbol{R}$.

Remark 4.1.2

Condition (1) is equivalent to

$$\begin{cases} T(\boldsymbol{v}_1 + \boldsymbol{v}_2) = T(\boldsymbol{v}_1) + T(\boldsymbol{v}_2) \\ T(\alpha \boldsymbol{v}_1) = \alpha T(\boldsymbol{v}_1) \end{cases}$$

for all $\boldsymbol{v}_1, \boldsymbol{v}_2 \in V$ and for all scalars $\alpha \in \boldsymbol{R}$.

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Remark 4.1.3

If $T : V \to W$ is a linear transformation from $V$ to $W$, then

$$T(\alpha_1 \boldsymbol{v}_1 + \alpha_2 \boldsymbol{v}_2 + \cdots + \alpha_n \boldsymbol{v}_n) = \alpha_1 T(\boldsymbol{v}_1) + \alpha_2 T(\boldsymbol{v}_2) + \cdots + \alpha_n T(\boldsymbol{v}_n).$$

for all $\boldsymbol{v}_1, \cdots, \boldsymbol{v}_n \in V$ and for all scalars $\alpha_1, \cdots, \alpha_n \in \boldsymbol{R}$.

Remark 4.1.4

If $T : V \to W$ is a linear transformation from $V$ to $W$, then $T(\mathbf{0}_V) = \mathbf{0}_W$.

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Remark 4.1.5

If $V = W$, the linear transformation $T$ is called the linear operator on $V$.

Example 4.1.6

Let $T : V \to W$ be zero mapping defined by $T(\boldsymbol{v}) = \mathbf{0}_W$ for all $\boldsymbol{v} \in V$. Then $T$ is linear.

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Example 4.1.7

Let $I : V \to V$ be identity mapping defined by $I(\boldsymbol{v}) = \boldsymbol{v}$ for all $\boldsymbol{v} \in V$. Then $I$ is linear.

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Example 4.1.8

Let $T : \boldsymbol{R}^2 \to \boldsymbol{R}^2$ be mapping defined by $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + 2y \\ 3x + 4y \end{pmatrix}$ for all $\begin{pmatrix} x \\ y \end{pmatrix} \in \boldsymbol{R}^2$. Then $T$ is linear.

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Example 4.1.9

The **trace** of an *n*-by-*n* square matrix $A=\left(a_{ij}\right)_{n\times n}$ is defined to be $\mathrm{Tr}(A)=a_{11}+a_{22}+\cdots+a_{nn}$. Let $T:M_{n\times n}\to \boldsymbol{R}$ be mapping defined by $T(A)=Tr(A)$ for all $A\in M_{n\times n}$. Then $T$ is linear.

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Example 4.1.10

Let $P_n=\{a_0+a_1x+\cdots+a_{n-1}x^{n-1}\mid a_0,\cdots,a_{n-1}\in \boldsymbol{R}\}$ where $n\geq 1,$ be set of polynomials having degree $\leq n-1$. It is an *n*-dimensional vector space. Let $T:P_3\to P_4$ be mapping defined by $T(p(x))=(1+x)p(x)$ for all $p(x)\in P_3,$ i.e., $T(a_0+a_1x+a_2x^2)=(1+x)(a_0+a_1x+a_2x^2)$ $=a_0+(a_0+a_1)x+(a_1+a_2)x^2+a_2x^3$. Then $T$ is linear.

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Example 4.1.11

Let $T:P_3\to P_2$ be differentiation mapping defined by $T(p(x))=p'(x)$ for all $p(x)\in P_3,$ i.e., $T(a_0+a_1x+a_2x^2)=a_1+2a_2x.$ Then $T$ is linear.

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Example 4.1.12

Let $T : P_3 \to P_4$ be integration mapping defined by $T(p(x)) = \int_1^t p(x)dx$ for all $p(x) \in P_3$, i.e., $T(a_0 + a_1x + a_2x^2) = \int_1^t a_0 + a_1x + a_2x^2 dx = \left[ a_0x + a_1\frac{x^2}{2} + a_2\frac{x^3}{3} \right]_1^t = -\left( a_0 + \frac{a_1}{2} + \frac{a_2}{3} \right) + a_0t + a_1\frac{t^2}{2} + a_2\frac{t^3}{3}$.

Then $T$ is linear.

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Example 4.1.13

Let $T : P_3 \to P_3$ be mapping defined by $T(p(x)) = p(x+1)$ for all $p(x) \in P_3$, i.e., $T(a_0 + a_1x + a_2x^2) = a_0 + a_1(x+1) + a_2(x+1)^2 = (a_0 + a_1 + a_2) + (a_1 + 2a_2)x + a_2x^2$. Then $T$ is linear.

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Example 4.1.14

Let $T : P_3 \to \boldsymbol{R}^3$ be mapping defined by $T(p(x)) = (p(1), p(2), p(3))$ for all $p(x) \in P_3$, i.e., $T(a_0 + a_1x + a_2x^2) = (a_0 + a_1 + a_2, a_0 + 2a_1 + 2^2a_2, a_0 + 3a_1 + 3^2a_2)$. Then $T$ is linear.

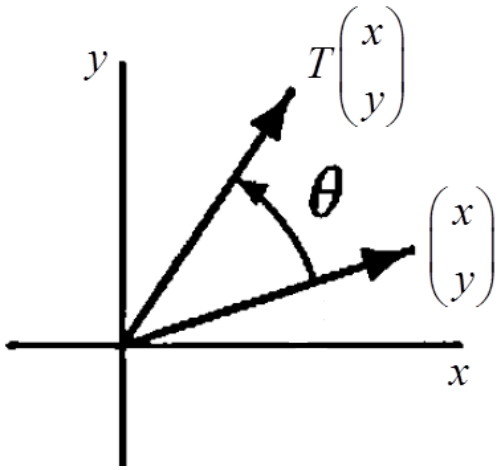
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Example 4.1.15

Let $M_{2\times 3}=\left\{\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix}\middle| a_{11},a_{12},a_{13},a_{21},a_{22},a_{23}\in \boldsymbol{R}\right\}$ consists of $2\times 3$ matrices. It is a 6-dimensional vector space. Let $T:M_{2\times 3}\to M_{2\times 3}$ be mapping defined by $T(A)=\begin{pmatrix} a & b \\ c & d \end{pmatrix}A.$ for all $A\in M_{2\times 3}$. Then $T$ is linear.

Example 4.1.16 (Rotation)

Let $T:\boldsymbol{R}^2\to\boldsymbol{R}^2$ be rotation mapping defined by counterclockwise rotating vector $\begin{pmatrix} x \\ y \end{pmatrix}$ by angle $\theta$. Then $T$ is linear.




Example 4.1.17

Recall complex number $C = \{z = x + yi \mid x, y \in \boldsymbol{R}\}$ is a vector space over $\boldsymbol{R}$. Let $T : \boldsymbol{C} \to \boldsymbol{C}$ be mapping defined by $T(z) = (a + bi)z$ for all $z \in \boldsymbol{C}$, i.e., $T(x + yi) = (a + bi)(x + yi)$ $= (ax - by) + (ay + bx)i$. Then $T$ is linear.

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Example 4.1.18 (Translation)

Let $T : \boldsymbol{R}^2 \to \boldsymbol{R}^2$ be mapping defined by $T(x, y) = (x + 1, y + 2)$ for all $(x, y) \in \boldsymbol{R}^2$. Is $T$ linear?

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Example 4.1.19

Is it possible to find a linear transformation $T : \boldsymbol{R}^2 \to \boldsymbol{R}^2$ such that $T(1,2) = (3,4)$ and $T(2,4) = (5,6)$?

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Example 4.1.20

Let $T : span\{(-1,0,1),(0,1,0)\} \to \boldsymbol{R}^2$ be linear transformation defined by $T(-1,0,1) = (1,3)$ and $T(0,1,0) = (2,4)$. What is $T(2,3,-2)$?

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Definition 4.1.21

Let $T : V \to W$ be a linear transformation. The **kernel** of $T$, denoted $\ker(T)$, is defined by

$$\ker(T) = \{\boldsymbol{v} \in V \mid T(\boldsymbol{v}) = \boldsymbol{0}_W\}.$$

Remark 4.1.22

Let $T : V \to W$ be a linear transformation. Then $T$ is injective if and only if $\ker(T) = \{\boldsymbol{0}_V\}$.

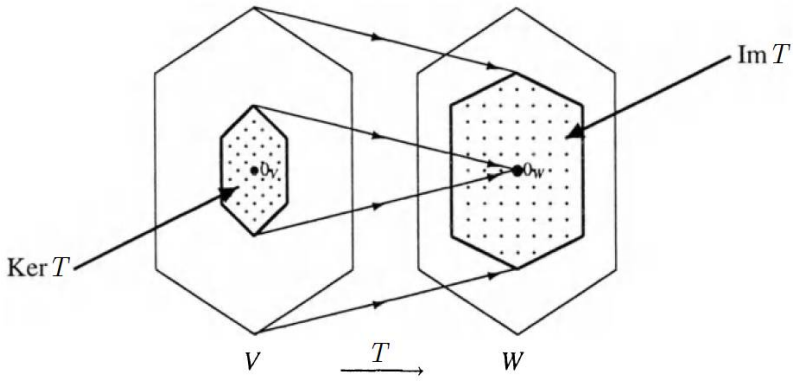
Definition 4.1.23

Let $T : V \to W$ be linear. The **image** or **range** of $T$, denoted $T(V)$, is defined by

$$T(V) = \{T(\boldsymbol{v}) \in W \mid \boldsymbol{v} \in V\}.$$

Theorem 4.1.24

Let $T : V \to W$ be a linear transformation. Then $\ker(T)$ and $T(V)$ are subspaces of $V$ and $W$ respectively.




Example 4.1.25

Let $T:V\to W$ be zero mapping defined by $T(\boldsymbol{v})=\mathbf{0}_W$ for all $\boldsymbol{v}\in V$. Then $\ker(T)=V$ and $T(V)=\{\mathbf{0}_W\}$.

Example 4.1.26

Let $T:V\to V$ be identity mapping defined by $T(\boldsymbol{v})=\boldsymbol{v}$ for all $\boldsymbol{v}\in V$. Then $\ker(T)=\{\mathbf{0}_V\}$ and $T(V)=V$.

Theorem 4.1.27 (Dimension Theorem)

Let $T:V\to W$ be a linear transformation. If $\dim V$ is finite, then

$$\dim V=\dim\ker(T)+\dim T(V).$$

In particular, if $V=\boldsymbol{R}^n$ and $W=\boldsymbol{R}^m$, this is exactly rank-nullity theorem.

Outline of proof

Let $\{\boldsymbol{v}_1,\boldsymbol{v}_2,\cdots,\boldsymbol{v}_m\}$ be the basis of ker($T$). Extend this basis to form a basis of $V$.

$$\{\overbrace{\boldsymbol{v}_1,\boldsymbol{v}_2,\cdots,\boldsymbol{v}_m}^{\text{basis of }\ker(T)},\underbrace{\boldsymbol{u}_1,\boldsymbol{u}_2,\cdots,\boldsymbol{u}_k}_{\text{basis of }V}\}$$

Then $\{T(\boldsymbol{u}_1),T(\boldsymbol{u}_2),\cdots,T(\boldsymbol{u}_k)\}$ is a basis of $T(V)$.

Theorem 4.1.28

Let $V$ and $W$ be vector spaces of equal finite dimension, and let $T:V\to W$ be linear. Then the following are equivalent.

(a) $T$ is one-to-one ($\ker(T)=\{\mathbf{0}_V\}$).

(b) $T$ is onto ($\dim T(V)=\dim W$).

Example 4.1.29

Let $T:\boldsymbol{R}^3\to\boldsymbol{R}^3$ be linear transformation defined by $T\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}x+2y+3z\\4x+5y+6z\\7x+8y+9z\end{pmatrix}$ for all $\begin{pmatrix}x\\y\\z\end{pmatrix}\in\boldsymbol{R}^3$.

Consider $\ker(T)$. By elementary row operation, $\begin{pmatrix}1&2&3\\4&5&6\\7&8&9\end{pmatrix}\to\begin{pmatrix}1&2&3\\0&-3&-6\\0&-6&-12\end{pmatrix}\to\begin{pmatrix}1&2&3\\0&1&2\\0&0&0\end{pmatrix}$.

Let $\begin{pmatrix}x\\y\\z\end{pmatrix}\in\ker(T)$. We have $x+2y+3z=y+2z=0$. Then $y=-2z$ and $x=-2y-3z$

$=-2(-2z)-3z=z$.

$$\ker(T)=\left\{\begin{pmatrix}x\\y\\z\end{pmatrix}\right\}=\left\{\begin{pmatrix}z\\-2z\\z\end{pmatrix}\right\}=span\left\{\begin{pmatrix}1\\-2\\1\end{pmatrix}\right\}.$$

Consider $T(\boldsymbol{R}^3)$.

Method 1: By elementary column operation, $\begin{pmatrix}1&2&3\\4&5&6\\7&8&9\end{pmatrix}\to\begin{pmatrix}1&0&0\\4&-3&-6\\7&-6&-12\end{pmatrix}\to\begin{pmatrix}1&0&0\\4&1&0\\7&2&0\end{pmatrix}$. Then

$$T(\boldsymbol{R}^3)=span\left\{\begin{pmatrix}1\\4\\7\end{pmatrix},\begin{pmatrix}2\\5\\8\end{pmatrix}\right\}.$$

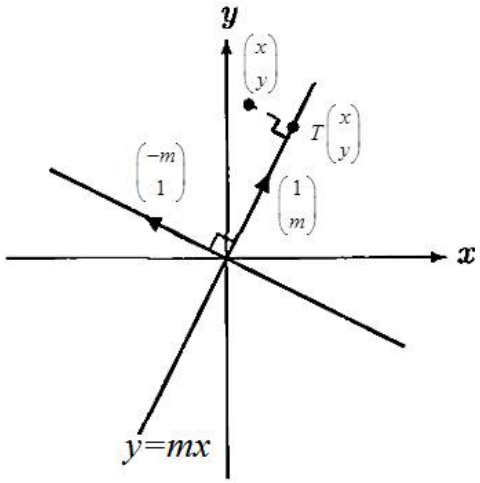
Method 2: $\left\{\begin{pmatrix}1\\-2\\1\end{pmatrix}\right\}$ is a basis of $\ker(T)$. Extend it to form a basis of $\boldsymbol{R}^3$, $\left\{\begin{pmatrix}1\\-2\\1\end{pmatrix},\begin{pmatrix}1\\0\\0\end{pmatrix},\begin{pmatrix}0\\1\\0\end{pmatrix}\right\}$. Then

$\left\{T\begin{pmatrix}1\\0\\0\end{pmatrix},T\begin{pmatrix}0\\1\\0\end{pmatrix}\right\}=\left\{\begin{pmatrix}1\\4\\7\end{pmatrix},\begin{pmatrix}2\\5\\8\end{pmatrix}\right\}$ form a basis of $T(V)$. Hence $T(V)=span\left\{\begin{pmatrix}1\\4\\7\end{pmatrix},\begin{pmatrix}2\\5\\8\end{pmatrix}\right\}$.

$\dim \boldsymbol{R}^3 = 3 = 1+2 = \dim\ker(T)+\dim T(\boldsymbol{R}^3)$.

Example 4.1.30 (Projection)

Let $T:\boldsymbol{R}^2\to\boldsymbol{R}^2$ be linear transformation defined by $T\begin{pmatrix}1\\m\end{pmatrix}=\begin{pmatrix}1\\m\end{pmatrix}$ and $T\begin{pmatrix}-m\\1\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix}$.




Method 1: Clearly, $\ker(T)=span\left\{\begin{pmatrix}-m\\1\end{pmatrix}\right\}$ and

$$T(\boldsymbol{R}^2)=\left\{T\left(\alpha\begin{pmatrix}1\\m\end{pmatrix}+\beta\begin{pmatrix}-m\\1\end{pmatrix}\right)\right\}=\left\{\alpha T\begin{pmatrix}1\\m\end{pmatrix}+\beta T\begin{pmatrix}-m\\1\end{pmatrix}\right\}=\left\{\alpha\begin{pmatrix}1\\m\end{pmatrix}\right\}=span\left\{\begin{pmatrix}1\\m\end{pmatrix}\right\}.$$

Method 2: _____

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$\dim \boldsymbol{R}^2 = 2 = 1+1 = \dim\ker(T)+\dim T(\boldsymbol{R}^2)$.

Example 4.1.31

Let $T: P_3 \to P_2$ be differentiation mapping defined by $T(p(x)) = p'(x)$ for all $p(x) \in P_3$, i.e., $T(a_0 + a_1x + a_2x^2) = a_1 + 2a_2x.$

Method 1: $T(a_0 + a_1x + a_2x^2) = 0$ if and only if $a_1 = a_2 = 0.$ $\ker(T) = \{a_0\}$, constant function. Clearly, $T(P_3) = \{a_1 + 2a_2x\} = span\{1, x\} = P_2.$

Method 2:_____

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$\dim P_3 = 3 = 1 + 2 = \dim \ker(T) + \dim T(P_3).$

Example 4.1.32

Let $T: P_3 \to \boldsymbol{R}^3$ be mapping defined by $T(p(x)) = (p(1), p(2), p(3))$ for all $p(x) \in P_3$, i.e., $T(a_0 + a_1x + a_2x^2) = (a_0 + a_1 + a_2, a_0 + 2a_1 + 2^2a_2, a_0 + 3a_1 + 3^2a_2).$

Consider $\ker(T) = \{p(x) = a_0 + a_1x + a_2x^2 \in P_3 \mid (p(1), p(2), p(3)) = (0, 0, 0)\}.$

Method 1: We need to solve

$$\begin{cases} a_0 + a_1 + a_2 = 0 \\ a_0 + 2a_1 + 2^2a_2 = 0 \\ a_0 + 3a_1 + 3^2a_2 = 0 \end{cases}$$

$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2^2 \\ 1 & 3 & 3^2 \end{pmatrix}$ is called Vandermonde matrix. Its determinant $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 2^2 \\ 1 & 3 & 3^2 \end{vmatrix} = (3-1)(3-2)(2-1) \neq 0.$

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Hence $a_0 = a_1 = a_2 = 0,$ i.e., $\ker(T) = \{0\},$ zero polynomial.

Method 2: $\ker(T) = \{p(x) = a_0 + a_1x + a_2x^2 \in P_3 \mid (p(1), p(2), p(3)) = (0,0,0)\}$.

That means the quadratic equation $p(x) = a_0 + a_1x + a_2x^2 = 0$ has 3 roots, 1, 2 and 3. Such polynomial $p(x)$ must be a zero polynomial. (By factor theorem, $p(x)$ must contain factor $(x-1)$ and $(x-2)$. Hence $p(x) = a_2(x-1)(x-2)$. Since $p(3) = 0$, $a_2(3-1)(3-2) = 0$. Then $a_2 = 0$, i.e., $p(x)$ is a zero polynomial and $\ker(T) = \{0\}$.)

Remark 4.1.33

Let $T : P_n \to \boldsymbol{R}^3$ be mapping defined by $T(p(x)) = (p(1), p(2), p(3))$ for all $p(x) \in P_n$, where $n \geq 4$. Then $\ker(T) = \{(x-1)(x-2)(x-3)q(x) \mid q(x) \in P_{n-3}\}$. Hence $\dim\ker(T) = n-3$.

Example 4.1.34

Find $a_0 + a_1x + a_2x^2$ so that the graph of $y = a_0 + a_1x + a_2x^2$ passes through $(1,-1),(2,-3),(3,-3)$.

Solution

Method 1: We need to solve

$$\begin{cases} a_0 + a_1 + a_2 = -1 \\ a_0 + 2a_1 + 2^2a_2 = -3 \\ a_0 + 3a_1 + 3^2a_2 = -3 \end{cases} \quad \text{or} \quad \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2^2 \\ 1 & 3 & 3^2 \end{pmatrix}\begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} -1 \\ -3 \\ -3 \end{pmatrix}$$

Finding $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2^2 \\ 1 & 3 & 3^2 \end{pmatrix}^{-1}$ is not easy or you may use Cramer's rule. (Do it yourself.)

Method 2: The question is:

Find $p(x) = a_0 + a_1x + a_2x^2 \in P_3$ so that $T(p(x)) = (p(1), p(2), p(3)) = (-1,-3,-3)\}$.

Since $\ker(T) = \{0\}$ and $\dim T(P_3) = \dim(P_3) = 3$, by Theorem 4.1.28, $T(P_3) = \boldsymbol{R}^3$. Let $p_1(x), p_2(x), p_3(x) \in P_3$ such that $T(p_1(x)) = (1,0,0)$, $T(p_2(x)) = (0,1,0)$, $T(p_3(x)) = (0,0,1)$. Then $p_1(x) = c_1(x-2)(x-3)$. Since $p_1(1) = 1, c_1(1-2)(1-3) = 1$. Then $c_1 = \dfrac{1}{(1-2)(1-3)}$ and $p_1(x) = \dfrac{(x-2)(x-3)}{(1-2)(1-3)}$. Similarly, $p_2(x) = \dfrac{(x-1)(x-3)}{(2-1)(2-3)}$ and $p_3(x) = \dfrac{(x-1)(x-2)}{(3-1)(3-2)}$.

$$\begin{aligned}(-1,-3,-3) &= -1(1,0,0)+(-3)(0,1,0)+(-3)(0,0,1)\\ &= -1T(p_1(x))+(-3)T(p_2(x))+(-3)T(p_3(x))\\ &= T\left((-1)\frac{(x-2)(x-3)}{(1-2)(1-3)}+(-3)\frac{(x-1)(x-3)}{(2-1)(2-3)}+(-3)\frac{(x-1)(x-2)}{(3-1)(3-2)}\right).\end{aligned}$$

The required polynomial is

$$(-1)\frac{(x-2)(x-3)}{(1-2)(1-3)}+(-3)\frac{(x-1)(x-3)}{(2-1)(2-3)}+(-3)\frac{(x-1)(x-2)}{(3-1)(3-2)} = x^2-5x+3.$$

Remark 4.1.35

In general, the degree 2 polynomial passes through $(x_1,y_1),(x_2,y_2),(x_3,y_3)$ is

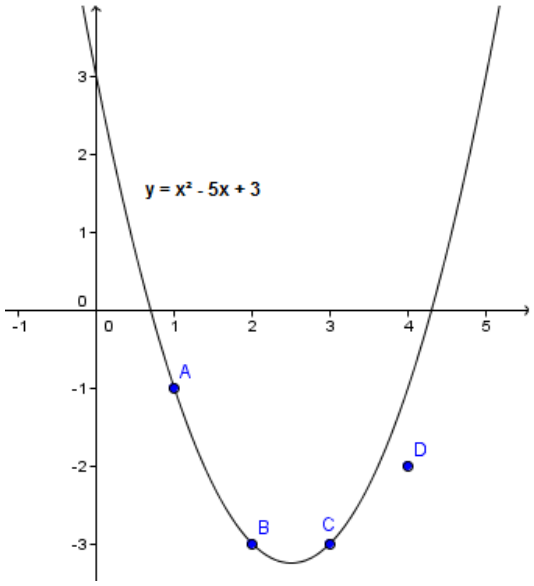
$$y_1\frac{(x-x_2)(x-x_3)}{(x_1-x_2)(x_1-x_3)}+y_2\frac{(x-x_1)(x-x_3)}{(x_2-x_1)(x_2-x_3)}+y_3\frac{(x-x_1)(x-x_2)}{(x_3-x_1)(x_3-x_2)}.$$

Example 4.1.36

Is it possible to find $a_0+a_1x+a_2x^2$ so that the graph of $y=a_0+a_1x+a_2x^2$ passes through $(1,-1),(2,-3),(3,-3),(4,-2)$?

Solution

No. $x^2-5x+3$ is the only polynomial so that the graph of $y=a_0+a_1x+a_2x^2$ passes through $(1,-1),(2,-3),(3,-3)$. When $x=4$, $y=4^2-5(4)+3=-1\neq -2$. It is impossible.



## Section 4.2 Matrix representations of Linear Transformations

Let $\beta = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \cdots, \boldsymbol{v}_n\}$ and $\gamma = \{\boldsymbol{w}_1, \boldsymbol{w}_2, \cdots, \boldsymbol{w}_m\}$ be ordered bases of finite dimensional vector spaces $V$ and $W$ respectively. Let $T : V \to W$ be linear. Then

$$\begin{cases} T(\boldsymbol{v}_1) = a_{11}\boldsymbol{w}_1 + a_{21}\boldsymbol{w}_2 + \cdots + a_{m1}\boldsymbol{w}_m \\ T(\boldsymbol{v}_2) = a_{12}\boldsymbol{w}_1 + a_{22}\boldsymbol{w}_2 + \cdots + a_{m2}\boldsymbol{w}_m \\ \vdots \\ T(\boldsymbol{v}_n) = a_{1n}\boldsymbol{w}_1 + a_{2n}\boldsymbol{w}_2 + \cdots + a_{mn}\boldsymbol{w}_m \end{cases}$$

Definition 4.2.1

We call the $m \times n$ matrix

$$A = \left(\begin{array}{c|c|c|c} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{array}\right)$$

the **matrix representation of $T$ in the ordered bases $\beta$ and $\gamma$** and write $A = [T]_\beta^\gamma$. If $V = W$ and $\beta = \gamma$, then we write $A = [T]_\beta$.

The following theorem is special case of Definition 4.2.1 by taking $V = \boldsymbol{R}^n, W = \boldsymbol{R}^m$, and $\beta$ and $\gamma$ the standard bases of $\boldsymbol{R}^n$ and $\boldsymbol{R}^m$ respectively.

Theorem 4.2.2

If $T : \boldsymbol{R}^n \to \boldsymbol{R}^m$ is linear, then there is an $m \times n$ matrix $A$ such that

$$T\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \underbrace{\left(\begin{array}{c|c|c|c} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{array}\right)}_{A} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

Indeed, $A = [T]_{\beta}^{\gamma}$ and $\ker(T) = N(A)$.

Example 4.2.3

Let $T : \boldsymbol{R}^2 \to \boldsymbol{R}^2$ be linear transformation defined by $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x-2y \\ -2x+y \end{pmatrix}$ for all $\begin{pmatrix} x \\ y \end{pmatrix} \in \boldsymbol{R}^2$. Let $\beta = \left\{\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 4 \end{pmatrix}\right\}, \beta' = \left\{\begin{pmatrix} 3 \\ 4 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix}\right\}, \gamma = \left\{\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right\}$ be ordered bases of $\boldsymbol{R}^2$. We have

$$T\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -3 \\ 0 \end{pmatrix} = -3\begin{pmatrix} 1 \\ 0 \end{pmatrix} + 0\begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \text{and} \quad T\begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} -5 \\ -2 \end{pmatrix} = -5\begin{pmatrix} 1 \\ 0 \end{pmatrix} - 2\begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Then $[T]_{\beta}^{\gamma} = \left(\begin{array}{c|c} -3 & -5 \\ 0 & -2 \end{array}\right)$ and $[T]_{\beta}^{\gamma}, = \left(\begin{array}{c|c} -5 & -3 \\ -2 & 0 \end{array}\right)$.

Switching order of basis changes the matrix representation of $T$.

Example 4.2.4

Let $T : \boldsymbol{R}^2 \to \boldsymbol{R}^2$ be linear transformation defined by $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+2y \\ 3x+4y \end{pmatrix}$ for all $\begin{pmatrix} x \\ y \end{pmatrix} \in \boldsymbol{R}^2$. Let $\beta = \left\{\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right\}$ be the standard basis of $\boldsymbol{R}^2$. We have

$$T\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix} = 1\begin{pmatrix} 1 \\ 0 \end{pmatrix} + 3\begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \text{and} \quad T\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} = 2\begin{pmatrix} 1 \\ 0 \end{pmatrix} + 4\begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Then $A = [T]_{\beta} =$ _____ and $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+2y \\ 3x+4y \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$.

Example 4.2.5

Let $T : \boldsymbol{R}^2 \to \boldsymbol{R}^2$ be linear transformation defined by $T\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $T\begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$. Find $T\begin{pmatrix} x \\ y \end{pmatrix}$?

Solution

Since $\det\begin{pmatrix}2 & 3\\ 1 & 2\end{pmatrix}=1\neq 0$, $\left\{\begin{pmatrix}2\\1\end{pmatrix},\begin{pmatrix}3\\2\end{pmatrix}\right\}$ is linearly independent. Hence $\left\{\begin{pmatrix}2\\1\end{pmatrix},\begin{pmatrix}3\\2\end{pmatrix}\right\}$ is a basis of $\boldsymbol{R}^2$. Consider

$$\begin{pmatrix}1\\0\end{pmatrix}=a\begin{pmatrix}2\\1\end{pmatrix}+b\begin{pmatrix}3\\2\end{pmatrix} \quad \text{and} \quad \begin{pmatrix}0\\1\end{pmatrix}=c\begin{pmatrix}2\\1\end{pmatrix}+d\begin{pmatrix}3\\2\end{pmatrix}.$$

That is

$$\begin{pmatrix}1\\0\end{pmatrix}=\begin{pmatrix}2 & 3\\ 1 & 2\end{pmatrix}\begin{pmatrix}a\\b\end{pmatrix} \quad \text{and} \quad \begin{pmatrix}0\\1\end{pmatrix}=\begin{pmatrix}2 & 3\\ 1 & 2\end{pmatrix}\begin{pmatrix}c\\d\end{pmatrix}.$$

By Cramer's rule,

$$a=\frac{\begin{vmatrix}1 & 3\\ 0 & 2\end{vmatrix}}{\begin{vmatrix}2 & 3\\ 1 & 2\end{vmatrix}}=2 \quad b=\frac{\begin{vmatrix}2 & 1\\ 1 & 0\end{vmatrix}}{\begin{vmatrix}2 & 3\\ 1 & 2\end{vmatrix}}=-1 \quad c=\frac{\begin{vmatrix}0 & 3\\ 1 & 2\end{vmatrix}}{\begin{vmatrix}2 & 3\\ 1 & 2\end{vmatrix}}=-3 \quad d=\frac{\begin{vmatrix}2 & 0\\ 1 & 1\end{vmatrix}}{\begin{vmatrix}2 & 3\\ 1 & 2\end{vmatrix}}=2.$$

Then

$$\begin{pmatrix}1\\0\end{pmatrix}=2\begin{pmatrix}2\\1\end{pmatrix}-\begin{pmatrix}3\\2\end{pmatrix} \quad \text{and} \quad \begin{pmatrix}0\\1\end{pmatrix}=-3\begin{pmatrix}2\\1\end{pmatrix}+2\begin{pmatrix}3\\2\end{pmatrix}.$$

$$\begin{aligned}T\begin{pmatrix}1\\0\end{pmatrix}&=T\left(2\begin{pmatrix}2\\1\end{pmatrix}-\begin{pmatrix}3\\2\end{pmatrix}\right)\\ &=2T\begin{pmatrix}2\\1\end{pmatrix}-T\begin{pmatrix}3\\2\end{pmatrix}\\ &=2\begin{pmatrix}1\\2\end{pmatrix}-\begin{pmatrix}3\\4\end{pmatrix}\\ &=\begin{pmatrix}-1\\0\end{pmatrix}\\ &=-1\begin{pmatrix}1\\0\end{pmatrix}+0\begin{pmatrix}0\\1\end{pmatrix}\end{aligned}$$

Let $\beta = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$ be the standard basis of $\boldsymbol{R}^2$. Then $A = [T]_\beta = \begin{pmatrix} -1 & 3 \\ 0 & 2 \end{pmatrix}$. Hence $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 & 3 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$.

Example 4.2.5

Define $T : span\{e^x \sin x, e^x \cos x\} \to span\{e^x \sin x, e^x \cos x\}$ by $T(f(x)) = f(x+\theta)$. Let $\beta = \{e^x \sin x, e^x \cos x\}$ be the basis.

We have $T(e^x \sin x) = e^{x+\theta} \sin(x+\theta) =$ _____

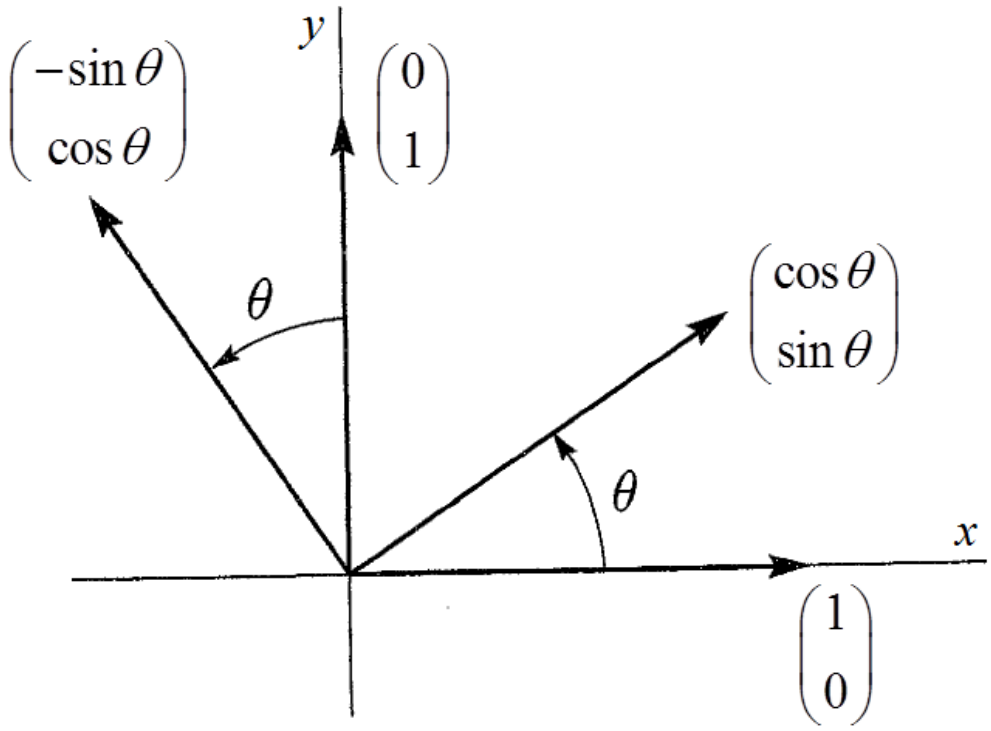
and $T(e^x \cos x) = e^{x+\theta} \cos(x+\theta) =$ _____.

Then $[T]_\beta = \begin{pmatrix} e^\theta \cos\theta & -e^\theta \sin\theta \\ e^\theta \sin\theta & e^\theta \cos\theta \end{pmatrix} = e^\theta \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.

Example 4.2.6 (Rotation)

Let $T : \boldsymbol{R}^2 \to \boldsymbol{R}^2$ be rotation mapping by counterclockwise rotating vector $\begin{pmatrix} x \\ y \end{pmatrix}$ by angle $\theta$. Let $\beta = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$ be the standard basis of $\boldsymbol{R}^2$. We have

$$T\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix} \quad \text{and} \quad T\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}.$$




Then $A=[T]_\beta=\left(\begin{array}{c|c}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{array}\right)$ and $T\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}$.

We call $\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}$ the rotation matrix.

Example 4.2.7

Let $T:\boldsymbol{C}\to\boldsymbol{C}$ be mapping defined by $T(z)=(a+bi)z$ for all $z\in\boldsymbol{C}$, i.e., $T(x+yi)=(a+bi)(x+yi)=(ax-by)+(ay+bx)i$. Let $\beta=\{1,i\}$ be the standard basis of $\boldsymbol{C}$. We have

$$T(1)=a+bi \quad \text{and} \quad T(i)=(a+bi)i=-b+ai.$$

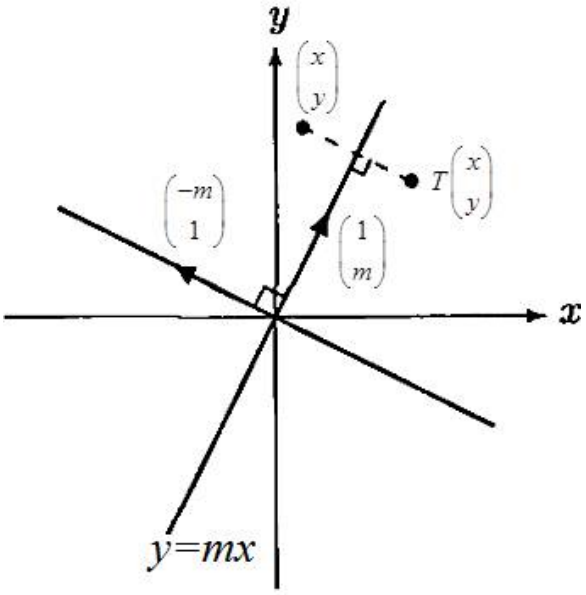
Then $A=[T]_\beta=\left(\begin{array}{c|c}a & -b\\ b & a\end{array}\right)$.

In particular, if $a+bi=\cos\theta+i\sin\theta$, then $A=[T]_\beta=\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}$ and $T$ is a linear transformation by counterclockwise rotating complex number $x+yi$ by angle $\theta$.

Example 4.2.8 (Reflection)

Let $T : \boldsymbol{R}^2 \to \boldsymbol{R}^2$ be linear transformation defined by $T\begin{pmatrix}1\\m\end{pmatrix} = \begin{pmatrix}1\\m\end{pmatrix}$ and $T\begin{pmatrix}-m\\1\end{pmatrix} = -\begin{pmatrix}-m\\1\end{pmatrix} = \begin{pmatrix}m\\-1\end{pmatrix}$.

We call $T$ **reflection along** $y = mx$.




Method 1: Since $\det\begin{pmatrix}1 & -m\\ m & 1\end{pmatrix} = 1+m^2 \neq 0$, $\left\{\begin{pmatrix}1\\m\end{pmatrix}, \begin{pmatrix}-m\\1\end{pmatrix}\right\}$ is linearly independent. Hence $\left\{\begin{pmatrix}1\\m\end{pmatrix}, \begin{pmatrix}-m\\1\end{pmatrix}\right\}$ is a basis of $\boldsymbol{R}^2$. Consider

$$\begin{pmatrix}1\\0\end{pmatrix} = a\begin{pmatrix}1\\m\end{pmatrix} + b\begin{pmatrix}-m\\1\end{pmatrix} \quad \text{and} \quad \begin{pmatrix}0\\1\end{pmatrix} = c\begin{pmatrix}1\\m\end{pmatrix} + d\begin{pmatrix}-m\\1\end{pmatrix}.$$

That is

$$\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}1 & -m\\ m & 1\end{pmatrix}\begin{pmatrix}a\\b\end{pmatrix} \quad \text{and} \quad \begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}1 & -m\\ m & 1\end{pmatrix}\begin{pmatrix}c\\d\end{pmatrix}.$$

By Cramer's rule,

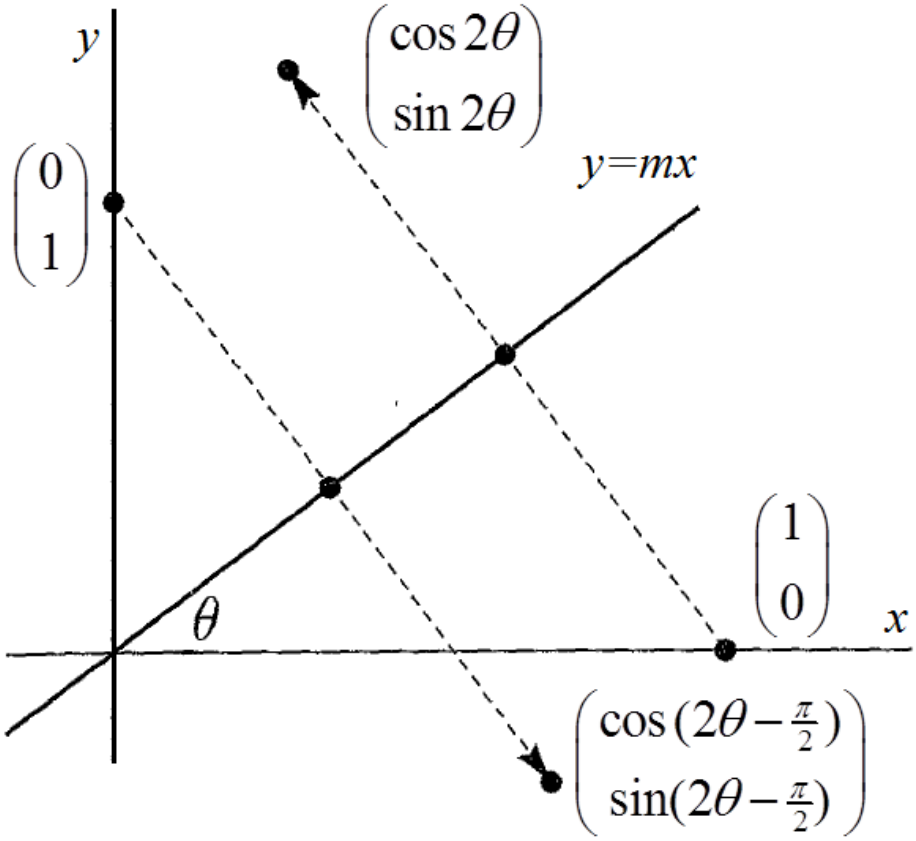
$$a = \frac{\begin{vmatrix}1 & -m\\ 0 & 1\end{vmatrix}}{\begin{vmatrix}1 & -m\\ m & 1\end{vmatrix}} = \frac{1}{1+m^2} \quad b = \frac{\begin{vmatrix}1 & 1\\ m & 0\end{vmatrix}}{\begin{vmatrix}1 & -m\\ m & 1\end{vmatrix}} = \frac{-m}{1+m^2} \quad c = \frac{\begin{vmatrix}0 & -m\\ 1 & 1\end{vmatrix}}{\begin{vmatrix}1 & -m\\ m & 1\end{vmatrix}} = \frac{m}{1+m^2} \quad d = \frac{\begin{vmatrix}1 & 0\\ m & 1\end{vmatrix}}{\begin{vmatrix}1 & -m\\ m & 1\end{vmatrix}} = \frac{1}{1+m^2}.$$

Then

$$\begin{pmatrix}1\\0\end{pmatrix}=\frac{1}{1+m^2}\begin{pmatrix}1\\m\end{pmatrix}-\frac{m}{1+m^2}\begin{pmatrix}-m\\1\end{pmatrix} \quad \text{and} \quad \begin{pmatrix}0\\1\end{pmatrix}=\frac{m}{1+m^2}\begin{pmatrix}1\\m\end{pmatrix}+\frac{1}{1+m^2}\begin{pmatrix}-m\\1\end{pmatrix}. \tag{2}$$

$$\begin{aligned}T\begin{pmatrix}1\\0\end{pmatrix}&=T\left(\frac{1}{1+m^2}\begin{pmatrix}1\\m\end{pmatrix}-\frac{m}{1+m^2}\begin{pmatrix}-m\\1\end{pmatrix}\right)\\&=\frac{1}{1+m^2}T\begin{pmatrix}1\\m\end{pmatrix}-\frac{m}{1+m^2}T\begin{pmatrix}-m\\1\end{pmatrix}\\&=\frac{1}{1+m^2}\begin{pmatrix}1\\m\end{pmatrix}-\frac{m}{1+m^2}\begin{pmatrix}m\\-1\end{pmatrix}\\&=\begin{pmatrix}\frac{1-m^2}{1+m^2}\\\frac{2m}{1+m^2}\end{pmatrix}\\&=\frac{1-m^2}{1+m^2}\begin{pmatrix}1\\0\end{pmatrix}+\frac{2m}{1+m^2}\begin{pmatrix}0\\1\end{pmatrix}\end{aligned}$$

$$\begin{aligned}T\begin{pmatrix}0\\1\end{pmatrix}&=T\left(\frac{m}{1+m^2}\begin{pmatrix}1\\m\end{pmatrix}+\frac{1}{1+m^2}\begin{pmatrix}-m\\1\end{pmatrix}\right)\\&=\frac{m}{1+m^2}T\begin{pmatrix}1\\m\end{pmatrix}+\frac{1}{1+m^2}T\begin{pmatrix}-m\\1\end{pmatrix}\\&=\frac{m}{1+m^2}\begin{pmatrix}1\\m\end{pmatrix}+\frac{1}{1+m^2}\begin{pmatrix}m\\-1\end{pmatrix}\\&=\begin{pmatrix}\frac{2m}{1+m^2}\\-\frac{1-m^2}{1+m^2}\end{pmatrix}\\&=\frac{2m}{1+m^2}\begin{pmatrix}1\\0\end{pmatrix}-\frac{1-m^2}{1+m^2}\begin{pmatrix}0\\1\end{pmatrix}\end{aligned}$$



Method 2: Since $m = \tan\theta,$ we have $\sin\theta = \frac{m}{\sqrt{1+m^2}}$ and $\cos\theta = \frac{1}{\sqrt{1+m^2}}$.

$$T\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}\cos 2\theta\\ \sin 2\theta\end{pmatrix} = \begin{pmatrix}\cos^2\theta - \sin^2\theta\\ 2\sin\theta\cos\theta\end{pmatrix} = \begin{pmatrix}\frac{1-m^2}{1+m^2}\\ \frac{2m}{1+m^2}\end{pmatrix}$$

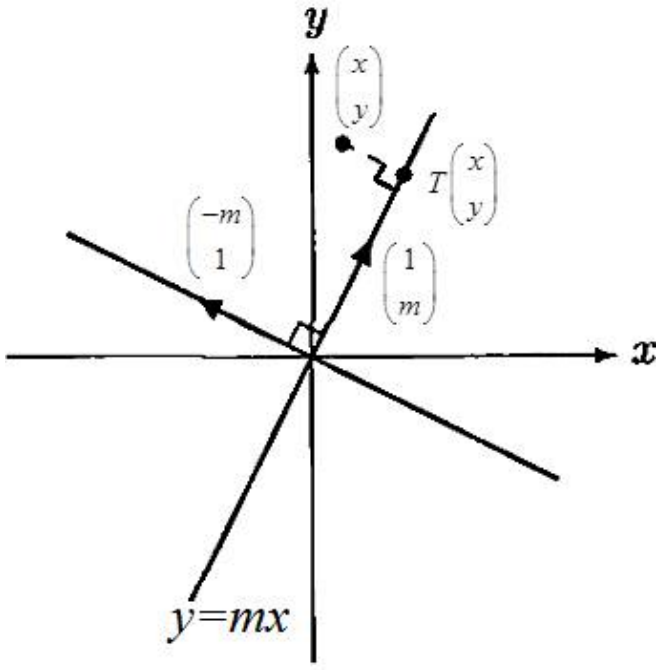
$$T\begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}\cos(2\theta - \frac{\pi}{2})\\ \sin(2\theta - \frac{\pi}{2})\end{pmatrix} = \begin{pmatrix}\sin 2\theta\\ -\cos 2\theta\end{pmatrix} = \begin{pmatrix}\frac{2m}{1+m^2}\\ -\frac{1-m^2}{1+m^2}\end{pmatrix}$$

Let $\beta = \left\{\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix}\right\}$ be the standard basis of $\boldsymbol{R}^2$. Then

$A = [T]_\beta = \begin{pmatrix}\cos 2\theta & \sin 2\theta\\ \sin 2\theta & -\cos 2\theta\end{pmatrix} = \begin{pmatrix}\frac{1-m^2}{1+m^2} & \frac{2m}{1+m^2}\\ \frac{2m}{1+m^2} & -\frac{1-m^2}{1+m^2}\end{pmatrix}$. Hence $T\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}\frac{1-m^2}{1+m^2} & \frac{2m}{1+m^2}\\ \frac{2m}{1+m^2} & -\frac{1-m^2}{1+m^2}\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}$.

Example 4.2.9 (Projection)

Let $T : \boldsymbol{R}^2 \to \boldsymbol{R}^2$ be linear transformation defined by $T\begin{pmatrix}1\\m\end{pmatrix} = \begin{pmatrix}1\\m\end{pmatrix}$ and $T\begin{pmatrix}-m\\1\end{pmatrix} = \begin{pmatrix}0\\0\end{pmatrix}$. We call $T$ **projection along** $y = mx$.



Method 1: Let $\beta = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$ be the standard basis of $\boldsymbol{R}^2$. From (2),

$$T\begin{pmatrix} 1 \\ 0 \end{pmatrix} = T\left( \frac{1}{1+m^2}\begin{pmatrix} 1 \\ m \end{pmatrix} - \frac{m}{1+m^2}\begin{pmatrix} -m \\ 1 \end{pmatrix} \right)$$

$$= \frac{1}{1+m^2} T\begin{pmatrix} 1 \\ m \end{pmatrix} - \frac{m}{1+m^2} T\begin{pmatrix} -m \\ 1 \end{pmatrix}$$

$$= \frac{1}{1+m^2}\begin{pmatrix} 1 \\ m \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{1+m^2} \\ \frac{m}{1+m^2} \end{pmatrix}$$

$$= \frac{1}{1+m^2}\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{m}{1+m^2}\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

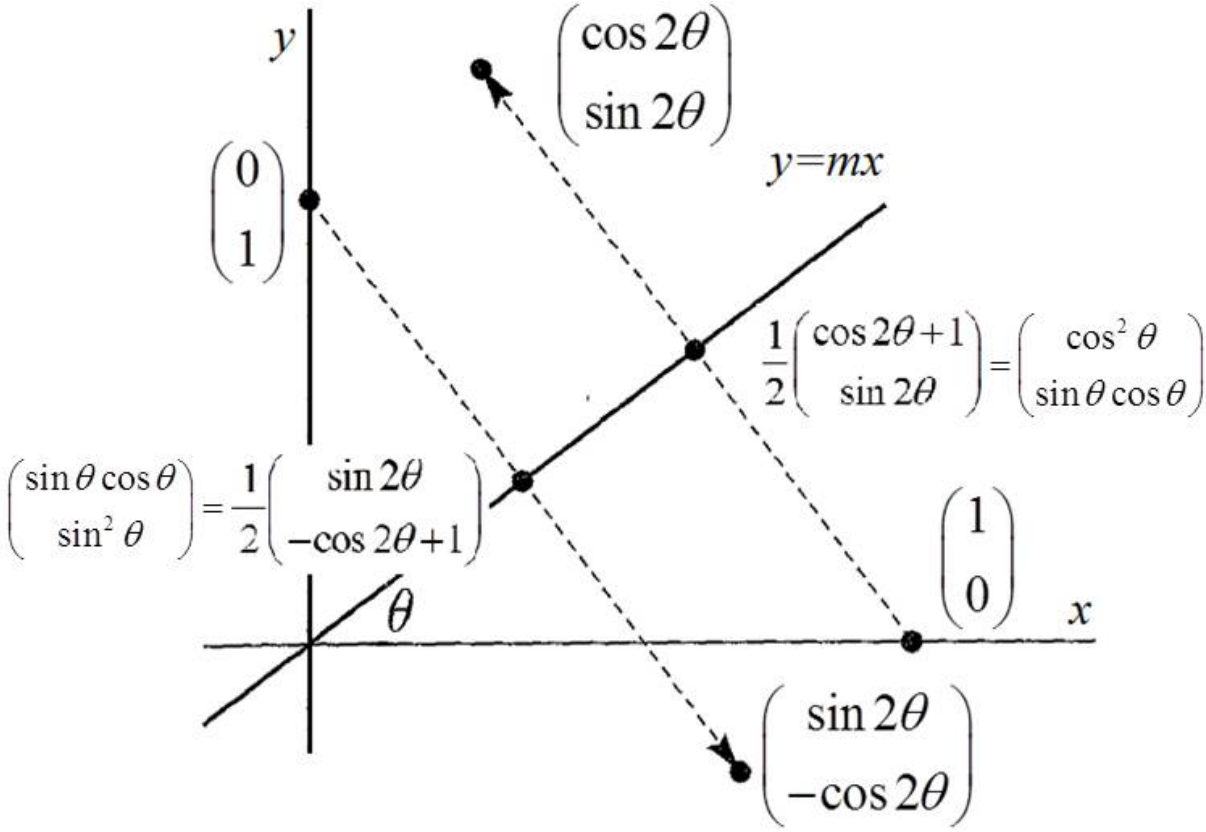
$$T\begin{pmatrix} 0 \\ 1 \end{pmatrix} = T\left( \frac{m}{1+m^2}\begin{pmatrix} 1 \\ m \end{pmatrix} + \frac{1}{1+m^2}\begin{pmatrix} -m \\ 1 \end{pmatrix} \right)$$

$$= \frac{m}{1+m^2} T\begin{pmatrix} 1 \\ m \end{pmatrix} + \frac{1}{1+m^2} T\begin{pmatrix} -m \\ 1 \end{pmatrix}$$

$$= \frac{m}{1+m^2}\begin{pmatrix} 1 \\ m \end{pmatrix}$$

$$= \begin{pmatrix} \frac{m}{1+m^2} \\ \frac{m^2}{1+m^2} \end{pmatrix}$$

$$= \frac{m}{1+m^2}\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{m^2}{1+m^2}\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$




Method 2: Since $m = \tan\theta,$ we have $\sin\theta = \frac{m}{\sqrt{1+m^2}}$ and $\cos\theta = \frac{1}{\sqrt{1+m^2}}.$

$$T\begin{pmatrix}1\\0\end{pmatrix}=\begin{pmatrix}\cos^2\theta\\ \sin\theta\cos\theta\end{pmatrix}=\begin{pmatrix}\frac{1}{1+m^2}\\ \frac{m}{1+m^2}\end{pmatrix} \quad \text{and} \quad T\begin{pmatrix}0\\1\end{pmatrix}=\begin{pmatrix}\sin\theta\cos\theta\\ \sin^2\theta\end{pmatrix}=\begin{pmatrix}\frac{m}{1+m^2}\\ \frac{m^2}{1+m^2}\end{pmatrix}$$

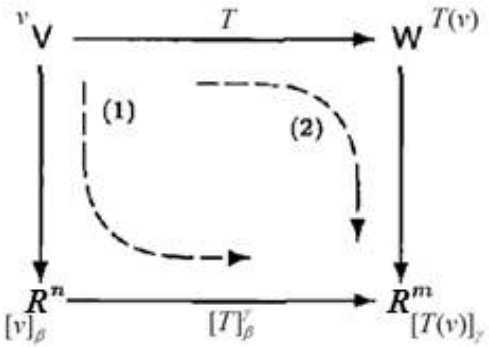
Then $A=[T]_\beta=\begin{pmatrix}\cos^2\theta & \sin\theta\cos\theta\\ \sin\theta\cos\theta & \sin^2\theta\end{pmatrix}=\begin{pmatrix}\frac{1}{1+m^2} & \frac{m}{1+m^2}\\ \frac{m}{1+m^2} & \frac{m^2}{1+m^2}\end{pmatrix}$. Hence

$$T\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}\frac{1}{1+m^2} & \frac{m}{1+m^2}\\ \frac{m}{1+m^2} & \frac{m^2}{1+m^2}\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}.$$

Theorem 4.2.10

Let $V$ and $W$ be finite dimensional vector spaces having ordered bases $\beta$ and $\gamma$ respectively, and let $T:V\to W$ be linear. Then, for each $\boldsymbol{v}\in V$, we have

$$[T(\boldsymbol{v})]_\gamma=[T]_\beta^\gamma[\boldsymbol{v}]_\beta.$$




Example 4.2.11

Let $\beta=\{\boldsymbol{x}_1,\boldsymbol{x}_2,\boldsymbol{x}_3\}$ be a basis of $\boldsymbol{R}^3$ where $\boldsymbol{x}_1=\begin{pmatrix}1\\1\\1\end{pmatrix},\boldsymbol{x}_2=\begin{pmatrix}1\\1\\0\end{pmatrix},\boldsymbol{x}_3=\begin{pmatrix}1\\0\\0\end{pmatrix}$. Define $T:\boldsymbol{R}^3\to\boldsymbol{R}^3$ the linear transformation by $T(c_1\boldsymbol{x}_1+c_2\boldsymbol{x}_2+c_3\boldsymbol{x}_3)=(c_1+c_2+c_3)\boldsymbol{x}_1+(2c_1+c_3)\boldsymbol{x}_2-(2c_2+c_3)\boldsymbol{x}_3$.

We have

$$T(\boldsymbol{x}_1)=\boldsymbol{x}_1+2\boldsymbol{x}_2+0\boldsymbol{x}_3,\quad T(\boldsymbol{x}_2)=\boldsymbol{x}_1+0\boldsymbol{x}_2-2\boldsymbol{x}_3,\quad T(\boldsymbol{x}_3)=\boldsymbol{x}_1+\boldsymbol{x}_2-\boldsymbol{x}_3$$

Then $[T]_\beta = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 0 & 1 \\ 0 & -2 & -1 \end{pmatrix}$.

$$\begin{pmatrix} 7 \\ 5 \\ 2 \end{pmatrix} = 2\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + 3\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + 2\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad \Rightarrow \quad \left[\begin{pmatrix} 7 \\ 5 \\ 2 \end{pmatrix}\right]_\beta = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix}$$

$$\left[T\begin{pmatrix} 7 \\ 5 \\ 2 \end{pmatrix}\right]_\beta = [T]_\beta \left[\begin{pmatrix} 7 \\ 5 \\ 2 \end{pmatrix}\right]_\beta = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 0 & 1 \\ 0 & -2 & -1 \end{pmatrix}\begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 7 \\ 6 \\ -8 \end{pmatrix}$$

$T\begin{pmatrix} 7 \\ 5 \\ 2 \end{pmatrix} \neq \begin{pmatrix} 7 \\ 6 \\ -8 \end{pmatrix}$ but $T\begin{pmatrix} 7 \\ 5 \\ 2 \end{pmatrix} = 7\boldsymbol{x}_1 + 6\boldsymbol{x}_2 - 8\boldsymbol{x}_3$.

On the other hand, you may calculate $T\begin{pmatrix} 7 \\ 5 \\ 2 \end{pmatrix}$ directly.

$$\begin{aligned} T\begin{pmatrix} 7 \\ 5 \\ 2 \end{pmatrix} &= T\left(2\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + 3\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + 2\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}\right) \\ &= 2T(\boldsymbol{x}_1) + 3T(\boldsymbol{x}_2) + 2T(\boldsymbol{x}_3) \\ &= 2(\boldsymbol{x}_1 + 2\boldsymbol{x}_2) + 3(\boldsymbol{x}_1 - 2\boldsymbol{x}_3) + 2(\boldsymbol{x}_1 + \boldsymbol{x}_2 - \boldsymbol{x}_3) \\ &= 7\boldsymbol{x}_1 + 6\boldsymbol{x}_2 - 8\boldsymbol{x}_3 \end{aligned}$$

Theorem 4.2.12

Let $V$, $W$ and $Z$ be vector spaces. Let $T: V \to W$ and $U: W \to Z$ be linear transformations. Then $\ker(T) \subseteq \ker(UT)$. Equality holds if $U$ is injective (by Remark 4.1.22, i.e., $\ker(U) = \{\mathbf{0}_W\}$).

Theorem 4.2.13

Let $V$, $W$ and $Z$ be finite dimensional vector spaces with ordered bases $\alpha, \beta$ and $\gamma$ respectively. Let $T: V \to W$ and $U: W \to Z$ be linear transformations. Then,

$$[UT]_\alpha^\gamma = [U]_\beta^\gamma [T]_\alpha^\beta.$$

Corollary 4.2.14

Let $V$ be finite dimensional vector spaces with an ordered basis $\beta$, and let $T,U:V\to V$ be linear. Then $[UT]_\beta=[U]_\beta[T]_\beta$ and $\left[T^n\right]_\beta=\left([T]_\beta\right)^n$ for all positive integer $n$.

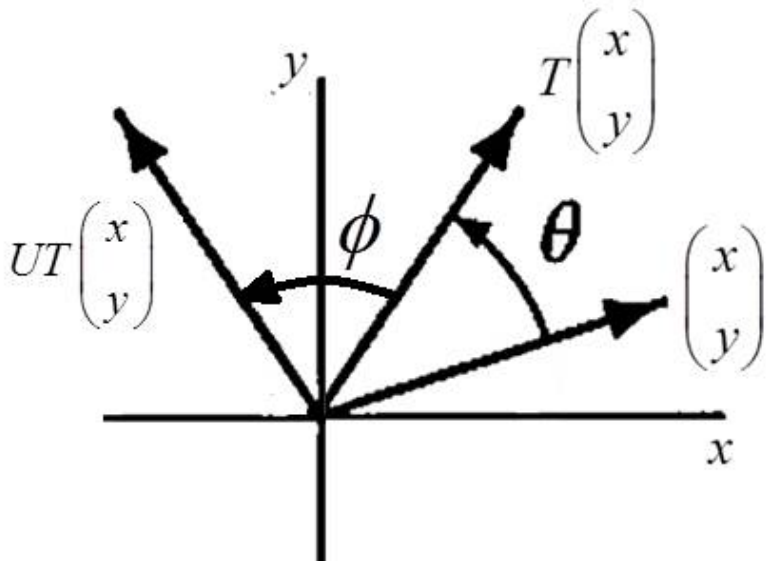
Example 4.2.15 (Rotation)

Let $T,U:\boldsymbol{R}^2\to\boldsymbol{R}^2$ be rotation mapping defined by counterclockwise rotating vector $\begin{pmatrix}x\\y\end{pmatrix}$ by angle $\theta$ and $\phi$ respectively. Let $\beta=\left\{\begin{pmatrix}1\\0\end{pmatrix},\begin{pmatrix}0\\1\end{pmatrix}\right\}$ be the standard basis of $\boldsymbol{R}^2$. We have

$$T\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}\quad\text{and}\quad U\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}\cos\phi & -\sin\phi\\ \sin\phi & \cos\phi\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}.$$

Clearly, $UT:\boldsymbol{R}^2\to\boldsymbol{R}^2$ is rotation mapping defined by counterclockwise rotating vector $\begin{pmatrix}x\\y\end{pmatrix}$ by angle $\theta+\phi$.

$$\boldsymbol{R}^2\xrightarrow[\text{rotate }\theta]{T}\boldsymbol{R}^2\xrightarrow[\text{rotate }\phi]{U}\boldsymbol{R}^2$$




Then $UT\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}\cos(\theta+\phi) & -\sin(\theta+\phi)\\ \sin(\theta+\phi) & \cos(\theta+\phi)\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}$.

$$
\begin{aligned}
[UT]_\beta &= \begin{pmatrix} \cos(\theta+\phi) & -\sin(\theta+\phi) \\ \sin(\theta+\phi) & \cos(\theta+\phi) \end{pmatrix} \\
&= \begin{pmatrix} \cos\theta\cos\phi - \sin\theta\sin\phi & -\sin\theta\cos\phi - \cos\theta\sin\phi \\ \sin\theta\cos\phi + \cos\theta\sin\phi & \cos\theta\cos\phi - \sin\theta\sin\phi \end{pmatrix} \\
&= \begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix} \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \\
&= [U]_\beta [T]_\beta .
\end{aligned}
$$

Example 4.2.16

Let $T: P_4 \to P_3$ be differentiation mapping at $t$ defined by $T(p(x)) = p'(t) = \left.\frac{dp}{dx}\right|_{x=t}$ for all $p(x) \in P_4$, and $U: P_3 \to P_4$ be integration mapping defined by $U(q(t)) = \int_1^x q(t)dt$ for all $q(t) \in P_3$. Let $\beta = \{1, x, x^2, x^3\}$ and $\gamma = \{1, t, t^2\}$ be the standard bases of $P_4$ and $P_3$ respectively. We have

$$T(1) = 0, \quad T(x) = 1, \quad T(x^2) = 2t, \quad T(x^3) = 3t^2$$

$$U(1) = \int_1^x 1dt = -1 + x, \quad U(t) = \int_1^x tdt = \left[\frac{t^2}{2}\right]_1^x = -\frac{1}{2} + \frac{1}{2}x^2, \quad U(t^2) = \int_1^x t^2 dt = \left[\frac{t^3}{3}\right]_1^x = -\frac{1}{3} + \frac{1}{3}x^3$$

Then $[T]_\beta^\gamma =$ _____ and $[U]_\gamma^\beta =$ _____.

Consider $UT: P_4 \to P_4$.

$$P_4 \xrightarrow{T} P_3 \xrightarrow{U} P_4$$

$UT(1) = U(0) =$ _____,

$UT(x) = U(1) =$ _____,

$UT(x^2) = U(2t) = 2U(t) =$ _____,

$UT(x^3) = U(3t^2) = 3U(t^2) =$ _____

On the other hand, you may get the answers by Fundamental theorem of Calculus.

$$UT(1)=\int_1^x \frac{d}{dt}1dt=0,\quad UT(x)=\int_1^x \frac{d}{dt}tdt=\int_1^x 1dt=-1+x,\quad UT(x^2)=\int_1^x \frac{d}{dt}t^2dt=\int_1^x 1dt^2=-1+x^2$$

$$UT(x^2)=\int_1^x \frac{d}{dt}t^3dt=\int_1^x 1dt^3=-1+x^3$$

$[UT]_\beta =$

---

Consider $TU: P_3 \to P_3$.

$$P_3 \xrightarrow{U} P_4 \xrightarrow{T} P_3$$

$$TU(1)=T(-1+x)=1,\quad TU(t)=T\left(-\frac{1}{2}+\frac{1}{2}x^2\right)=t,\quad TU(t^2)=T\left(-\frac{1}{3}+\frac{1}{3}x^3\right)=t^2$$

On the other hand, you may get the answers by Fundamental theorem of Calculus.

$$TU(1)=\frac{d}{dx}\int_1^x 1dt=1,\quad TU(t)=\frac{d}{dx}\int_1^x tdt=x\,|_t=t,\quad TU(t^2)=\frac{d}{dx}\int_1^x t^2dt=x^2\,|_t=t^2$$

Then $[TU]_\gamma=\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}$.

It is easy to see that $[T]_\beta^\gamma[U]_\gamma^\beta=\begin{pmatrix}0&1&0&0\\0&0&2&0\\0&0&0&3\end{pmatrix}\begin{pmatrix}-1&-\frac{1}{2}&-\frac{1}{3}\\1&0&0\\0&\frac{1}{2}&0\\0&0&\frac{1}{3}\end{pmatrix}=\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}=[TU]_\gamma$ and

$$[U]_\gamma^\beta[T]_\beta^\gamma=\begin{pmatrix}-1&-\frac{1}{2}&-\frac{1}{3}\\1&0&0\\0&\frac{1}{2}&0\\0&0&\frac{1}{3}\end{pmatrix}\begin{pmatrix}0&1&0&0\\0&0&2&0\\0&0&0&3\end{pmatrix}=\begin{pmatrix}0&-1&-1&-1\\0&1&0&0\\0&0&1&0\\0&0&0&1\end{pmatrix}=[UT]_\beta.$$

Clearly, $\ker(T)=\boldsymbol{R}$, $\ker(U)=0$. Also, $\ker(UT)=\boldsymbol{R}$. This verifies Theorem 4.2.12.

Example 4.2.17

Let $T : P_3 \to P_3$ be differentiation mapping defined by $T(p(x)) = p'(x)$ for all $p(x) \in P_3$, i.e., $T(a_0 + a_1x + a_2x^2) = a_1 + 2a_2x$. Let $\beta = \{1, x, x^2\}$ be the standard basis of $P_3$. We have

$$T(1) = 0, \quad T(x) = 1, \quad T(x^2) = 2x$$

Then $[T]_\beta = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$.

Hence $\left[T^3\right]_\beta = \left([T]_\beta\right)^3 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}^3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ and

$$[T^3(a_0 + a_1x + a_2x^2)]_\beta = \left[T^3\right]_\beta [a_0 + a_1x + a_2x^2]_\beta = \left([T]_\beta\right)^3 \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

That means $T^3$ is a zero transformation and it sends all degree 2 polynomials to 0. This can be explained by Calculus. Derivative of a degree 2 polynomial 3 times must be 0, i.e.,

$$\frac{d^3}{dx^3}(a_0 + a_1x + a_2x^2) = 0.$$

Example 4.2.18 (Rotation)

Let $T : \boldsymbol{R}^2 \to \boldsymbol{R}^2$ be rotation mapping defined by counterclockwise rotating vector $\begin{pmatrix} x \\ y \end{pmatrix}$ by angle $\theta$. Let $\beta = \left\{\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right\}$ be the standard basis of $\boldsymbol{R}^2$. We have

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Let $U = T^n : \boldsymbol{R}^2 \to \boldsymbol{R}^2$ be rotation mapping defined by counterclockwise rotating vector $\begin{pmatrix} x \\ y \end{pmatrix}$ by angle $\underbrace{\theta + \theta + \cdots + \theta}_{n \text{ times}} = n\theta$ where $n$ is a positive integer.

$$\underbrace{\boldsymbol{R}^2 \xrightarrow[\text{rotate } \theta]{T} \boldsymbol{R}^2 \xrightarrow[\text{rotate } \theta]{T} \cdots \xrightarrow[\text{rotate } \theta]{T} \boldsymbol{R}^2 \xrightarrow[\text{rotate } \theta]{T} \boldsymbol{R}^2}_{n \text{ times}} = \boldsymbol{R}^2 \xrightarrow[\text{rotate } n\theta]{U} \boldsymbol{R}^2$$

We have $U\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \cos n\theta & -\sin n\theta \\ \sin n\theta & \cos n\theta \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$.

Then $\begin{pmatrix} \cos n\theta & -\sin n\theta \\ \sin n\theta & \cos n\theta \end{pmatrix} = [U]_\beta = \left[T^n\right]_\beta = \left([T]_\beta\right)^n = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}^n$.

Example 4.2.19

Define $T : span\left\{e^x \sin x, e^x \cos x\right\} \to span\left\{e^x \sin x, e^x \cos x\right\}$ by $T(f(x)) = f'(x)$. Let $\beta = \left\{e^x \sin x, e^x \cos x\right\}$ be the basis. We have

$T(e^x \sin x) = (e^x \sin x)' = e^x \sin x + e^x \cos x$ and $T(e^x \cos x) = (e^x \cos x)' = -e^x \sin x + e^x \cos x$.

Then $[T]_\beta = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \sqrt{2}\begin{pmatrix} \cos\frac{\pi}{4} & -\sin\frac{\pi}{4} \\ \sin\frac{\pi}{4} & \cos\frac{\pi}{4} \end{pmatrix}$.

$$\begin{aligned}
\left[T^5\right]_\beta &= \left([T]_\beta\right)^5 \\
&= \left(\sqrt{2}\begin{pmatrix} \cos\frac{\pi}{4} & -\sin\frac{\pi}{4} \\ \sin\frac{\pi}{4} & \cos\frac{\pi}{4} \end{pmatrix}\right)^5 \\
&= \sqrt{2}^5\begin{pmatrix} \cos\frac{\pi}{4} & -\sin\frac{\pi}{4} \\ \sin\frac{\pi}{4} & \cos\frac{\pi}{4} \end{pmatrix}^5 \\
&= 4\sqrt{2}\begin{pmatrix} \cos\frac{5\pi}{4} & -\sin\frac{5\pi}{4} \\ \sin\frac{5\pi}{4} & \cos\frac{5\pi}{4} \end{pmatrix} \\
&= 4\sqrt{2}\begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \\
&= \begin{pmatrix} -4 & 4 \\ -4 & -4 \end{pmatrix}.
\end{aligned}$$

$$\left[T^5(e^x \sin x)\right]_\beta = \left[T^5\right]_\beta [e^x \sin x]_\beta = \begin{pmatrix} -4 & 4 \\ -4 & -4 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -4 \\ -4 \end{pmatrix},$$

i.e., $\dfrac{d^5}{dx^5}(e^x \sin x) = T^5(e^x \sin x) = -4e^x \sin x - 4e^x \cos x$

$$\left[T^5(e^x\cos x)\right]_\beta = \left[T^5\right]_\beta [e^x\cos x]_\beta = \begin{pmatrix} -4 & 4 \\ -4 & -4 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ -4 \end{pmatrix},$$

i.e., $\dfrac{d^5}{dx^5}(e^x\cos x) = T^5(e^x\cos x) = 4e^x\sin x - 4e^x\cos x.$

These can be explained by Calculus. Recall

$(\sin x)' = \cos x \quad (\sin x)'' = -\sin x \quad (\sin x)''' = -\cos x \quad (\sin x)^{(4)} = \sin x \quad (\sin x)^{(5)} = \cos x$ and
$(\cos x)' = -\sin x \quad (\cos x)'' = -\cos x \quad (\cos x)''' = \sin x \quad (\cos x)^{(4)} = \cos x \quad (\cos x)^{(5)} = -\sin x$

and Leibniz's rule

$$(f\cdot g)^{(n)} = \sum_{k=0}^{n}\binom{n}{k} f^{(n-k)}g^{(k)}.$$

$$\begin{aligned}
\frac{d^5(e^x\sin x)}{dx^5} &= \left(\frac{d^5e^x}{dx^5}\right)\sin x + 5\left(\frac{d^4e^x}{dx^4}\right)\left(\frac{d\sin x}{dx}\right) + 10\left(\frac{d^3e^x}{dx^3}\right)\left(\frac{d^2\sin x}{dx^2}\right) + 10\left(\frac{d^2e^x}{dx^2}\right)\left(\frac{d^3\sin x}{dx^3}\right) \\
&\quad + 5\left(\frac{de^x}{dx}\right)\left(\frac{d^4\sin x}{dx^4}\right) + e^x\left(\frac{d^5\sin x}{dx^5}\right) \\
&= e^x\sin x + 5e^x\cos x - 10e^x\sin x - 10e^x\cos x + 5e^x\sin x + e^x\cos x \\
&= -4e^x\sin x - 4e^x\cos x \\
\frac{d^5(e^x\cos x)}{dx^5} &= \left(\frac{d^5e^x}{dx^5}\right)\cos x + 5\left(\frac{d^4e^x}{dx^4}\right)\left(\frac{d\cos x}{dx}\right) + 10\left(\frac{d^3e^x}{dx^3}\right)\left(\frac{d^2\cos x}{dx^2}\right) + 10\left(\frac{d^2e^x}{dx^2}\right)\left(\frac{d^3\cos x}{dx^3}\right) \\
&\quad + 5\left(\frac{de^x}{dx}\right)\left(\frac{d^4\cos x}{dx^4}\right) + e^x\left(\frac{d^5\cos x}{dx^5}\right) \\
&= e^x\cos x - 5e^x\sin x - 10e^x\cos x + 10e^x\sin x + 5e^x\cos x - e^x\sin x \\
&= 4e^x\sin x - 4e^x\cos x
\end{aligned}$$

Example 4.2.20 (Reflection)

Let $T$ be **reflection along** $y = mx$, i.e., $T:\boldsymbol{R}^2 \to \boldsymbol{R}^2$ a linear transformation defined by $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \dfrac{1-m^2}{1+m^2} & \dfrac{2m}{1+m^2} \\ \dfrac{2m}{1+m^2} & -\dfrac{1-m^2}{1+m^2} \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$. Let $\beta = \left\{\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right\}$ be the standard basis of $\boldsymbol{R}^2$.

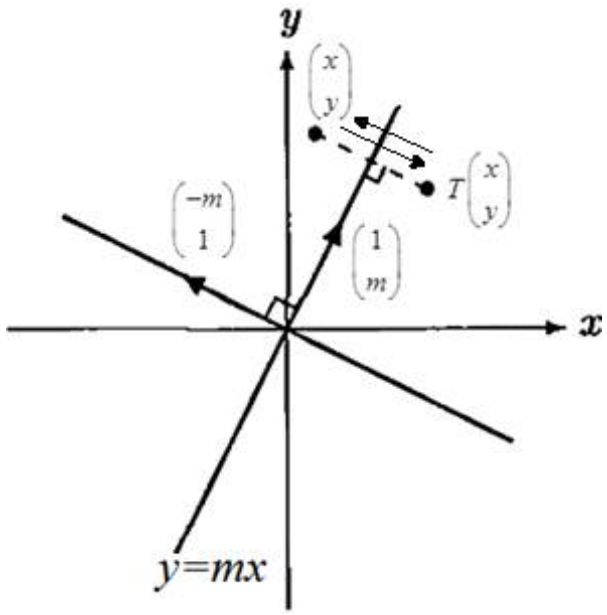
$$\left[T^2\right]_\beta = \left([T]_\beta\right)^2$$

$$= \begin{pmatrix} \frac{1-m^2}{1+m^2} & \frac{2m}{1+m^2} \\ \frac{2m}{1+m^2} & -\frac{1-m^2}{1+m^2} \end{pmatrix} \begin{pmatrix} \frac{1-m^2}{1+m^2} & \frac{2m}{1+m^2} \\ \frac{2m}{1+m^2} & -\frac{1-m^2}{1+m^2} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{(1-m^2)^2+(2m)^2}{(1+m^2)^2} & \frac{2m(1-m^2)-(1-m^2)2m}{(1+m^2)^2} \\ \frac{2m(1-m^2)-(1-m^2)2m}{(1+m^2)^2} & \frac{(1-m^2)^2+(2m)^2}{(1+m^2)^2} \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Hence $T^2 = I$ and $T = T^{-1}$. That means, reflection twice will go back to the original position.




Example 4.2.21 (Projection)

Let $T$ be **projection along** $y = mx$, i.e., $T : \boldsymbol{R}^2 \to \boldsymbol{R}^2$ a linear transformation defined by $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{1+m^2} & \frac{m}{1+m^2} \\ \frac{m}{1+m^2} & \frac{m^2}{1+m^2} \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$. Let $\beta = \left\{\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right\}$ be the standard basis of $\boldsymbol{R}^2$.

$$\left[T^2\right]_\beta = \left([T]_\beta\right)^2$$

$$= \begin{pmatrix} \frac{1}{1+m^2} & \frac{m}{1+m^2} \\ \frac{m}{1+m^2} & \frac{m^2}{1+m^2} \end{pmatrix} \begin{pmatrix} \frac{1}{1+m^2} & \frac{m}{1+m^2} \\ \frac{m}{1+m^2} & \frac{m^2}{1+m^2} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1+m^2}{(1+m^2)^2} & \frac{m+m^3}{(1+m^2)^2} \\ \frac{m+m^3}{(1+m^2)^2} & \frac{m^2+m^4}{(1+m^2)^2} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{1+m^2} & \frac{m}{1+m^2} \\ \frac{m}{1+m^2} & \frac{m^2}{1+m^2} \end{pmatrix} = [T]_\beta$$

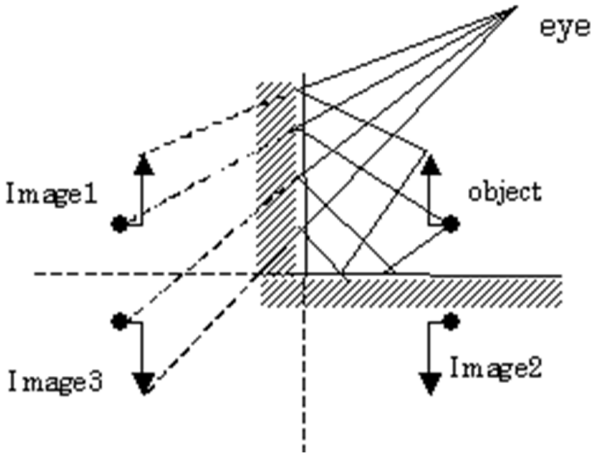
Hence $T^2 = T$. That means projection of a shadow is shadow itself.

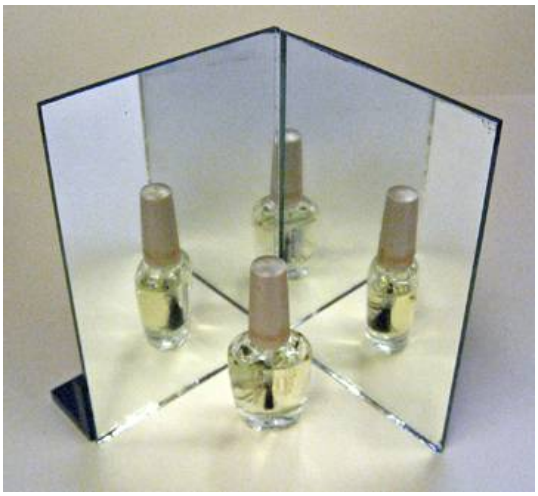
Example 4.2.22

Let $T : \boldsymbol{R}^2 \to \boldsymbol{R}^2$ be the reflection about $x$-axis and $U : \boldsymbol{R}^2 \to \boldsymbol{R}^2$ be the reflection about $y$-axis, i.e., $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ -y \end{pmatrix}$ and $U\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x \\ y \end{pmatrix}$.

Then $UT\begin{pmatrix} x \\ y \end{pmatrix} = U\begin{pmatrix} x \\ -y \end{pmatrix} = \begin{pmatrix} -x \\ -y \end{pmatrix}$ and $TU\begin{pmatrix} x \\ y \end{pmatrix} = T\begin{pmatrix} -x \\ y \end{pmatrix} = \begin{pmatrix} -x \\ -y \end{pmatrix}$. Hence $UT = TU$.

reflect about $x$-axis then reflect about $y$-axis = reflect about $y$-axis then reflect about $x$-axis



Theorem 4.2.23

Let $V$ and $W$ be finite dimensional vector spaces having ordered bases $\beta$ and $\gamma$ respectively, and let $T: V \to W$ be linear. Then $T$ is invertible if and only if $[T]_\beta^\gamma$ is invertible. Furthermore, if $T$ is invertible, then $T^{-1}: W \to V$ is linear, $\dim V = \dim W$ and $\left[T^{-1}\right]_\gamma^\beta = \left([T]_\beta^\gamma\right)^{-1}$.

Example 4.2.24

Define $T: span\left\{\begin{pmatrix}1\\0\\0\\0\end{pmatrix},\begin{pmatrix}1\\1\\1\\1\end{pmatrix}\right\} \to span\left\{\begin{pmatrix}1\\0\\0\end{pmatrix},\begin{pmatrix}1\\1\\1\end{pmatrix}\right\}$ by $T(\boldsymbol{v}) = \begin{pmatrix}1&1&1&1\\1&0&0&0\\1&0&0&0\end{pmatrix}\boldsymbol{v}$. Let $\beta = \left\{\begin{pmatrix}1\\0\\0\\0\end{pmatrix},\begin{pmatrix}1\\1\\1\\1\end{pmatrix}\right\}$ and $\gamma = \left\{\begin{pmatrix}1\\0\\0\end{pmatrix},\begin{pmatrix}1\\1\\1\end{pmatrix}\right\}$. Show that $T$ is invertible. Find $T^{-1}\begin{pmatrix}1\\0\\0\end{pmatrix}$, $T^{-1}\begin{pmatrix}1\\1\\1\end{pmatrix}$ and $T^{-1}\begin{pmatrix}a\\b\\b\end{pmatrix}$.

Solution

We have

$$T\begin{pmatrix}1\\0\\0\\0\end{pmatrix} = \begin{pmatrix}1&1&1&1\\1&0&0&0\\1&0&0&0\end{pmatrix}\begin{pmatrix}1\\0\\0\\0\end{pmatrix} = \begin{pmatrix}1\\1\\1\end{pmatrix} = 0\begin{pmatrix}1\\0\\0\end{pmatrix} + 1\begin{pmatrix}1\\1\\1\end{pmatrix} \quad \text{and} \quad T\begin{pmatrix}1\\1\\1\\1\end{pmatrix} = \begin{pmatrix}1&1&1&1\\1&0&0&0\\1&0&0&0\end{pmatrix}\begin{pmatrix}1\\1\\1\\1\end{pmatrix} = \begin{pmatrix}4\\1\\1\end{pmatrix} = 3\begin{pmatrix}1\\0\\0\end{pmatrix} + 1\begin{pmatrix}1\\1\\1\end{pmatrix}.$$

Then $[T]_\beta^\gamma = \begin{pmatrix}0&3\\1&1\end{pmatrix}$.

Since $\det[T]_\beta^\gamma = \det\begin{pmatrix}0&3\\1&1\end{pmatrix} = -3 \neq 0$, $T$ is invertible, i.e. $T^{-1}: span\left\{\begin{pmatrix}1\\0\\0\end{pmatrix},\begin{pmatrix}1\\1\\1\end{pmatrix}\right\} \to span\left\{\begin{pmatrix}1\\0\\0\\0\end{pmatrix},\begin{pmatrix}1\\1\\1\\1\end{pmatrix}\right\}$ exists.

By Theorem 4.2.23, $\left[T^{-1}\right]_\gamma^\beta = \left([T]_\beta^\gamma\right)^{-1} = \begin{pmatrix}0&3\\1&1\end{pmatrix}^{-1} = \frac{-1}{3}\begin{pmatrix}1&-3\\-1&0\end{pmatrix} = \begin{pmatrix}-\frac{1}{3}&1\\\frac{1}{3}&0\end{pmatrix}$.

Then $T^{-1}\begin{pmatrix}1\\0\\0\end{pmatrix}=-\frac{1}{3}\begin{pmatrix}1\\0\\0\\0\end{pmatrix}+\frac{1}{3}\begin{pmatrix}1\\1\\1\\1\end{pmatrix}=\begin{pmatrix}0\\\frac{1}{3}\\\frac{1}{3}\\\frac{1}{3}\end{pmatrix}$ and $T^{-1}\begin{pmatrix}1\\1\\1\end{pmatrix}=\begin{pmatrix}1\\0\\0\\0\end{pmatrix}$.

$$T^{-1}\begin{pmatrix}a\\b\\b\end{pmatrix}=T^{-1}\left(\begin{pmatrix}a-b\\0\\0\end{pmatrix}+\begin{pmatrix}b\\b\\b\end{pmatrix}\right)=(a-b)T^{-1}\begin{pmatrix}1\\0\\0\end{pmatrix}+bT^{-1}\begin{pmatrix}1\\1\\1\end{pmatrix}=\begin{pmatrix}0\\\frac{1}{3}(a-b)\\\frac{1}{3}(a-b)\\\frac{1}{3}(a-b)\end{pmatrix}+\begin{pmatrix}b\\0\\0\\0\end{pmatrix}=\begin{pmatrix}b\\\frac{1}{3}(a-b)\\\frac{1}{3}(a-b)\\\frac{1}{3}(a-b)\end{pmatrix}.$$

Corollary 4.2.25

Let $V$ be finite dimensional vector spaces with an ordered basis $\beta$, and let $T:V\to V$ be linear. Then $T$ is invertible if and only if $[T]_\beta$ is invertible. Furthermore, $\left[T^{-1}\right]_\beta=\left([T]_\beta\right)^{-1}$.

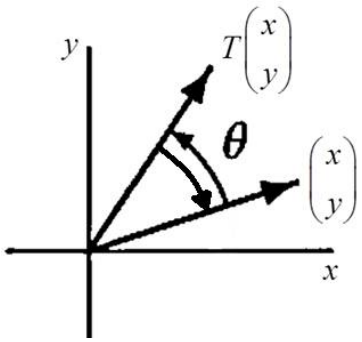
Example 4.2.26 (Rotation)

Let $T:\boldsymbol{R}^2\to\boldsymbol{R}^2$ be rotation mapping defined by counterclockwise rotating vector $\begin{pmatrix}x\\y\end{pmatrix}$ by angle $\theta$. Let $\beta=\left\{\begin{pmatrix}1\\0\end{pmatrix},\begin{pmatrix}0\\1\end{pmatrix}\right\}$ be the standard basis of $\boldsymbol{R}^2$. Then

$$T\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}\cos\theta&-\sin\theta\\\sin\theta&\cos\theta\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}.$$

Let $U:\boldsymbol{R}^2\to\boldsymbol{R}^2$ be rotation mapping defined by counterclockwise rotating vector $\begin{pmatrix}x\\y\end{pmatrix}$ by angle $-\theta$, i.e., clockwise rotating vector $\begin{pmatrix}x\\y\end{pmatrix}$ by angle $\theta$. Then

$$U\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}\cos(-\theta)&-\sin(-\theta)\\\sin(-\theta)&\cos(-\theta)\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}.$$




Clearly, $U = T^{-1}$.

$$\boldsymbol{R}^2 \xrightarrow[\text{rotate } \theta]{T} \boldsymbol{R}^2 \xrightarrow[\text{rotate } -\theta]{U} \boldsymbol{R}^2 \quad \text{and} \quad \boldsymbol{R}^2 \xrightarrow[\text{rotate } -\theta]{U} \boldsymbol{R}^2 \xrightarrow[\text{rotate } \theta]{T} \boldsymbol{R}^2$$

$$\left[T^{-1}\right]_\beta = [U]_\beta = \begin{pmatrix} \cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta) \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}^{-1} = \left([T]_\beta\right)^{-1}.$$

Example 4.2.27

Let $T : M_{2\times 2} \to M_{2\times 2}$ be linear transformation defined by $T(A) = \begin{pmatrix} a & b \\ c & d \end{pmatrix} A$ for all $A \in M_{2\times 2}$. Let $\beta = \left\{\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\right\}$ be the standard basis of $M_{2\times 2}$. We have

$$T\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \underline{\qquad\qquad}$$

$$T\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \underline{\qquad\qquad}$$

$$T\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \underline{\qquad\qquad}$$

$$T\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \underline{\qquad\qquad}$$

Then $[T]_\beta = \left(\begin{array}{cc|cc} a & b & 0 & 0 \\ c & d & 0 & 0 \\ \hline 0 & 0 & a & b \\ 0 & 0 & c & d \end{array}\right)$.

Suppose $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is invertible. Let $U : M_{2\times 2} \to M_{2\times 2}$ be linear transformation defined by $U(A) = \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} A = \begin{pmatrix} \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{pmatrix} A$ for all $A \in M_{2\times 2}$.

Clearly, $U = T^{-1}$.

$$\underset{A}{M_{2\times 2}} \xrightarrow{T} \underset{\begin{pmatrix} a & b \\ c & d \end{pmatrix} A}{M_{2\times 2}} \xrightarrow{U} \underset{\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1}\begin{pmatrix} a & b \\ c & d \end{pmatrix} A = A}{M_{2\times 2}} \quad \text{and}$$

$$\underset{A}{M_{2\times 2}} \xrightarrow{U} \underset{\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} A}{M_{2\times 2}} \xrightarrow{T} \underset{\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} A = A}{M_{2\times 2}}$$

$$\left[T^{-1}\right]_\beta = [U]_\beta = \left(\begin{array}{cc|cc} \frac{d}{ad-bc} & \frac{-b}{ad-bc} & 0 & 0 \\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} & 0 & 0 \\ \hline 0 & 0 & \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ 0 & 0 & \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{array}\right) = \left(\begin{array}{cc|cc} \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} & & 0 & 0 \\ & & 0 & 0 \\ \hline 0 & 0 & \multicolumn{2}{c}{\multirow{2}{*}{$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1}$}} \\ 0 & 0 & & \end{array}\right) = \left(\begin{array}{cc|cc} a & b & 0 & 0 \\ c & d & 0 & 0 \\ \hline 0 & 0 & a & b \\ 0 & 0 & c & d \end{array}\right)^{-1} = \left([T]_\beta\right)^{-1}.$$

Example 4.2.28

Let $T : P_3 \to P_3$ be linear transformation defined by $T(p(x)) = p(x+1)$ for all $p(x) \in P_3$, i.e., $T(a_0 + a_1 x + a_2 x^2) = a_0 + a_1(x+1) + a_2(x+1)^2 = (a_0 + a_1 + a_2) + (a_1 + 2a_2)x + a_2 x^2$. Let $\beta = \{1, x, x^2\}$ be the standard basis of $P_3$. We have

$$T(1) = 1, \quad T(x) = 1 + x, \quad T(x^2) = (1+x)^2 = 1 + 2x + x^2.$$

Then $[T]_\beta = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$.

Consider $T^2 : P_3 \to P_3$, $T(T(p(x))) = T(p(x+1)) = p((x+1)+1) = p(x+2)$

$$\underset{x}{P_3} \xrightarrow{T} \underset{x+1}{P_3} \xrightarrow{T} \underset{(x+1)+1=x+2}{P_3}$$

We have $T^2(1)=1,\quad T^2(x)=T(1+x)=2+x$ and

$$T^2(x^2)=T(1+2x+x^2)=T(1)+2T(x)+T(x^2)=1+2(2+x)+(1+2x+x^2)=4+4x+x^2.$$

Then $\left[T^2\right]_\beta=\begin{pmatrix}1&2&4\\0&1&4\\0&0&1\end{pmatrix}$.

$$\left([T]_\beta\right)^2=\begin{pmatrix}1&1&1\\0&1&2\\0&0&1\end{pmatrix}\begin{pmatrix}1&1&1\\0&1&2\\0&0&1\end{pmatrix}=\begin{pmatrix}1&2&4\\0&1&4\\0&0&1\end{pmatrix}=\left[T^2\right]_\beta.$$

You may find $\left([T]_\beta\right)^{-1}=\begin{pmatrix}1&1&1\\0&1&2\\0&0&1\end{pmatrix}^{-1}$ directly by row operation:

$$\left(\begin{array}{ccc|ccc}1&1&1&1&0&0\\0&1&2&0&1&0\\0&0&1&0&0&1\end{array}\right)\to\left(\begin{array}{ccc|ccc}1&1&0&1&0&-1\\0&1&0&0&1&-2\\0&0&1&0&0&1\end{array}\right)\to\left(\begin{array}{ccc|ccc}1&0&0&1&-1&1\\0&1&0&0&1&-2\\0&0&1&0&0&1\end{array}\right).$$

Hence $\left([T]_\beta\right)^{-1}=\begin{pmatrix}1&1&1\\0&1&2\\0&0&1\end{pmatrix}^{-1}=\begin{pmatrix}1&-1&1\\0&1&-2\\0&0&1\end{pmatrix}$.

On the other hand, you may try to find the inverse of $T$. Define $U:P_3\to P_3$ by $U(p(x))=p(x-1)$ for all $p(x)\in P_3$. Then

$$U(1)=1,\quad U(x)=-1+x,\quad U(x)=(-1+x)^2=1-2x+x^2.$$

Clearly, $U=T^{-1}$.

$$\underset{x}{P_3}\xrightarrow{T}\underset{x+1}{P_3}\xrightarrow{U}\underset{(x-1)+1=x}{P_3}\qquad\text{and}\qquad\underset{x}{P_3}\xrightarrow{U}\underset{x-1}{P_3}\xrightarrow{T}\underset{(x+1)-1=x}{P_3}$$

Hence $[U]_\beta=\left[T^{-1}\right]_\beta=\begin{pmatrix}1&-1&1\\0&1&-2\\0&0&1\end{pmatrix}$. Therefore, $\left[T^{-1}\right]_\beta=\left([T]_\beta\right)^{-1}$.

## Section 4.3 Similarity

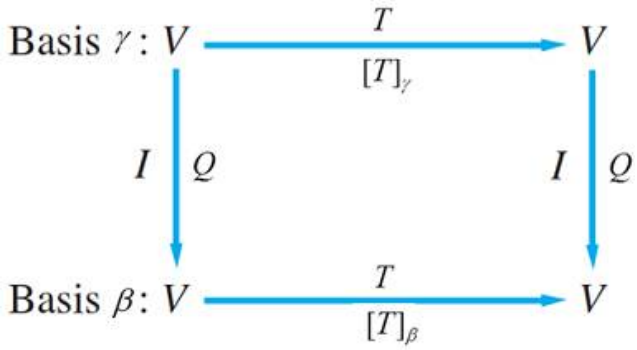
Definition 4.3.1

Let $A$ and $B$ be $n\times n$ matrices. $B$ is said to be **similar** to $A$ if there exists nonsingular matrix $Q$ such that $B=Q^{-1}AQ$.

Theorem 4.3.2

Let $T:V\to V$ be a linear operator on a finite dimensional vector space $V$, and let $\beta$ and $\gamma$ be bases of $V$. Suppose that $Q$ is the transition matrix from $\gamma$ to $\beta$, i.e., $Q=[I]_\gamma^\beta$. Then

$$[T]_\gamma=Q^{-1}[T]_\beta Q.$$

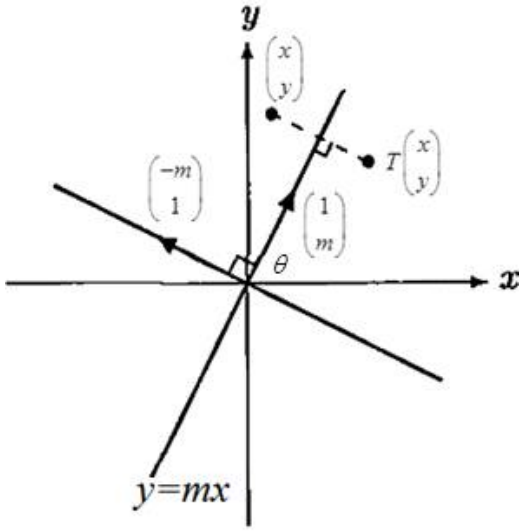



Proof

$Q[T]_\gamma=\underbrace{[I]_\gamma^\beta[T]_\gamma=[I\circ T]_\gamma^\beta}_{\text{Theorem 4.2.13}}=[T]_\gamma^\beta=\underbrace{[T\circ I]_\gamma^\beta=[T]_\beta[I]_\gamma^\beta}_{\text{Theorem 4.2.13}}=[T]_\beta Q.$ Hence $[T]_\gamma=Q^{-1}[T]_\beta Q.$

Example 4.3.3 (Reflection)

Let $T$ be **reflection along** $y=mx$, i.e., $T:\boldsymbol{R}^2\to\boldsymbol{R}^2$ a linear transformation defined by $T\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}\frac{1-m^2}{1+m^2} & \frac{2m}{1+m^2}\\ \frac{2m}{1+m^2} & -\frac{1-m^2}{1+m^2}\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}$. Let $\beta=\left\{\begin{pmatrix}\frac{1}{\sqrt{1+m^2}}\\ \frac{m}{\sqrt{1+m^2}}\end{pmatrix},\begin{pmatrix}-\frac{m}{\sqrt{1+m^2}}\\ \frac{1}{\sqrt{1+m^2}}\end{pmatrix}\right\}$ and $\gamma=\left\{\begin{pmatrix}1\\0\end{pmatrix},\begin{pmatrix}0\\1\end{pmatrix}\right\}$ be bases of $\boldsymbol{R}^2$.




Since $m = \tan\theta$, we have $\sin\theta = \dfrac{m}{\sqrt{1+m^2}}$ and $\cos\theta = \dfrac{1}{\sqrt{1+m^2}}$. Then $\beta = \left\{\begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}, \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}\right\}$.

Clearly, $Q^{-1} = [I]_\beta^\gamma = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ and $Q = [I]_\gamma^\beta = \begin{pmatrix} \cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta) \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$.

Clearly, $[T]_\gamma = \begin{pmatrix} \dfrac{1-m^2}{1+m^2} & \dfrac{2m}{1+m^2} \\ \dfrac{2m}{1+m^2} & -\dfrac{1-m^2}{1+m^2} \end{pmatrix} = \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$.

$$T\begin{pmatrix} \dfrac{1}{\sqrt{1+m^2}} \\ \dfrac{m}{\sqrt{1+m^2}} \end{pmatrix} = \frac{1}{\sqrt{1+m^2}} T\begin{pmatrix} 1 \\ m \end{pmatrix} = \frac{1}{\sqrt{1+m^2}}\begin{pmatrix} 1 \\ m \end{pmatrix} = \begin{pmatrix} \dfrac{1}{\sqrt{1+m^2}} \\ \dfrac{m}{\sqrt{1+m^2}} \end{pmatrix} \quad \text{and}$$

$$T\begin{pmatrix} -\dfrac{m}{\sqrt{1+m^2}} \\ \dfrac{1}{\sqrt{1+m^2}} \end{pmatrix} = \frac{1}{\sqrt{1+m^2}} T\begin{pmatrix} -m \\ 1 \end{pmatrix} = \frac{-1}{\sqrt{1+m^2}}\begin{pmatrix} -m \\ 1 \end{pmatrix} = -\begin{pmatrix} -\dfrac{m}{\sqrt{1+m^2}} \\ \dfrac{1}{\sqrt{1+m^2}} \end{pmatrix}.$$

Then $[T]_\beta = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

$$Q^{-1}[T]_\beta Q = \overbrace{\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}}^{\text{rotate } \theta \text{ counterclockwise}} \overbrace{\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}}^{\text{reflect along } x\text{-axis}} \overbrace{\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}}^{\text{rotate } \theta \text{ clockwise}}$$

$$= \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$$

$$= \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix} = [T]_\gamma .$$

Example 4.3.4

Let $A$ be a $2\times 2$ matrix and $\boldsymbol{v} \in \boldsymbol{R}^2$ such that $\boldsymbol{v}$ and $A\boldsymbol{v}$ are linearly independent in $\boldsymbol{R}^2$. Suppose $A^2\boldsymbol{v} = 3A\boldsymbol{v} - 2\boldsymbol{v}$.

(a) Show that $A^2 - 3A + 2I_2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$.

(b) Find $\det(A + I_2)$.

(c) Find $(A + I_2)^{-1}(A\boldsymbol{v})$ in terms of $\boldsymbol{v}$ and $A\boldsymbol{v}$.

Solution

(a) Since $A^2\boldsymbol{v} - 3A\boldsymbol{v} + 2\boldsymbol{v} = \boldsymbol{0}$, $A^2(A\boldsymbol{v}) - 3A(A\boldsymbol{v}) + 2(A\boldsymbol{v}) = \boldsymbol{0}$ and $\{A\boldsymbol{v}, \boldsymbol{v}\}$ form a basis of $\boldsymbol{R}^2$, $(A^2 - 3A + 2I_2)\boldsymbol{u} = \boldsymbol{0}$ for any $\boldsymbol{u} \in \boldsymbol{R}^2$. Then $A^2 - 3A + 2I_2$ is a zero transformation, that is,

$$A^2 - 3A + 2I_2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

(b) Define $T : \boldsymbol{R}^2 \to \boldsymbol{R}^2$ a linear transformation by $T\begin{pmatrix} x \\ y \end{pmatrix} = (A + I_2)\begin{pmatrix} x \\ y \end{pmatrix}$. Since $\dim \boldsymbol{R}^2 = 2$ and $\boldsymbol{v}$ and $A\boldsymbol{v}$ are linearly independent, $\{A\boldsymbol{v}, \boldsymbol{v}\}$ is a basis of $\boldsymbol{R}^2$. Let $\beta = \{A\boldsymbol{v}, \boldsymbol{v}\}$ and $\gamma = \left\{\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right\}$ be bases of $\boldsymbol{R}^2$. Clearly,

$$(A + I_2)(A\boldsymbol{v}) = A^2\boldsymbol{v} + A\boldsymbol{v} = 4A\boldsymbol{v} - 2\boldsymbol{v}.$$
$$(A + I_2)(\boldsymbol{v}) = A\boldsymbol{v} + \boldsymbol{v}.$$

Then $[A + I_2]_\beta = \begin{pmatrix} 4 & 1 \\ -2 & 1 \end{pmatrix}$

$$\det(A + I_2) = \det[T]_\gamma = \det\left(Q^{-1}[T]_\beta Q\right) = \det[T]_\beta = \det[A + I_2]_\beta = \det\begin{pmatrix} 4 & 1 \\ -2 & 1 \end{pmatrix} = 6.$$

(c) By Corollary 4.2.25, $\left[(A+I_2)^{-1}\right]_\beta = [A+I_2]_\beta^{-1} = \begin{pmatrix} 4 & 1 \\ -2 & 1 \end{pmatrix}^{-1} = \frac{1}{6}\begin{pmatrix} 1 & -1 \\ 2 & 4 \end{pmatrix}$. We have

$$\left[(A+I_2)^{-1}(A\boldsymbol{v})\right]_\beta = \left[(A+I_2)^{-1}\right]_\beta [A\boldsymbol{v}]_\beta = \frac{1}{6}\begin{pmatrix} 1 & -1 \\ 2 & 4 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{6}\begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

Hence $(A+I_2)^{-1}(A\boldsymbol{v}) = \frac{1}{6}A\boldsymbol{v} + \frac{1}{3}\boldsymbol{v}$.

Remark 4.3.5

$$\begin{aligned} A^2\boldsymbol{v} &= 3A\boldsymbol{v} - 2\boldsymbol{v} \\ A^2\boldsymbol{v} - 3A\boldsymbol{v} + 2\boldsymbol{v} &= \boldsymbol{0} \\ (A^2 - 3A + 2I_2)\boldsymbol{v} &= \boldsymbol{0} \\ (A^2 - 3A + 2I_2)\boldsymbol{v}\boldsymbol{v}^{-1} &= \boldsymbol{0}\boldsymbol{v}^{-1} \\ A^2 - 3A + 2I_2 &= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}. \end{aligned}$$

Such argument is not correct because there is no inverse for $\boldsymbol{v}$.

Even $A^2 - 3A + 2I_2 = (A - I_2)(A - 2I_2) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ together with $\boldsymbol{v}$ and $A\boldsymbol{v}$ linearly independent in $\boldsymbol{R}^2$, it does not mean $A = I_2$ or $A = 2I_2$. There are infinite many choices for $A$ and $\boldsymbol{v}$ satisfying such condition but $A \neq I_2$ and $A \neq 2I_2$.

Consider $A = \begin{pmatrix} 3 & 1 \\ -2 & 0 \end{pmatrix}$ and $\boldsymbol{v} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Clearly, $\boldsymbol{v} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $A\boldsymbol{v} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ are linearly independent in $\boldsymbol{R}^2$. Furthermore,

$$A^2\boldsymbol{v} = A \cdot A\boldsymbol{v} = \begin{pmatrix} 3 & 1 \\ -2 & 0 \end{pmatrix}\begin{pmatrix} 3 \\ -2 \end{pmatrix} = \begin{pmatrix} 7 \\ -6 \end{pmatrix} = 3\begin{pmatrix} 3 \\ -2 \end{pmatrix} - 2\begin{pmatrix} 1 \\ 0 \end{pmatrix} = 3A\boldsymbol{v} - 2\boldsymbol{v}.$$

But $A \neq I_2$ and $A \neq 2I_2$.

In (c), keep in mind that

$$(A+I_2)^{-1}(A\boldsymbol{v}) \neq (A^{-1} + I_2^{-1})(A\boldsymbol{v}) = \boldsymbol{v} + A\boldsymbol{v}.$$